

# Depression Connect Forum

Master Analysis Report

Dataset variant: New Data Only

EDA · Time series · CDS prevalence · LIWC · Longitudinal trends · LIWC-22 validation

# **1. EDA - All Users**

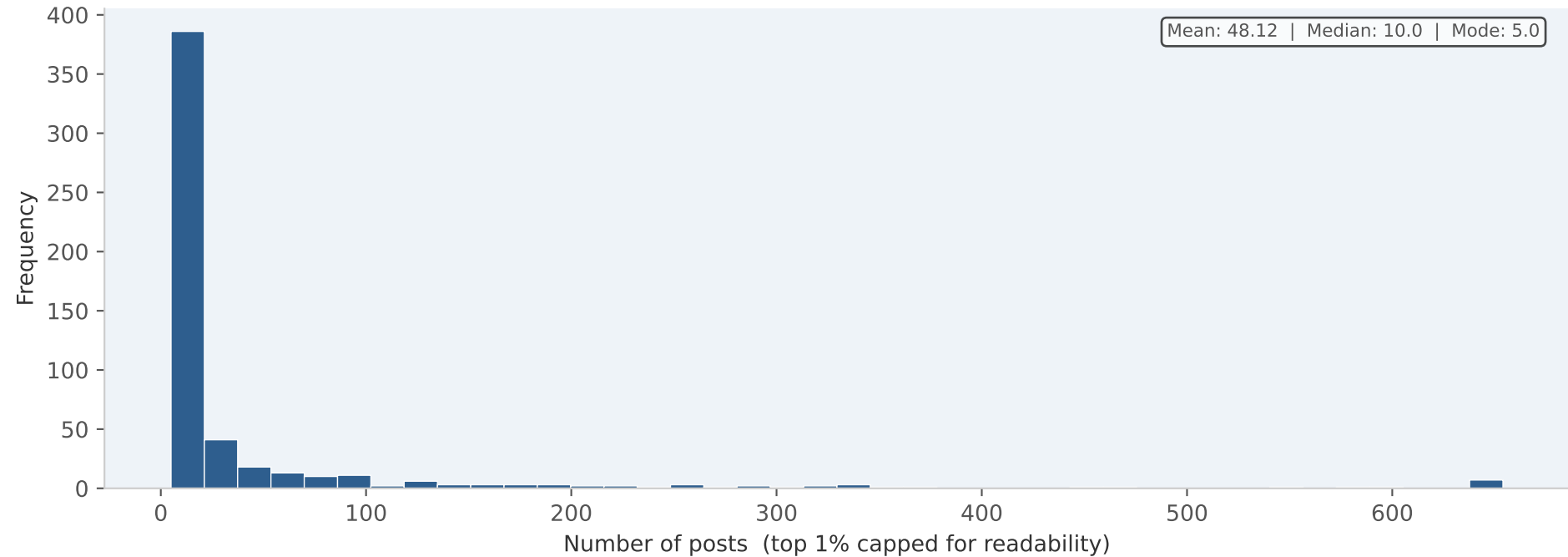
# **Depression Connect Forum**

Exploratory Data Analysis Report — All Users

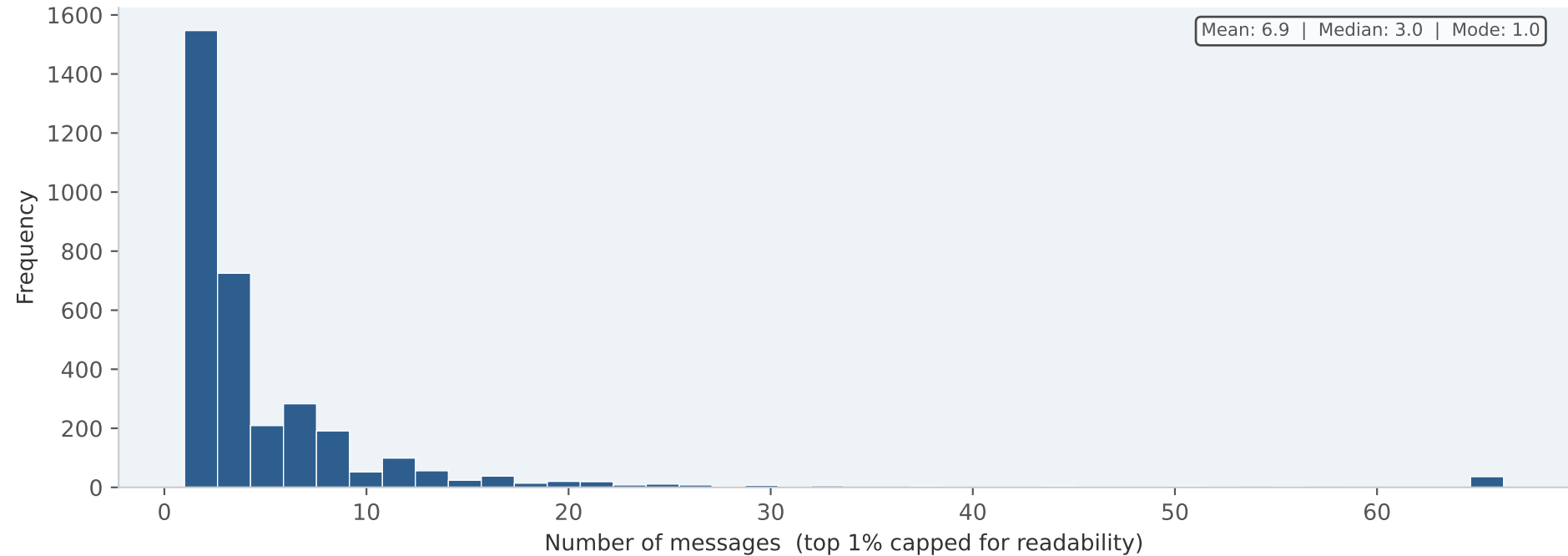
## Summary Statistics

| Metric                      | Mean    | Median | Mode  | Min | Max    |
|-----------------------------|---------|--------|-------|-----|--------|
| Posts per user              | 48.12   | 10.0   | 5.0   | 5   | 1486   |
| Messages per thread (total) | 6.9     | 3.0    | 1.0   | 1   | 605    |
| Replies per thread          | 8.13    | 3.0    | 1.0   | 1   | 604    |
| Activity span (days)        | 209.74  | 50.5   | 0.0   | 0   | 2493   |
| Active days per user        | 20.8    | 5.0    | 3.0   | 1   | 430    |
| Words per post              | 93.51   | 62.0   | 5.0   | 0   | 2486   |
| Words per user (total)      | 4499.97 | 980.0  | 213.0 | 75  | 127497 |
| 'ik' count per user         | 197.84  | 52.0   | 12.0  | 0   | 5382   |
| 'ik' % per user             | 5.16    | 5.2    | 4.0   | 0.0 | 11.24  |
| 'mijn' count per user       | 42.6    | 11.0   | 2.0   | 0   | 2230   |
| 'mijn' % per user           | 1.06    | 0.99   | 0.0   | 0.0 | 3.44   |

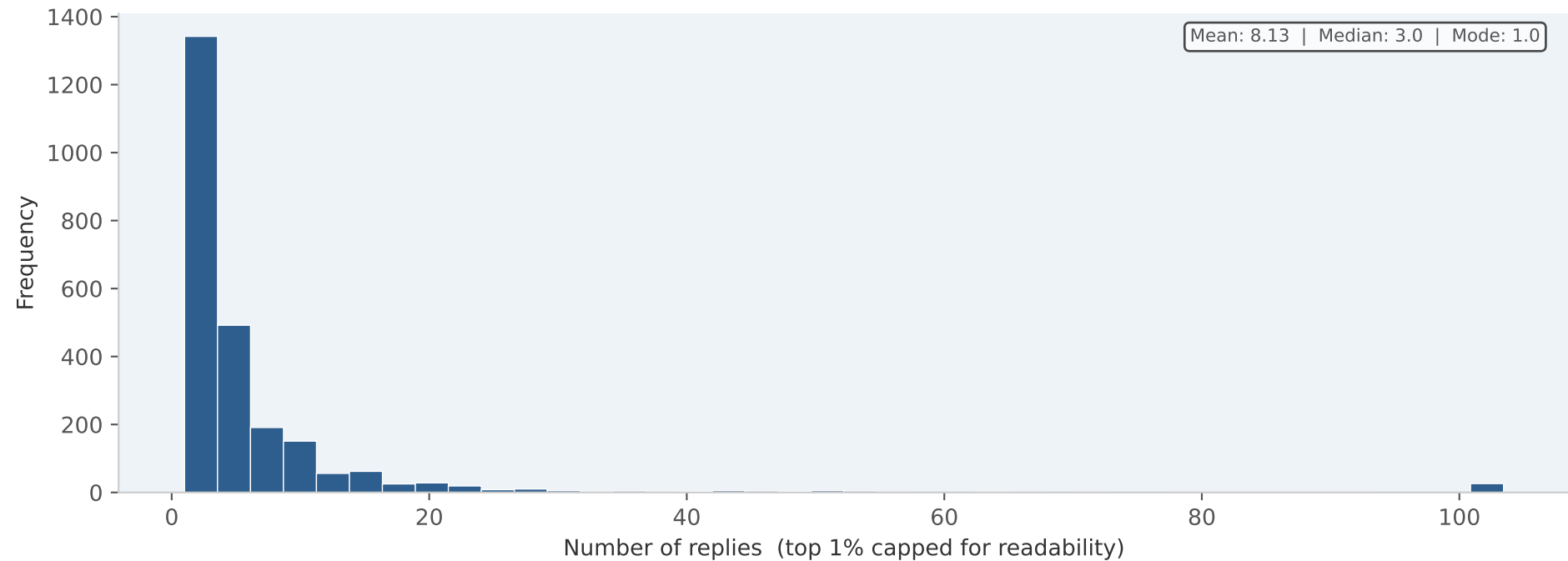
## Distribution: Posts per User



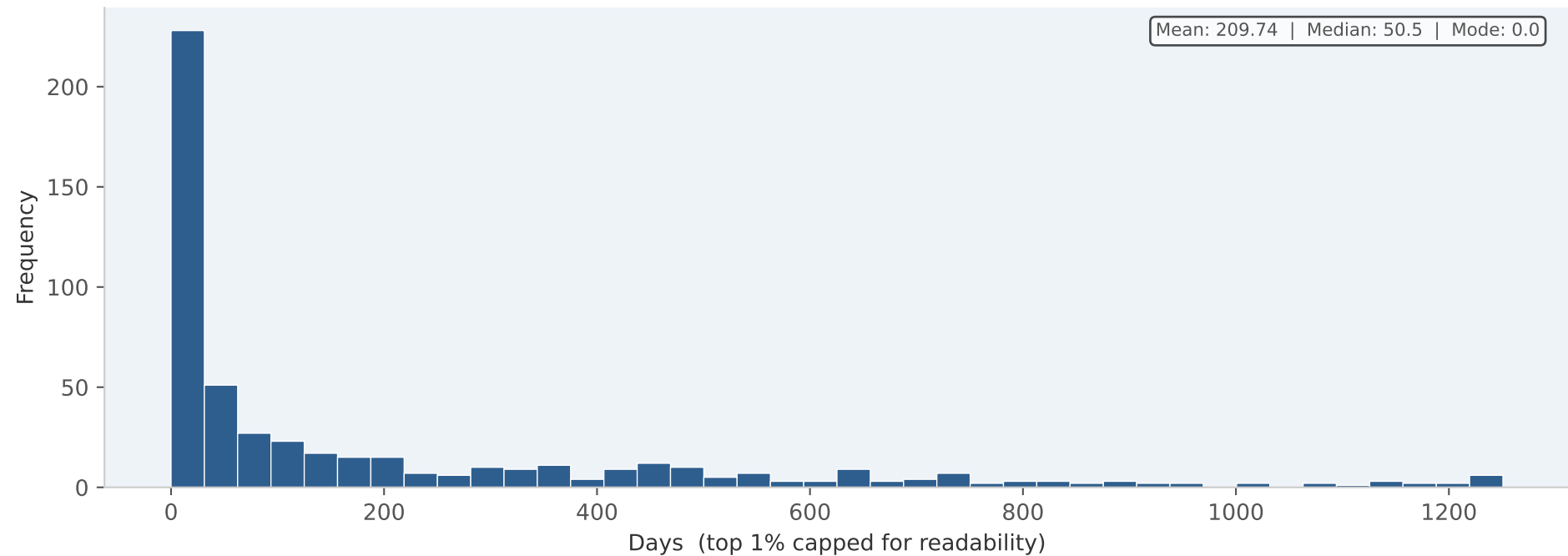
## Distribution: Total Messages per Thread



## Distribution: Replies per Thread (excluding opening post)

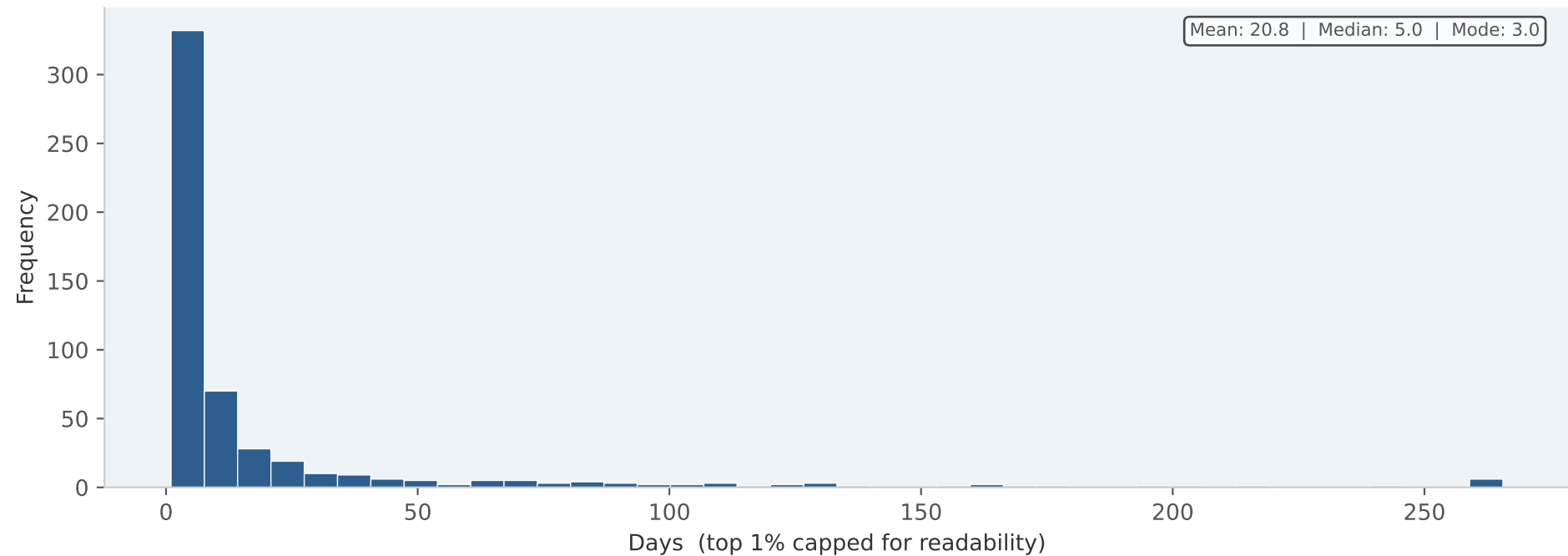


## Distribution: User Activity Span (days between first and last post)

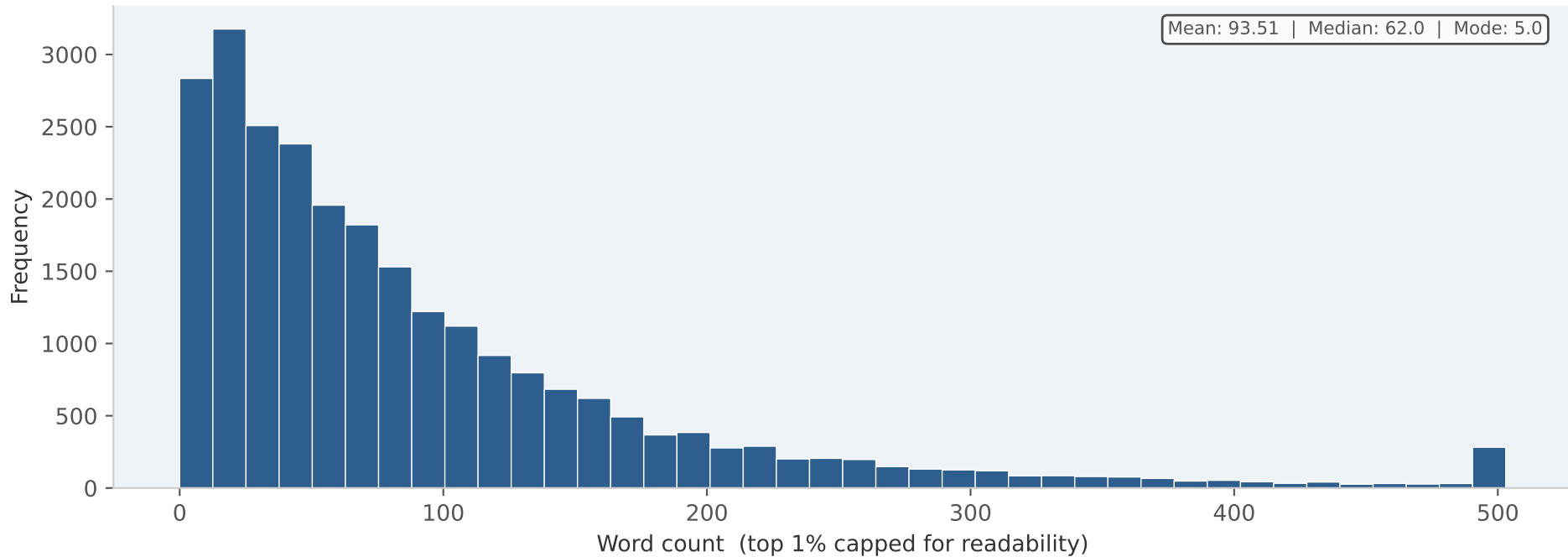




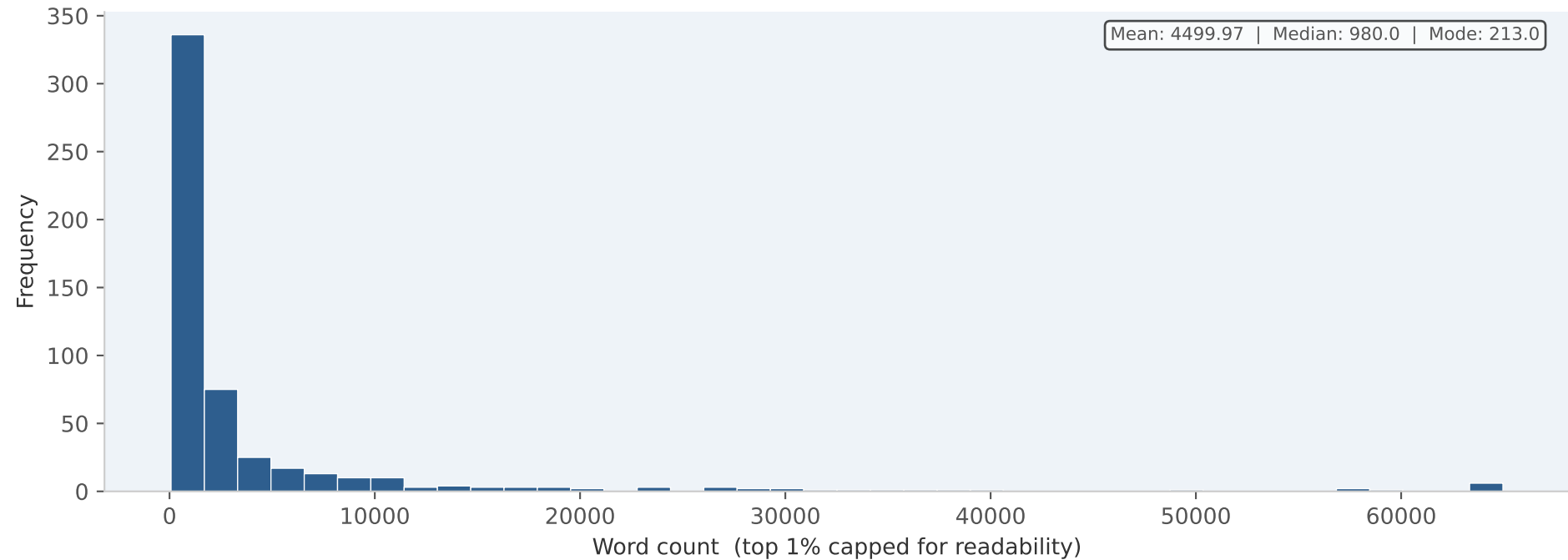
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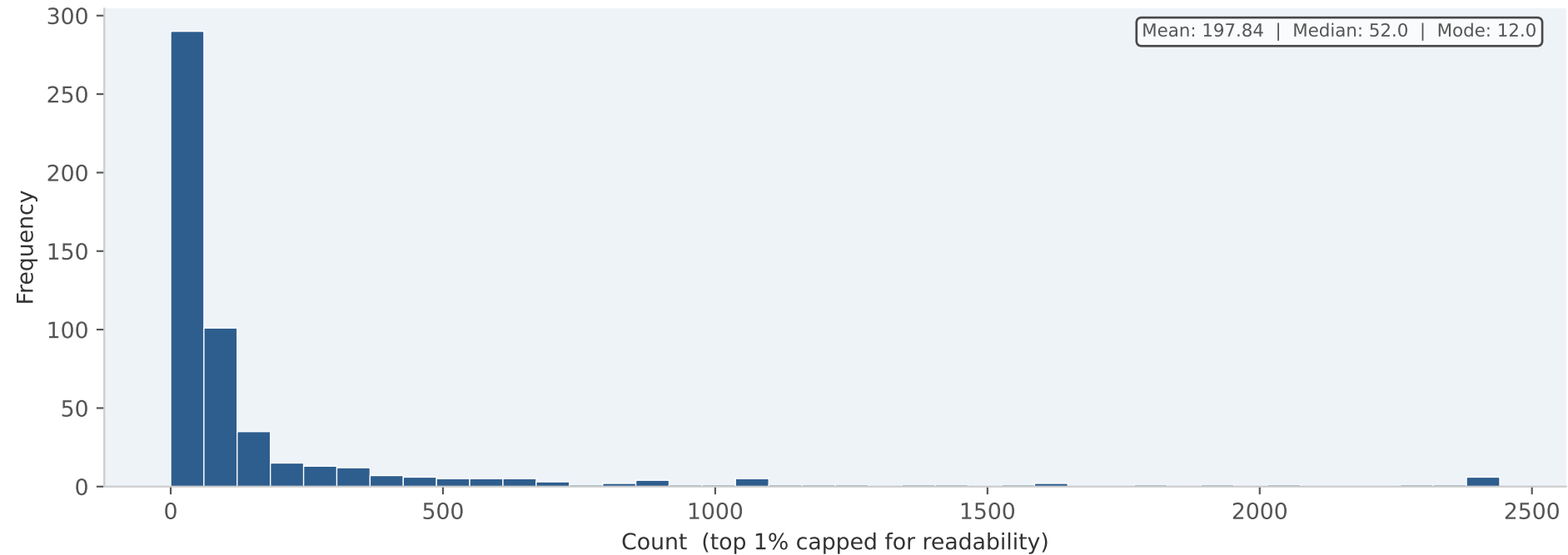
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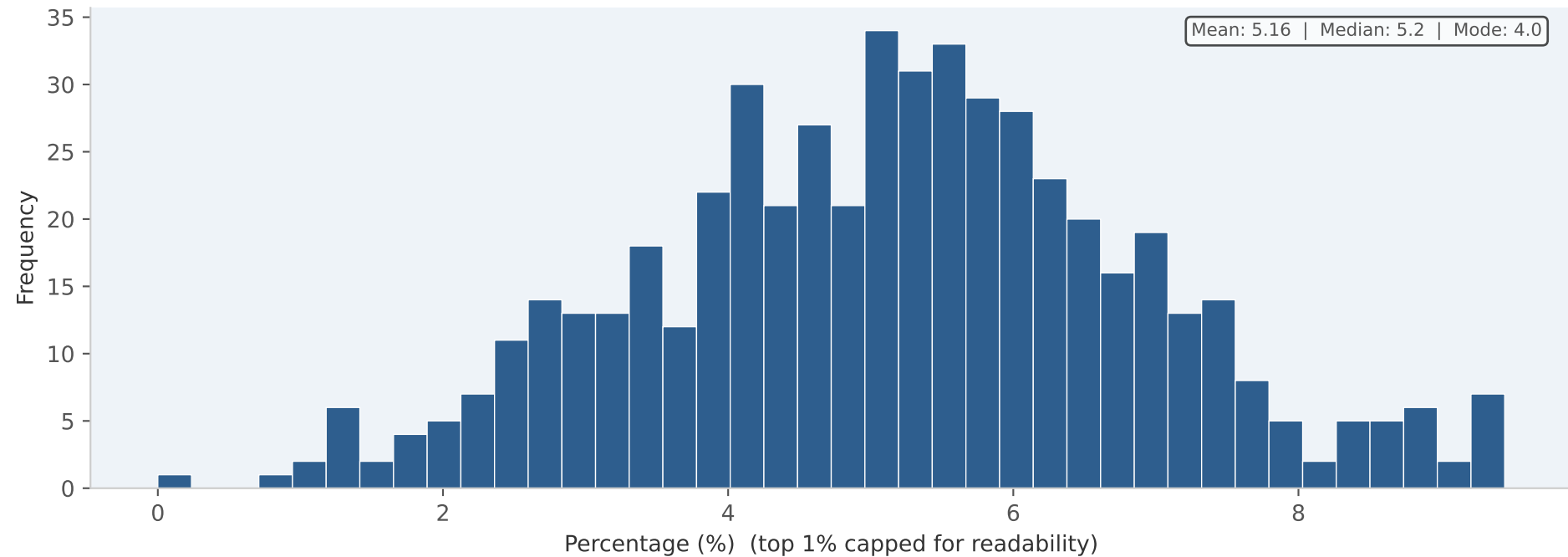
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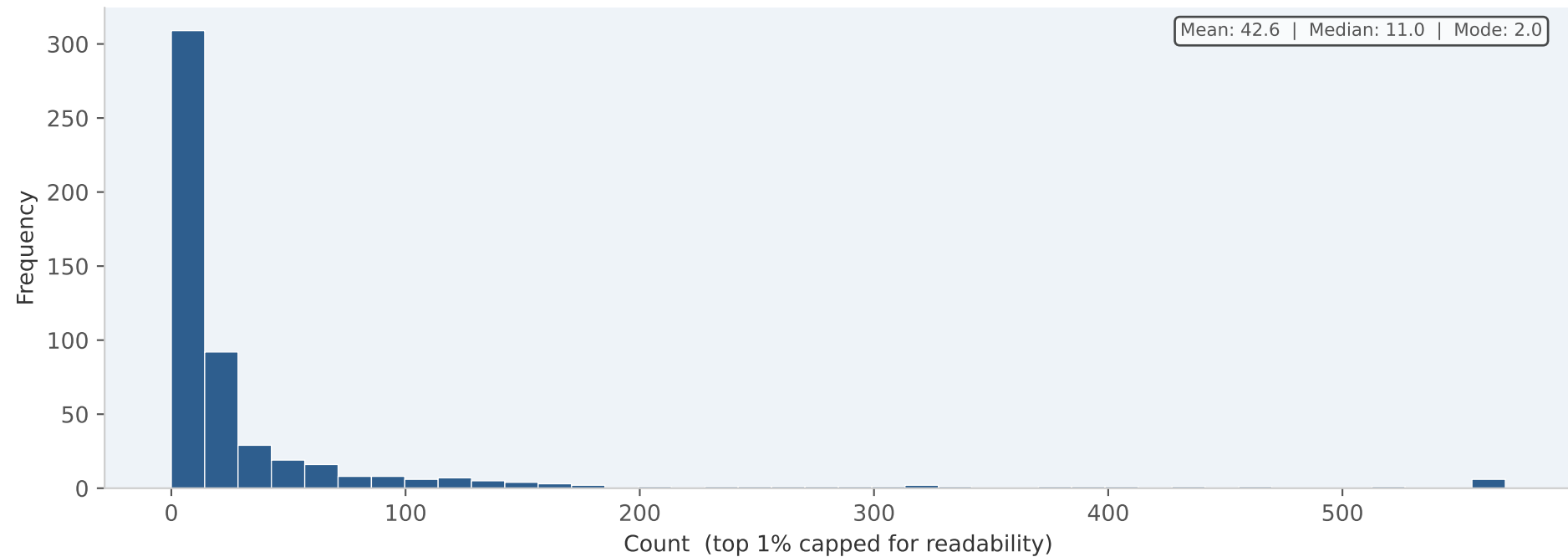
# Distribution: 'ik' Count per User



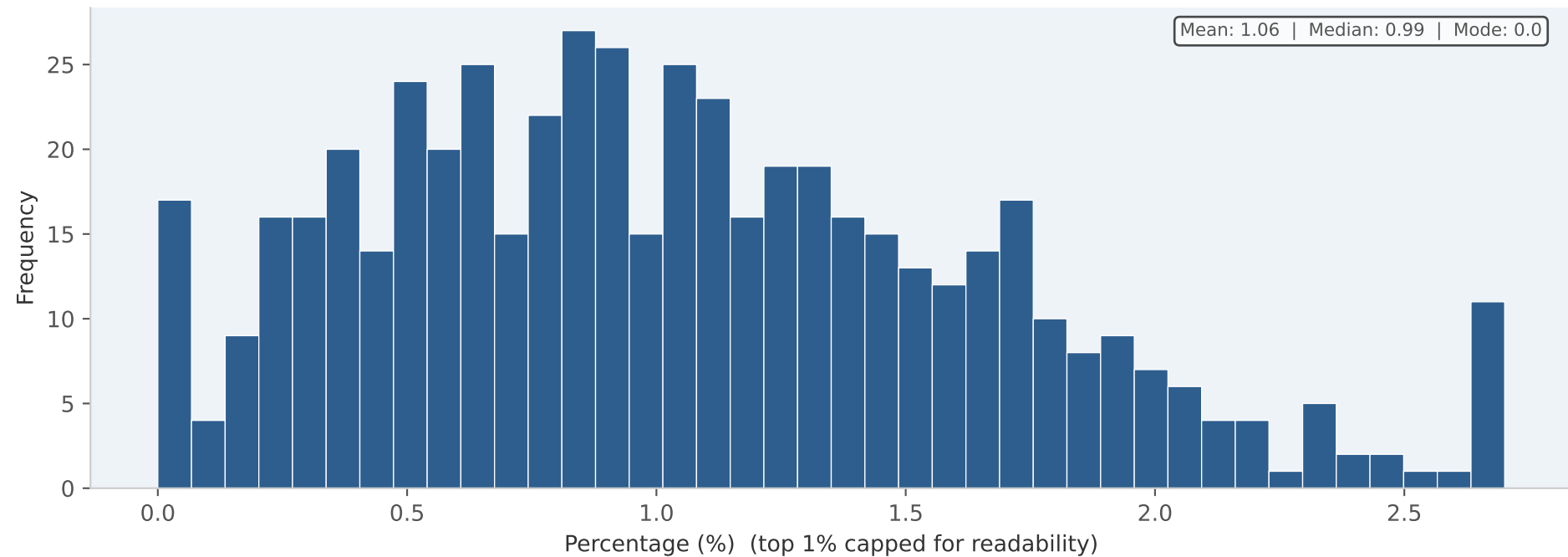
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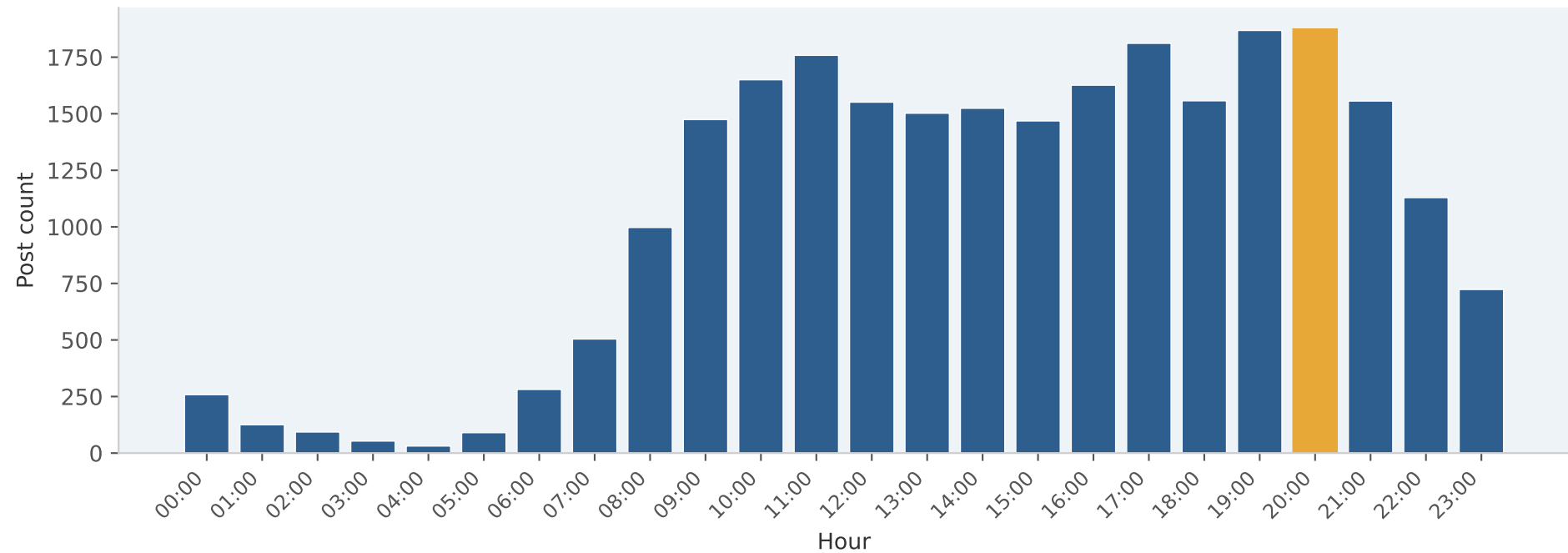
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# Distribution: 'mijn' as % of Total Words per User

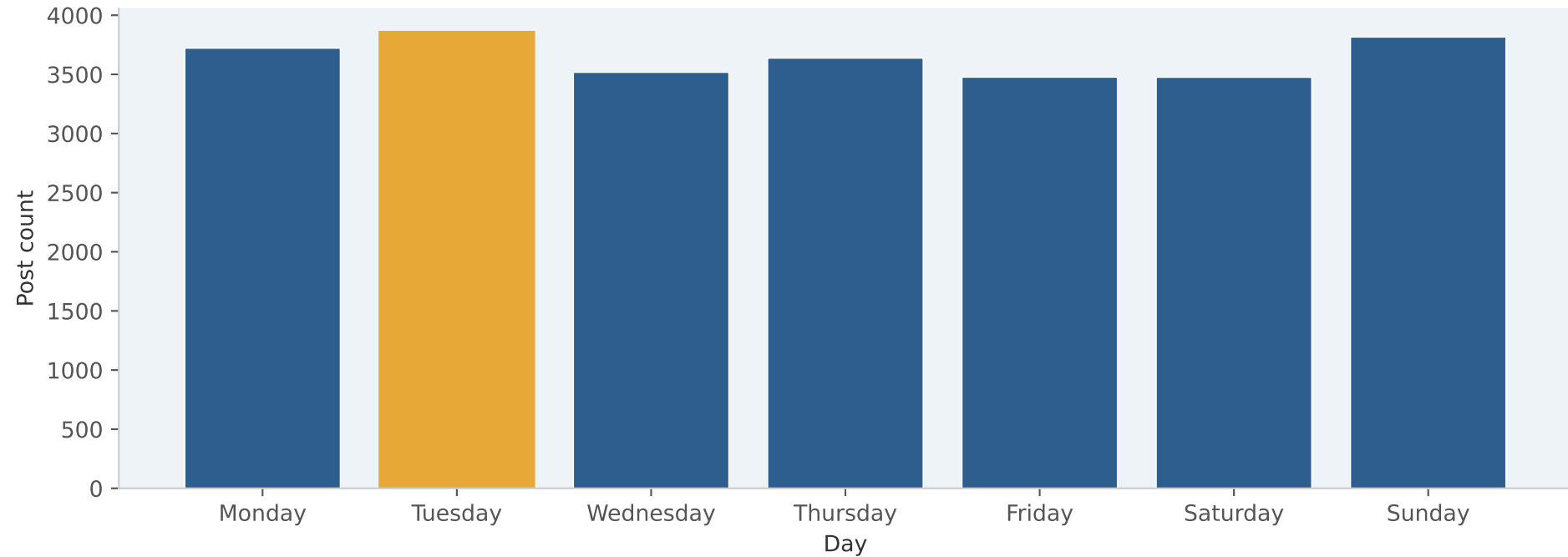


## Popular Hours of Day

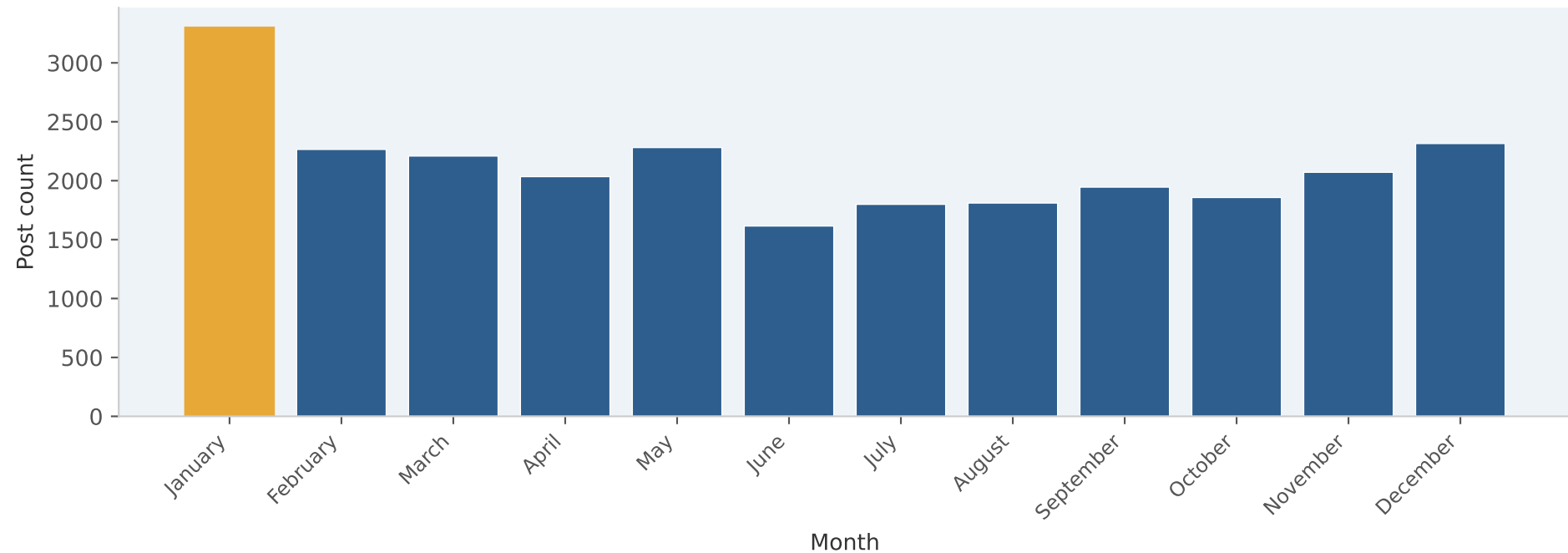




## Popular Days of Week

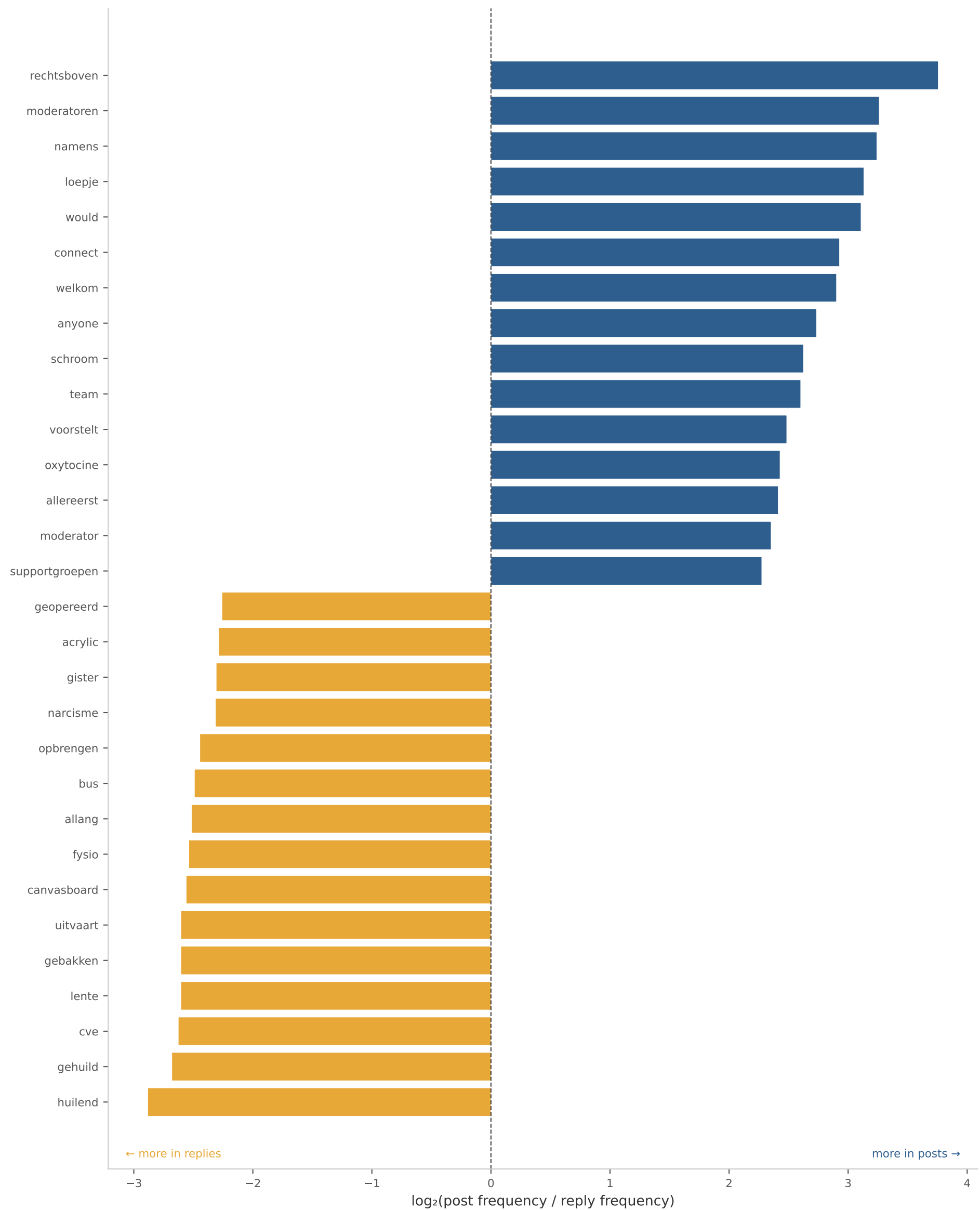


## Popular Months



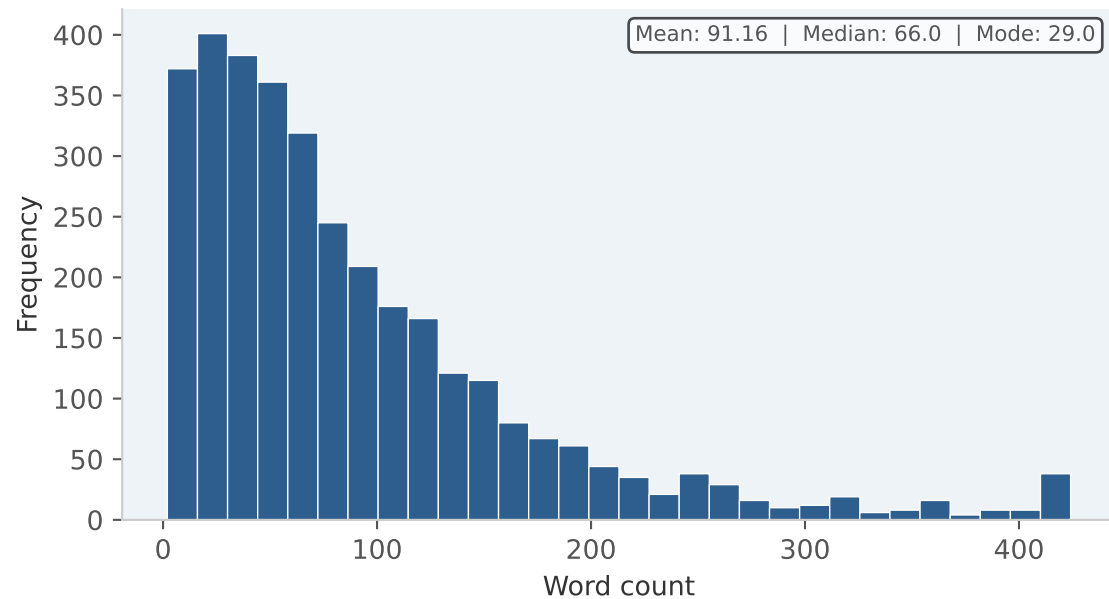
## **Role-Based Analysis — Posts vs Replies**

## Word Choice: Initial Posts vs Replies

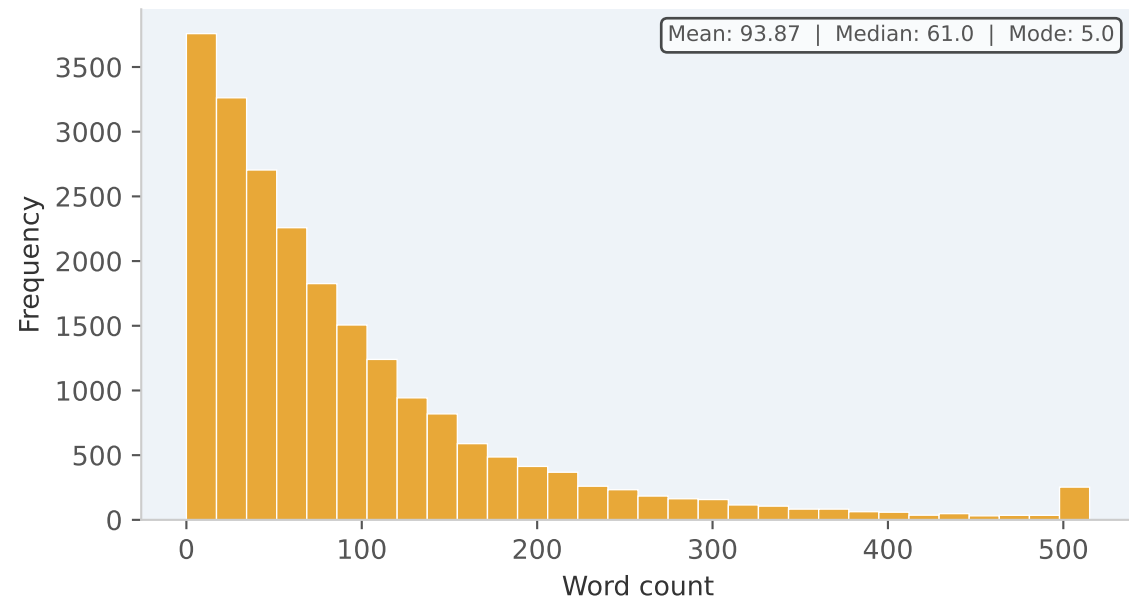


# Words per Message: Initial Posts vs Replies

## Initial Posts

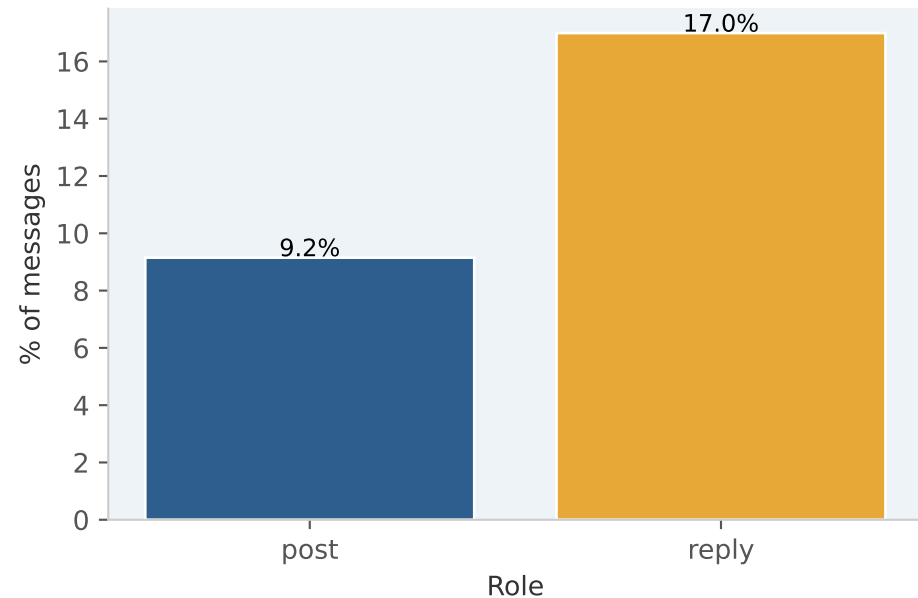


## Replies

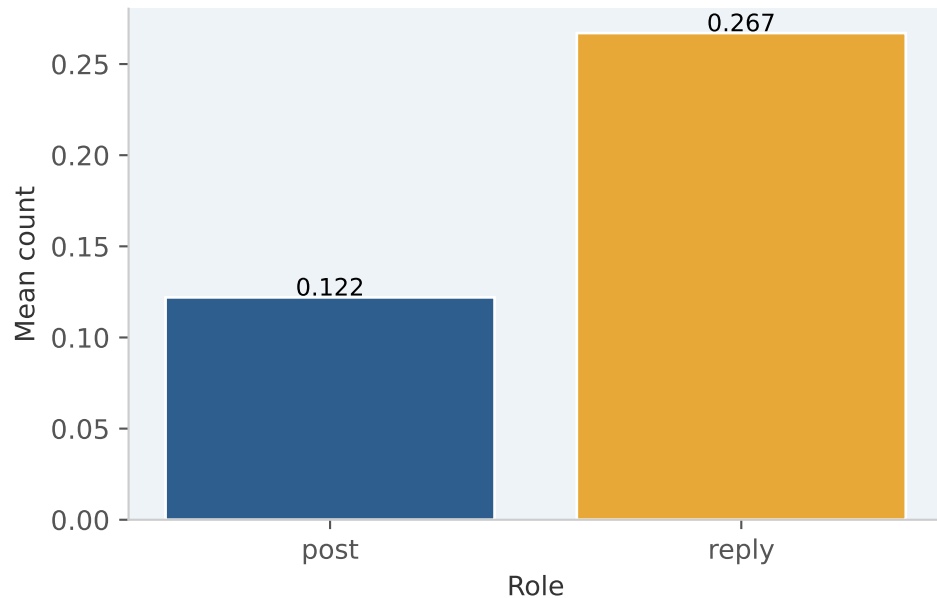


## Emoji Use by Message Role

### % Messages Containing Emoji



### Mean Emoji Count per Message



Top 20 Most Frequent Emojis — Posts vs Replies

| #  | Posts emoji                    | Count | %     | Replies emoji                  | Count | %    |
|----|--------------------------------|-------|-------|--------------------------------|-------|------|
| 1  | rose                           | 97    | 23.49 | rose                           | 967   | 16.4 |
| 2  | four leaf clover               | 34    | 8.23  | four leaf clover               | 395   | 6.7  |
| 3  | folded hands                   | 21    | 5.08  | face blowing a kiss            | 387   | 6.56 |
| 4  | people hugging                 | 18    | 4.36  | folded hands                   | 371   | 6.29 |
| 5  | sparkling heart                | 15    | 3.63  | heart suit                     | 332   | 5.63 |
| 6  | cherry blossom                 | 15    | 3.63  | people hugging                 | 210   | 3.56 |
| 7  | face blowing a kiss            | 13    | 3.15  | grinning face with sweat       | 196   | 3.32 |
| 8  | smiling face with open hands   | 12    | 2.91  | sparkling heart                | 190   | 3.22 |
| 9  | winking face                   | 11    | 2.66  | red heart                      | 162   | 2.75 |
| 10 | smiling face with smiling eyes | 11    | 2.66  | smiling face with smiling eyes | 146   | 2.48 |
| 11 | two hearts                     | 10    | 2.42  | cherry blossom                 | 129   | 2.19 |
| 12 | beaming face with smiling eyes | 7     | 1.69  | flexed biceps                  | 108   | 1.83 |
| 13 | thinking face                  | 6     | 1.45  | face with tears of joy         | 101   | 1.71 |
| 14 | egg                            | 6     | 1.45  | sleepy face                    | 99    | 1.68 |
| 15 | loudly crying face             | 6     | 1.45  | sad but relieved face          | 83    | 1.41 |
| 16 | pleading face                  | 6     | 1.45  | revolving hearts               | 83    | 1.41 |
| 17 | thumbs up                      | 5     | 1.21  | winking face                   | 81    | 1.37 |
| 18 | heart suit                     | 5     | 1.21  | pleading face                  | 77    | 1.31 |
| 19 | waving hand                    | 5     | 1.21  | smiling face with open hands   | 67    | 1.14 |
| 20 | red heart                      | 5     | 1.21  | crying face                    | 60    | 1.02 |

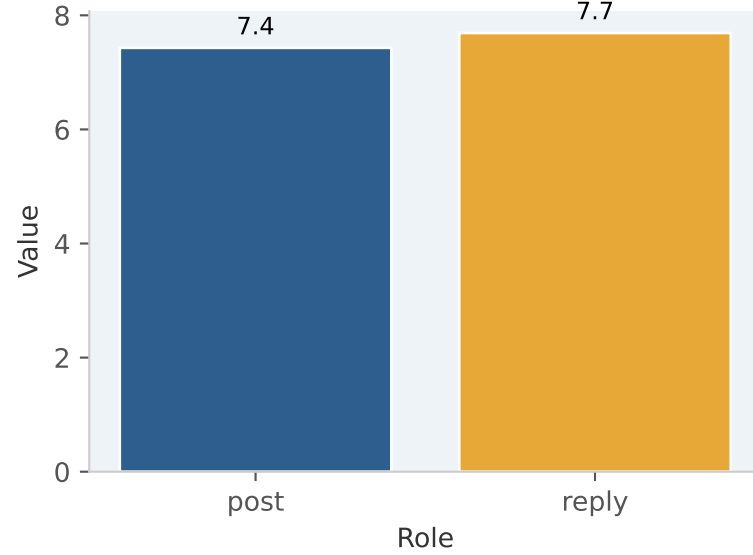
Top 20 Most Frequent Words — Posts vs Replies

| #  | Post word | Count | %     | Reply word | Count | %     |
|----|-----------|-------|-------|------------|-------|-------|
| 1  | mijn      | 1802  | 1.128 | mijn       | 20775 | 1.92  |
| 2  | goed      | 1524  | 0.954 | ben        | 12986 | 1.2   |
| 3  | ben       | 1439  | 0.901 | weer       | 10854 | 1.003 |
| 4  | hier      | 1327  | 0.831 | goed       | 9129  | 0.844 |
| 5  | veel      | 1308  | 0.819 | naar       | 8171  | 0.755 |
| 6  | heel      | 1287  | 0.806 | veel       | 8124  | 0.751 |
| 7  | depressie | 1228  | 0.769 | heel       | 7809  | 0.722 |
| 8  | hebt      | 1178  | 0.737 | over       | 6056  | 0.56  |
| 9  | weer      | 1164  | 0.729 | hoe        | 6041  | 0.558 |
| 10 | naar      | 1127  | 0.706 | geen       | 6004  | 0.555 |
| 11 | geen      | 857   | 0.536 | gaat       | 5890  | 0.544 |
| 12 | hoe       | 842   | 0.527 | doen       | 5596  | 0.517 |
| 13 | gaat      | 831   | 0.52  | moet       | 5507  | 0.509 |
| 14 | doen      | 813   | 0.509 | gaan       | 5302  | 0.49  |
| 15 | over      | 808   | 0.506 | echt       | 5295  | 0.489 |
| 16 | forum     | 790   | 0.495 | depressie  | 5272  | 0.487 |
| 17 | bent      | 785   | 0.491 | daar       | 5252  | 0.485 |
| 18 | welkom    | 771   | 0.483 | toch       | 5139  | 0.475 |
| 19 | iets      | 748   | 0.468 | dag        | 5132  | 0.474 |
| 20 | zelf      | 742   | 0.464 | dus        | 4953  | 0.458 |

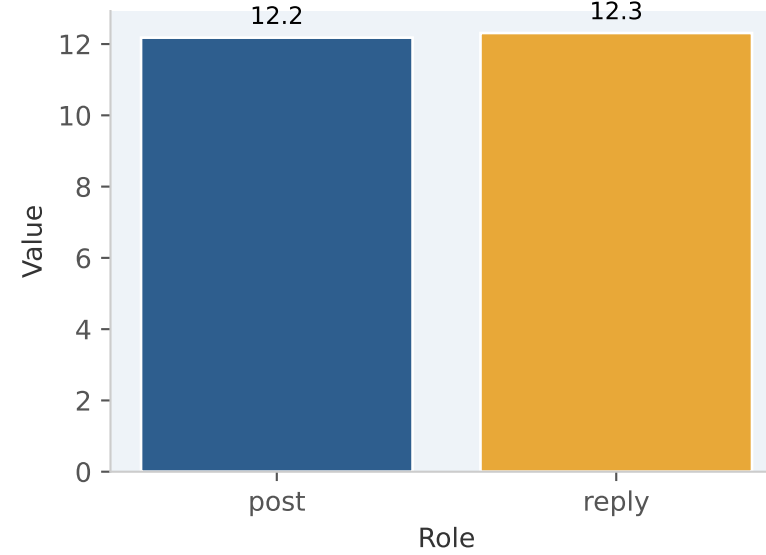


## Sentence Structure — Initial Posts vs Replies

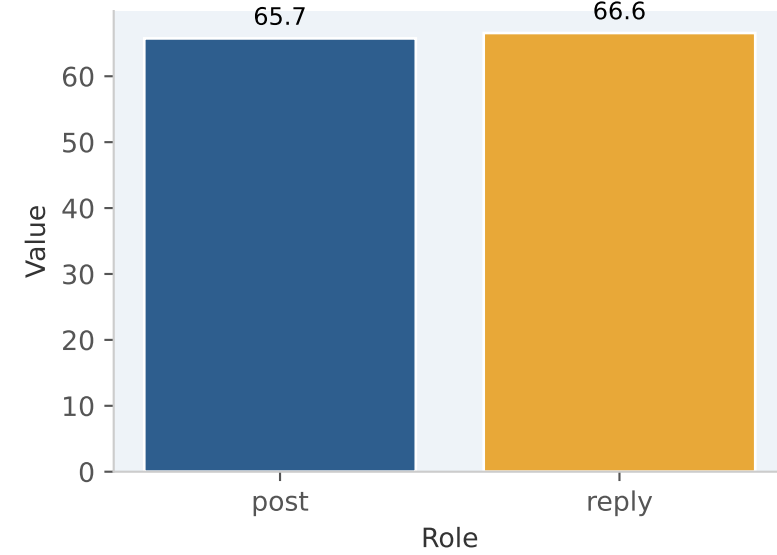
### Avg Sentences per Message



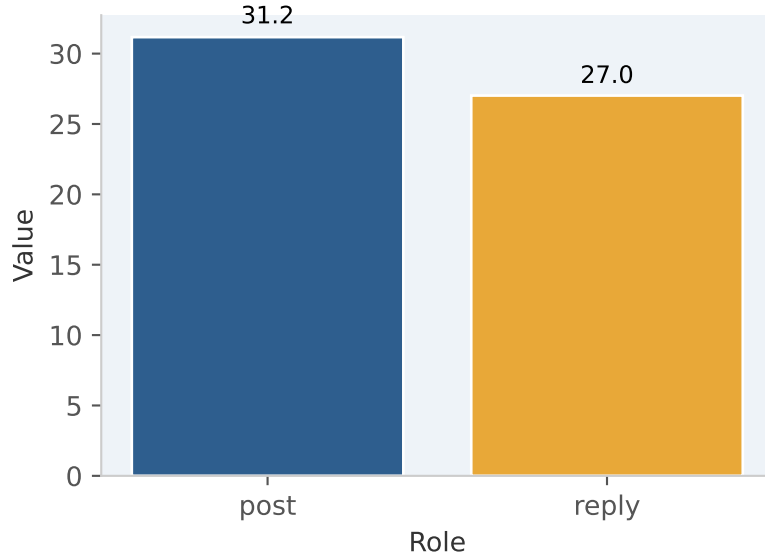
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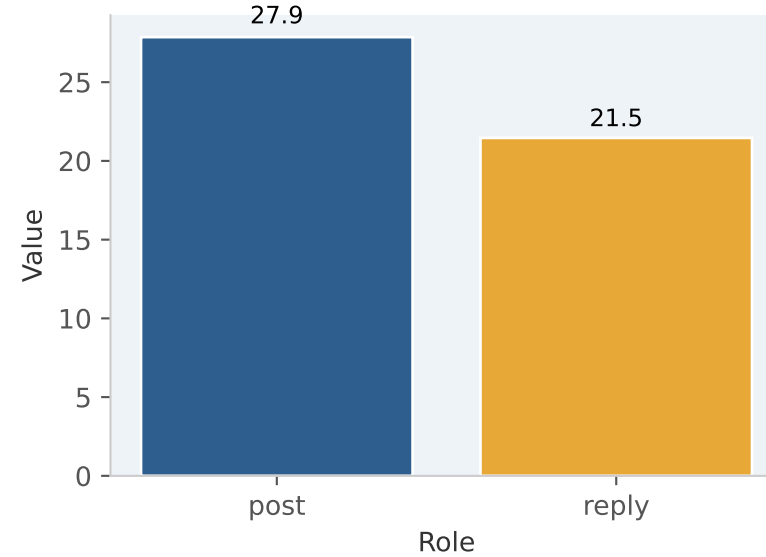
### Avg Chars per Sentence



### % Messages with '?'



### % Messages with '!'



### % Messages with '...'



## **2. EDA - Multi-Posters Only**

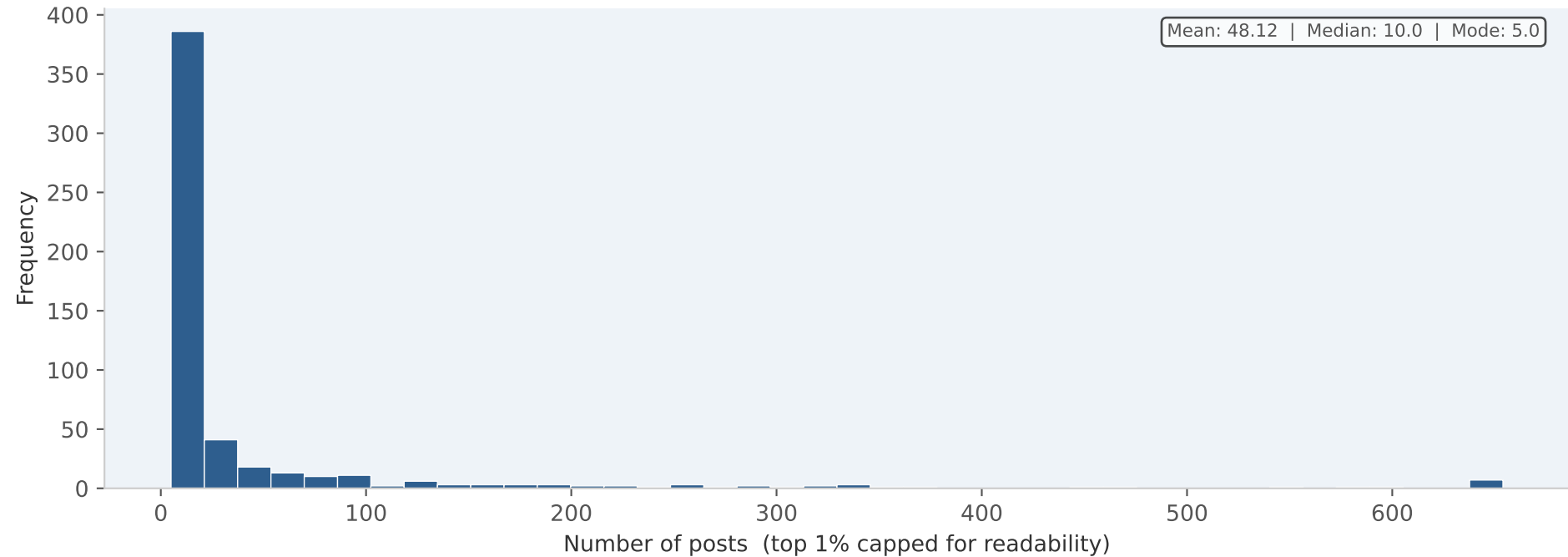
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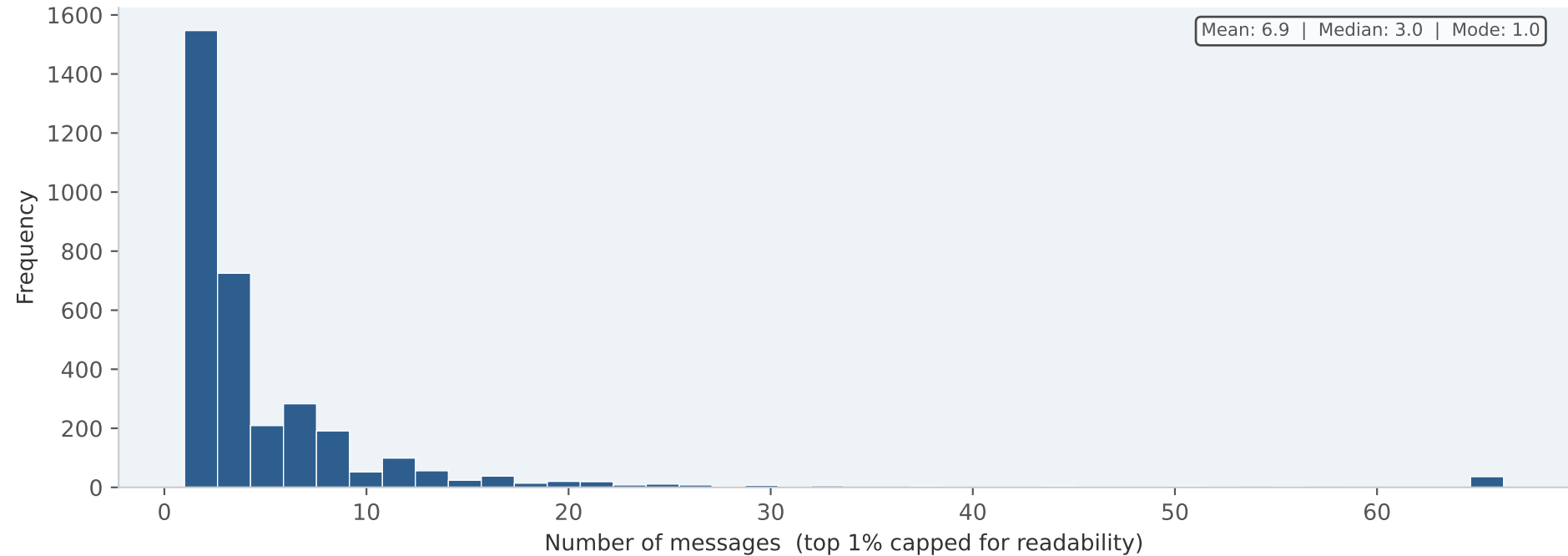
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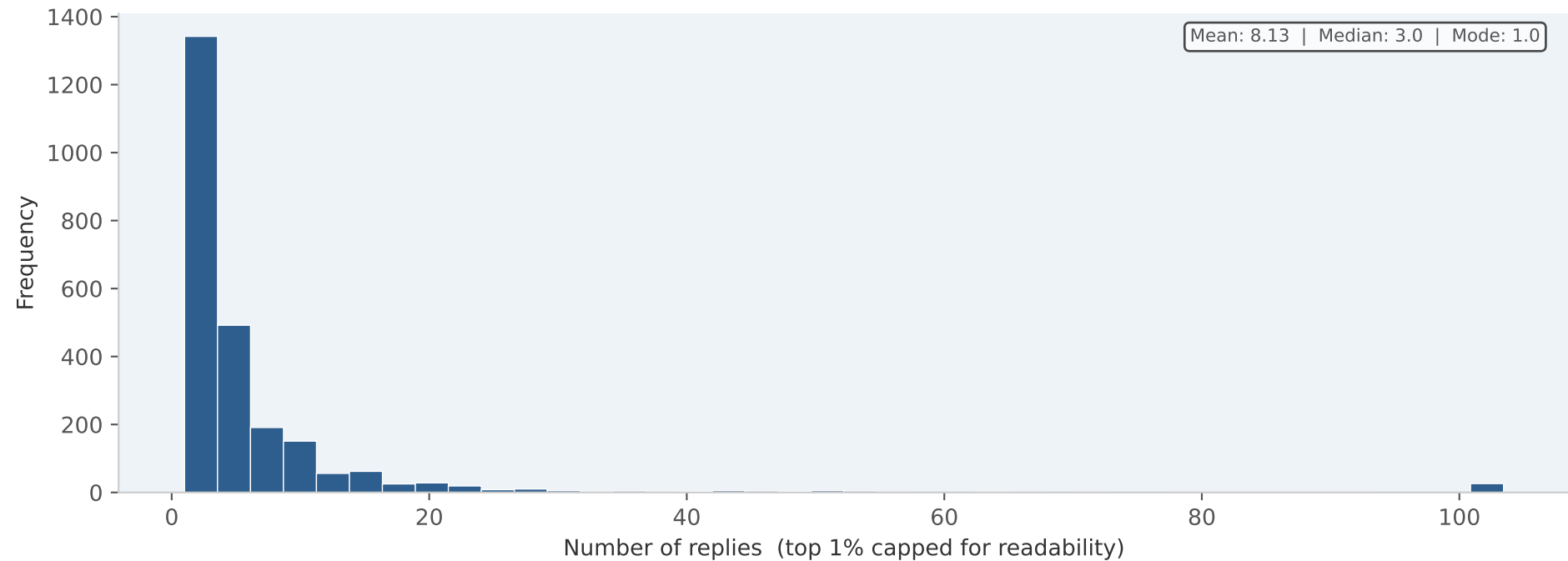
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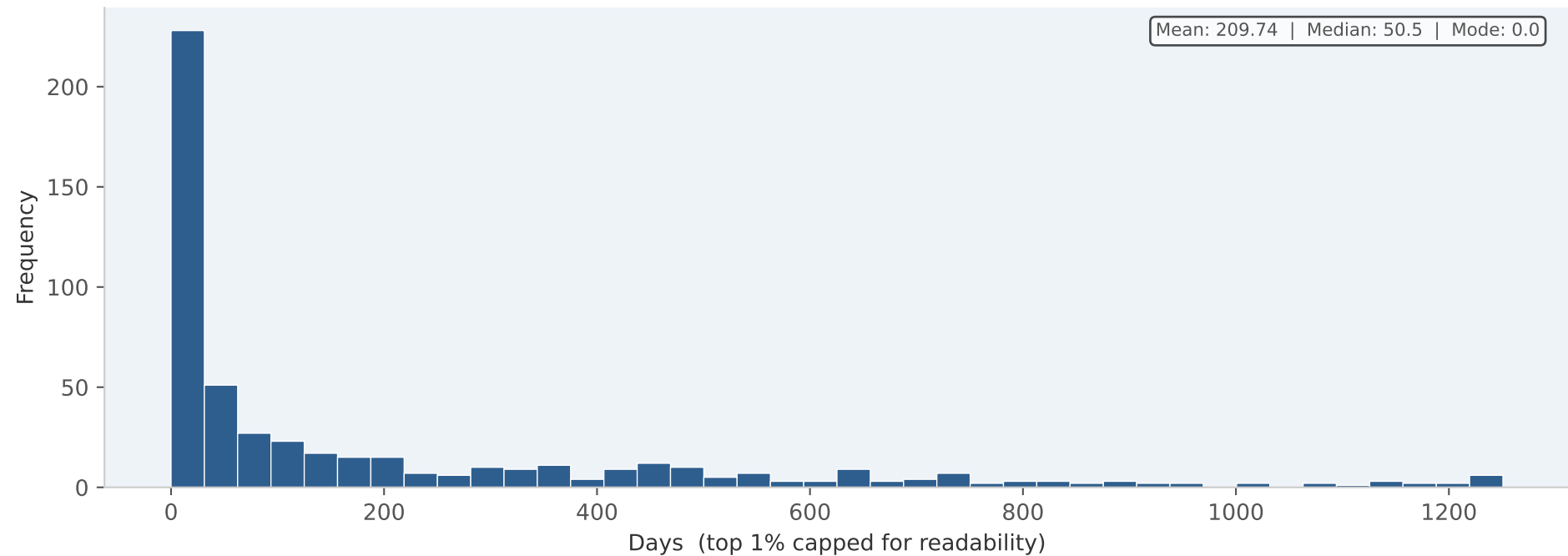
## Distribution: Total Messages per Thread



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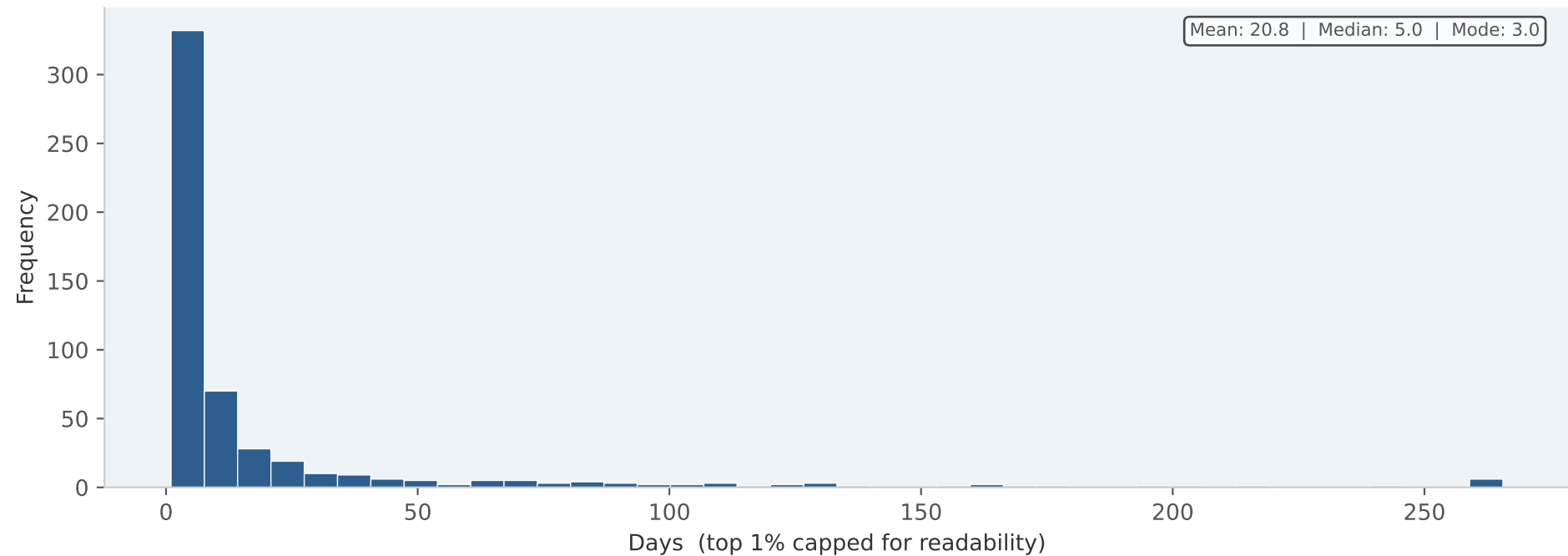


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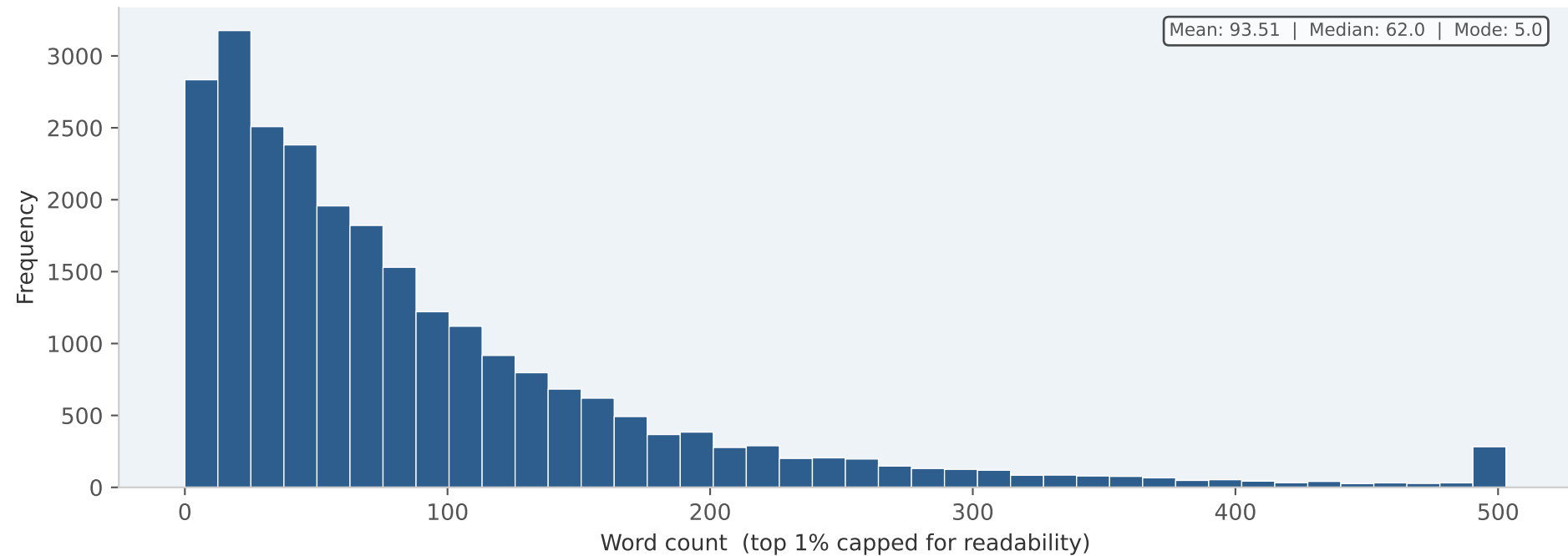




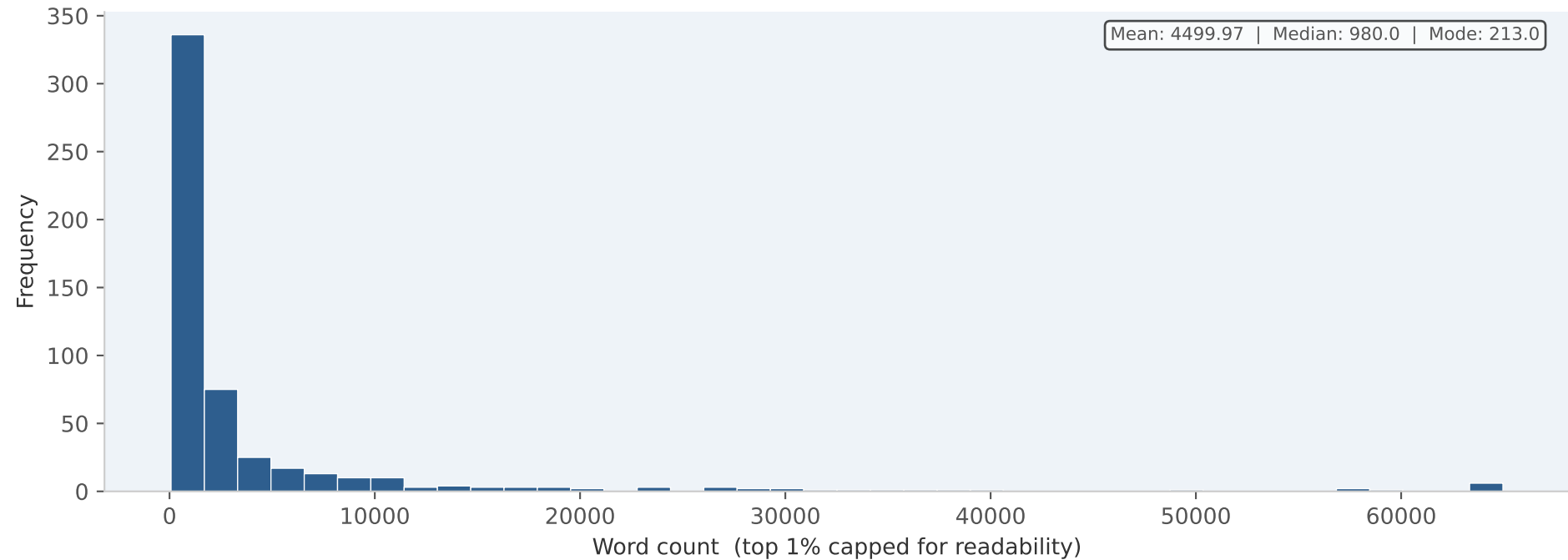
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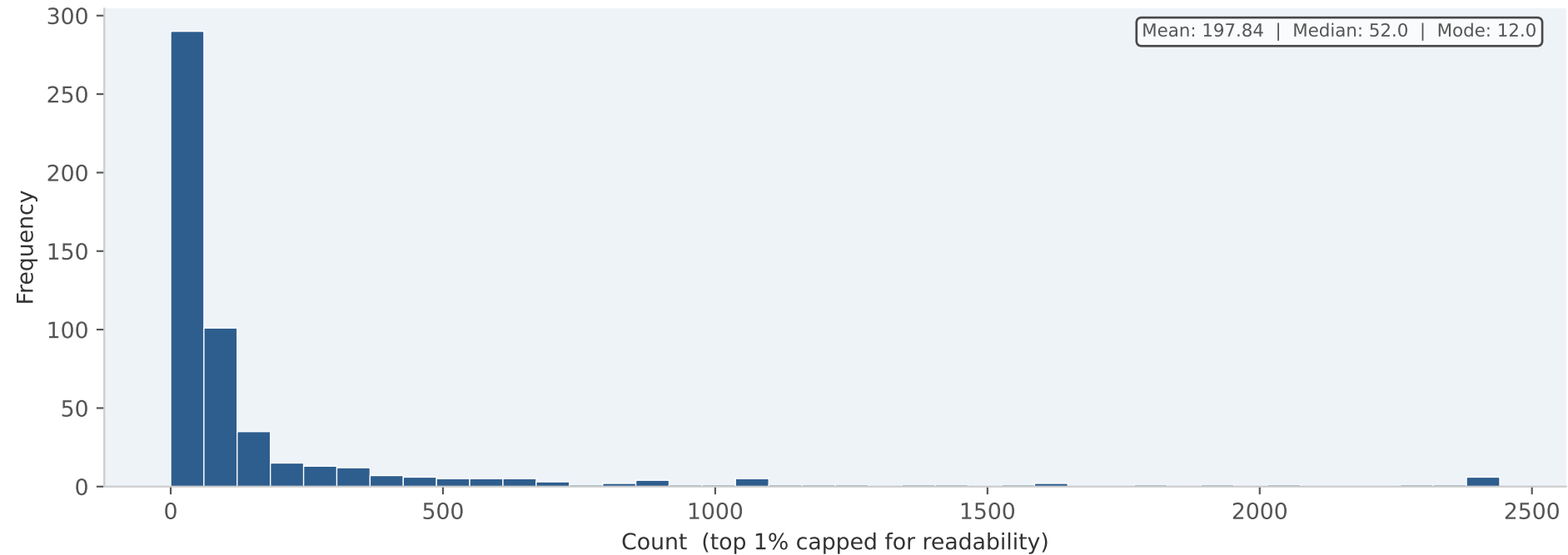
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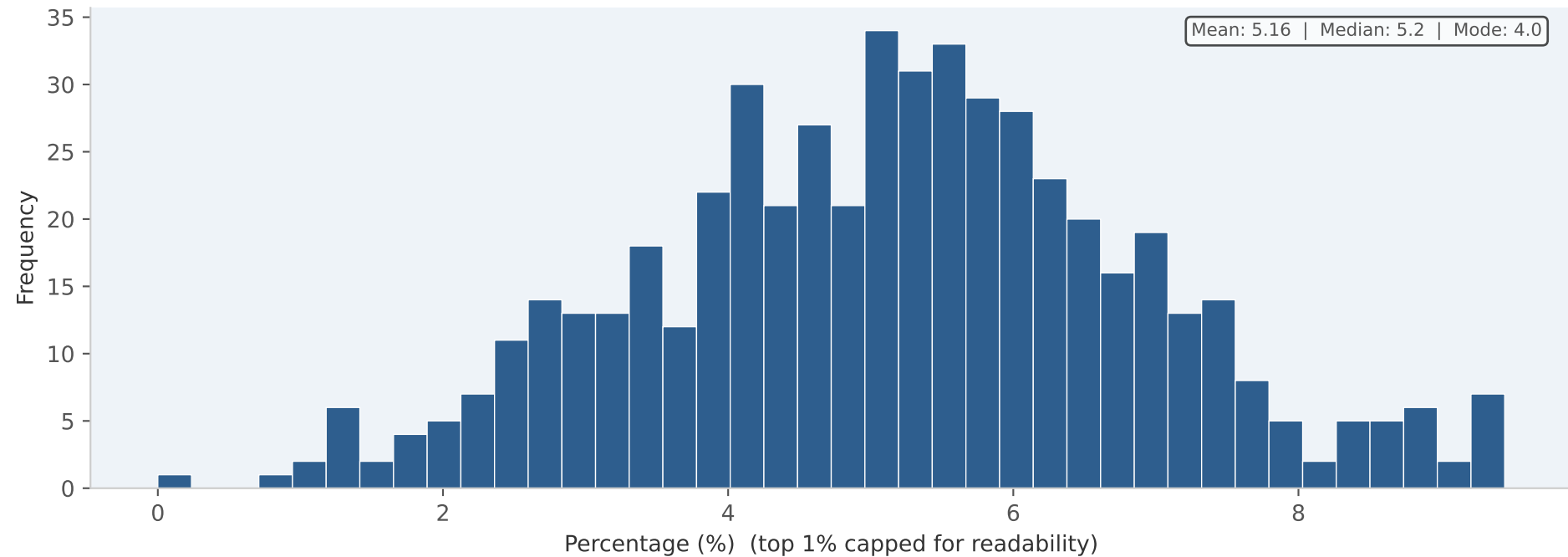
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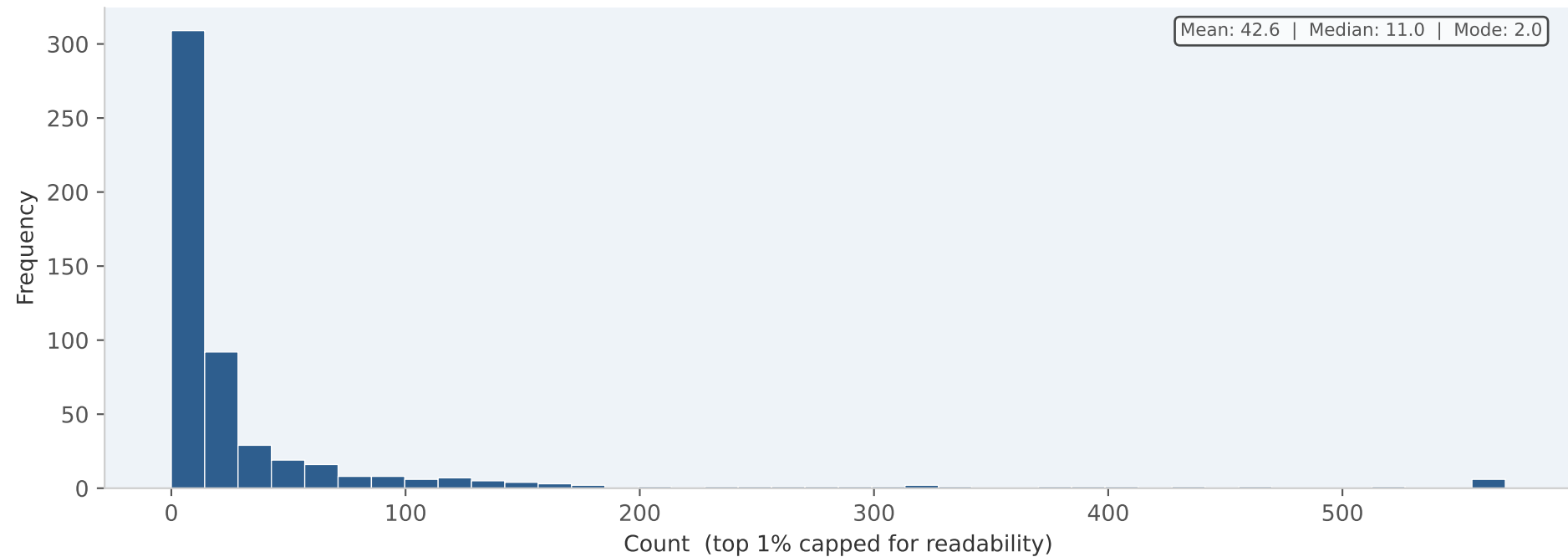
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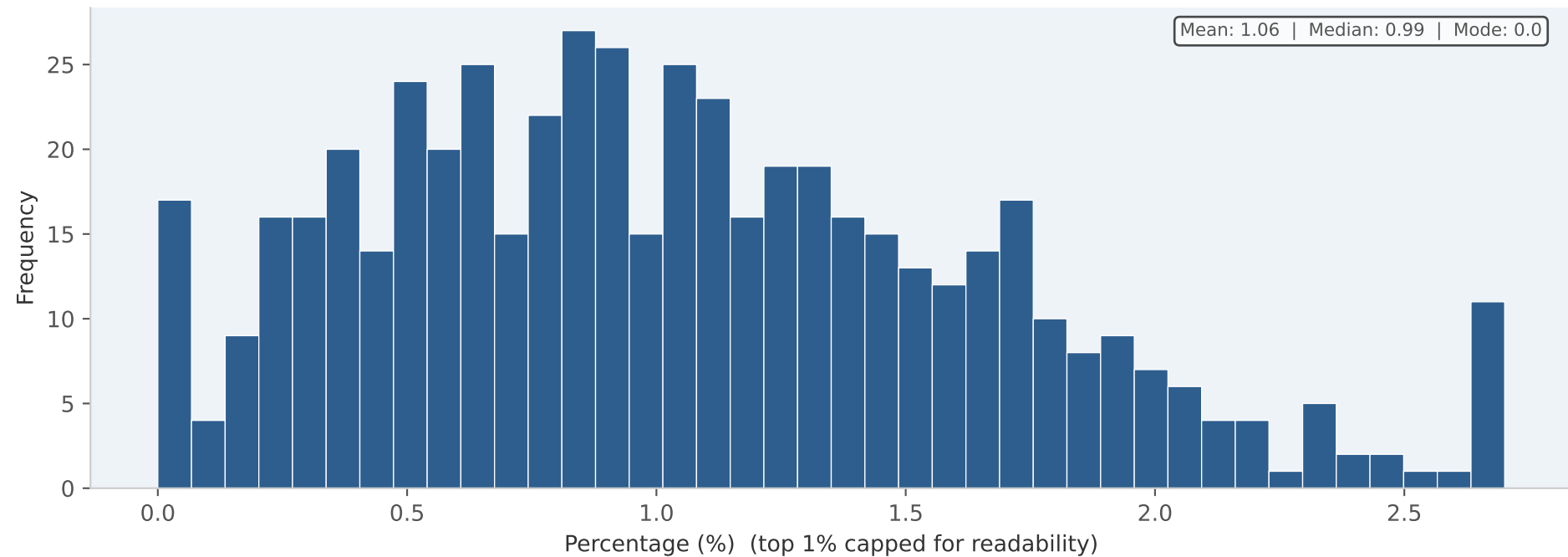
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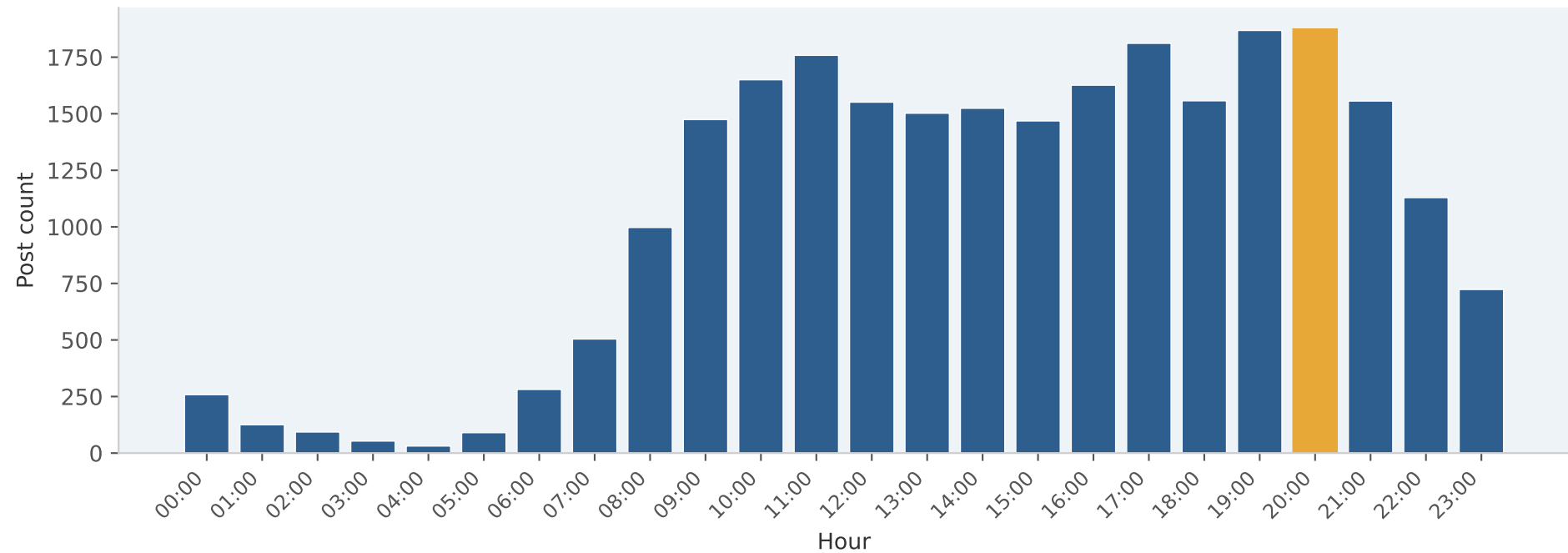
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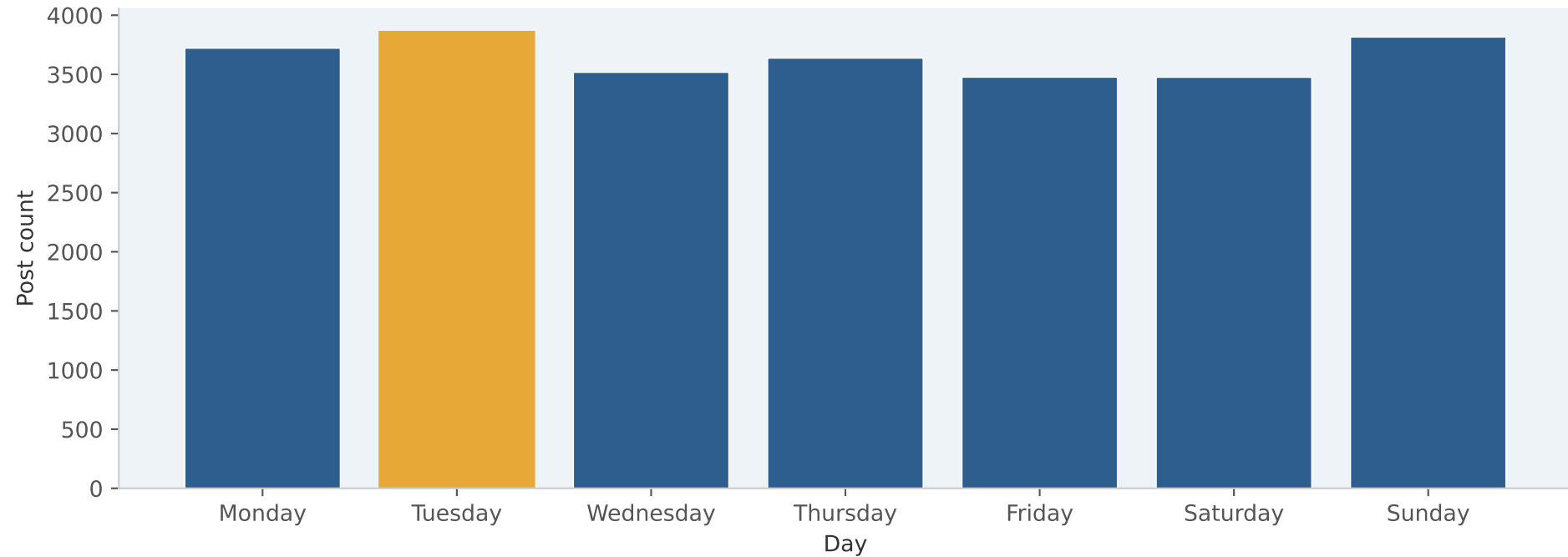


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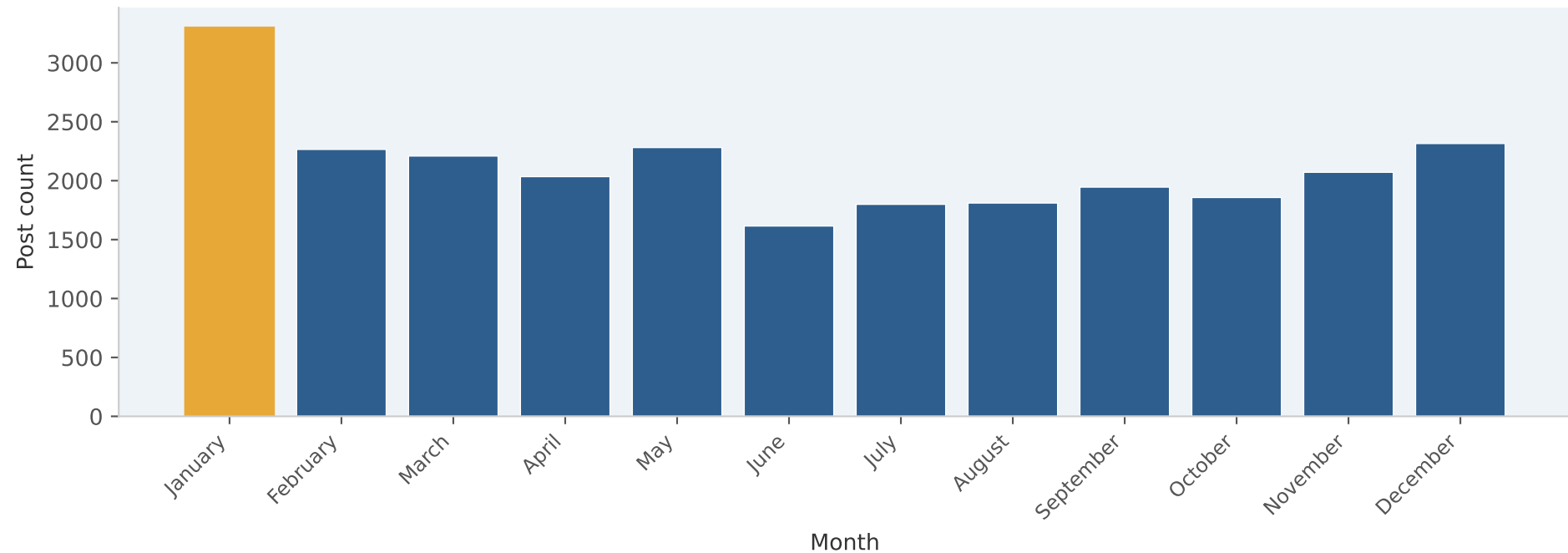




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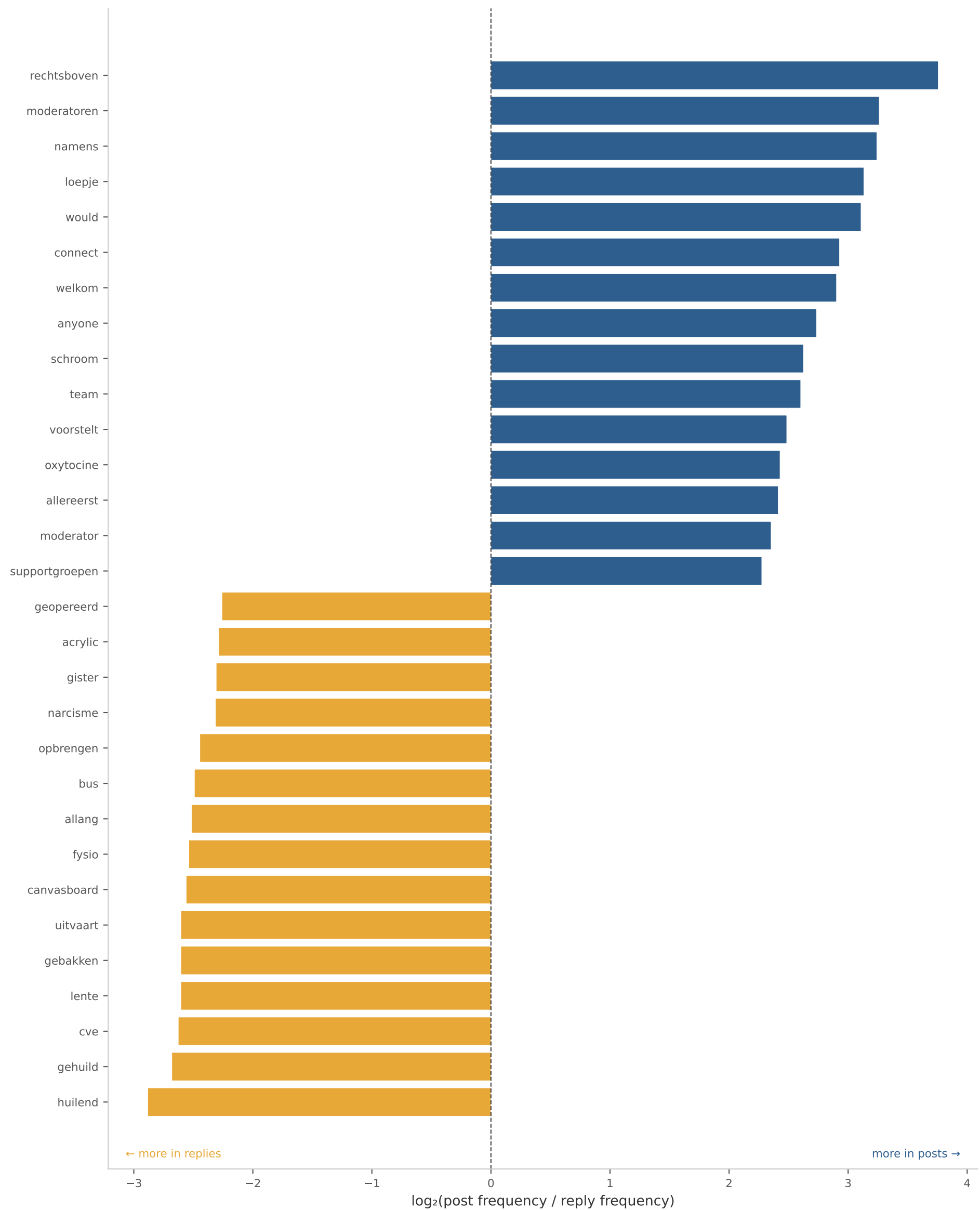


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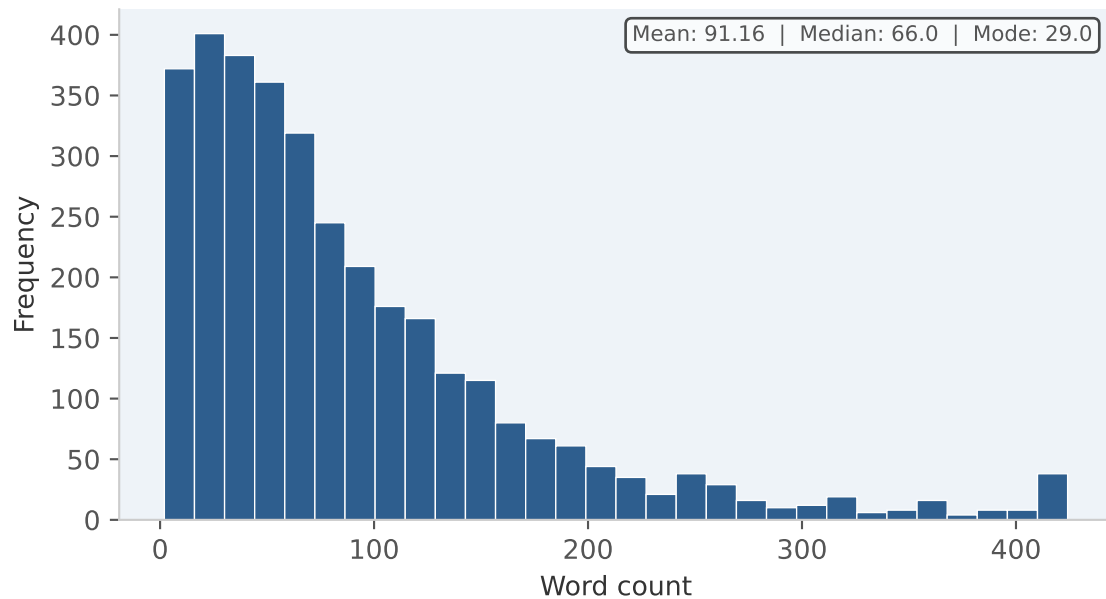
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## Word Choice: Initial Posts vs Replies

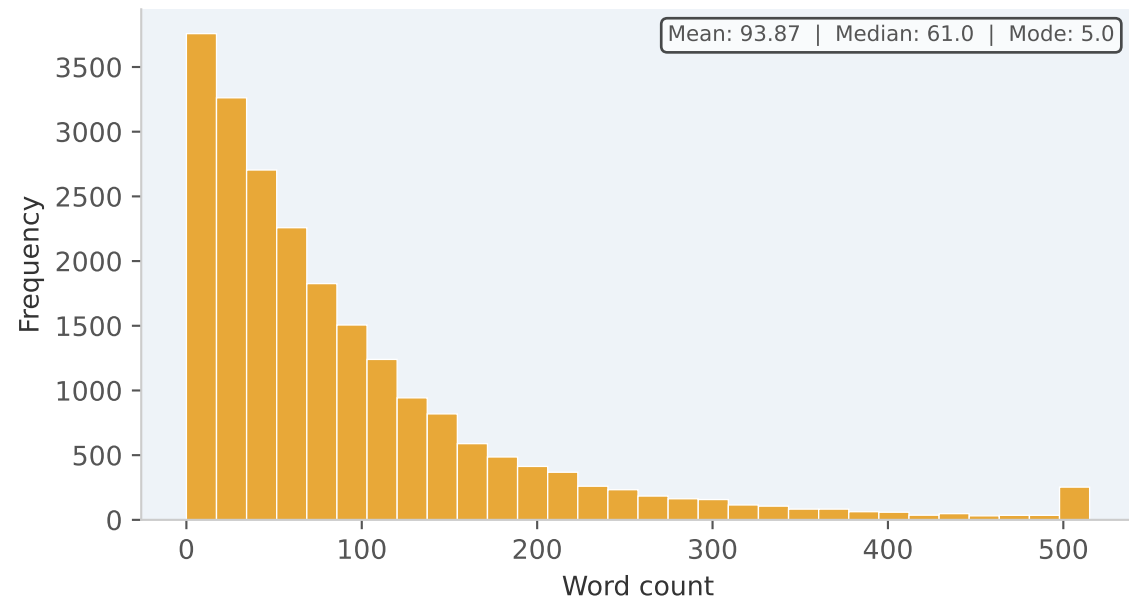


# Words per Message: Initial Posts vs Replies

## Initial Posts

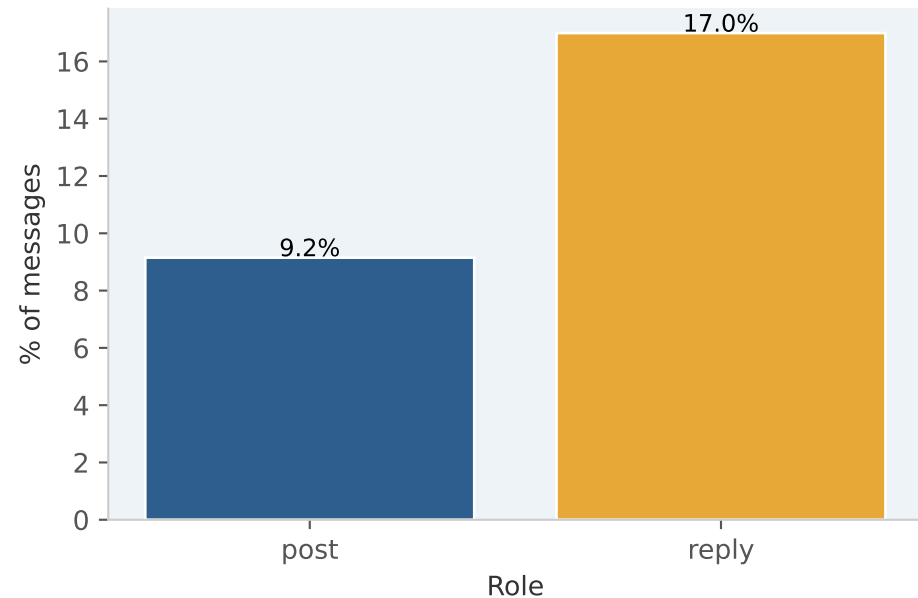


## Replies

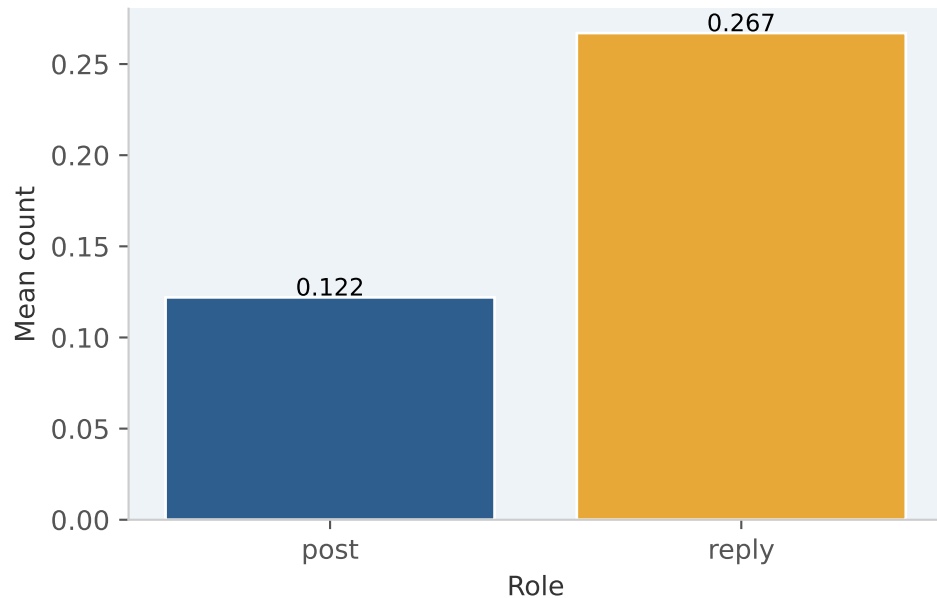


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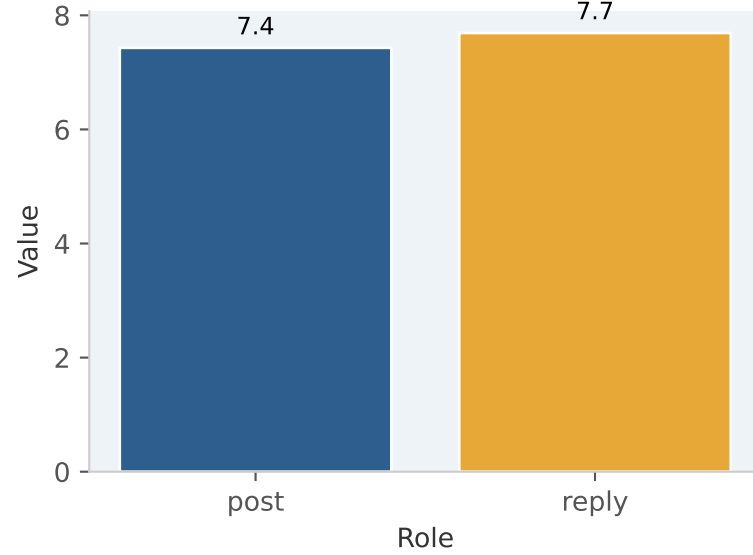
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| 11 | geen      | 857   | 0.536 | gaat       | 5890  | 0.544 |
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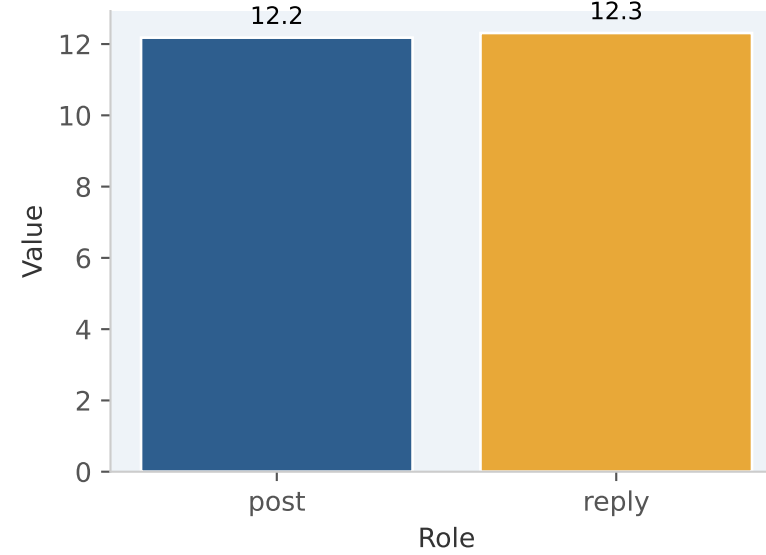


## Sentence Structure — Initial Posts vs Replies

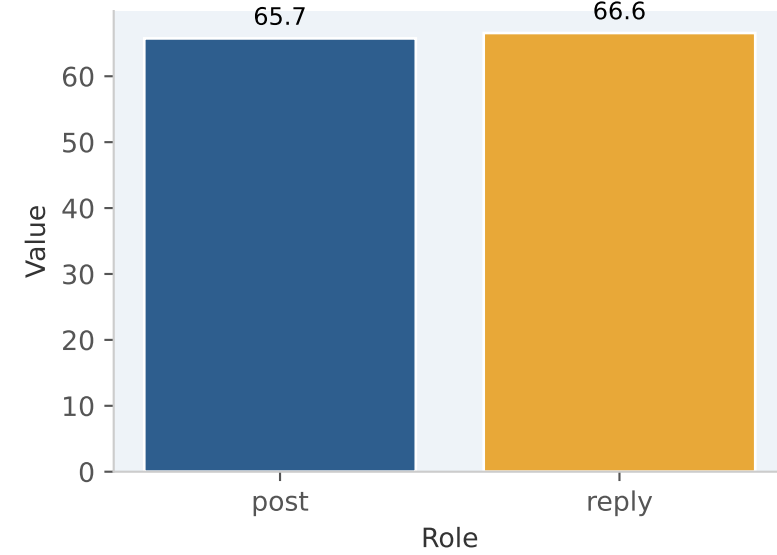
### Avg Sentences per Message



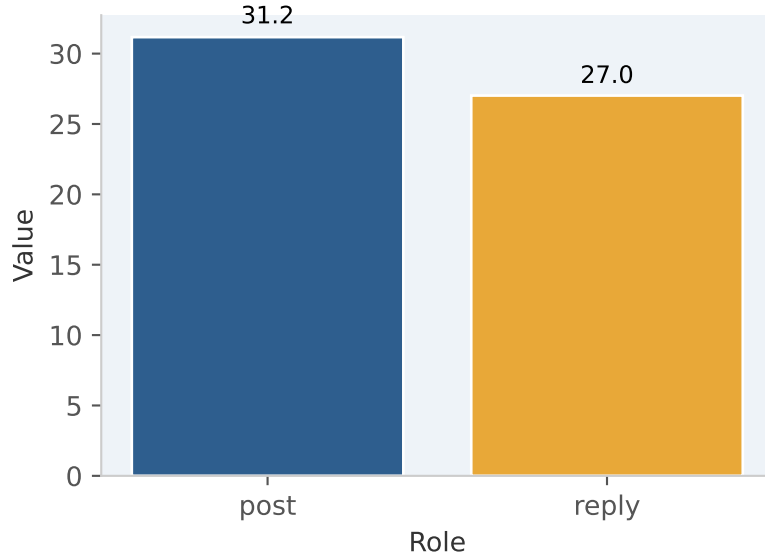
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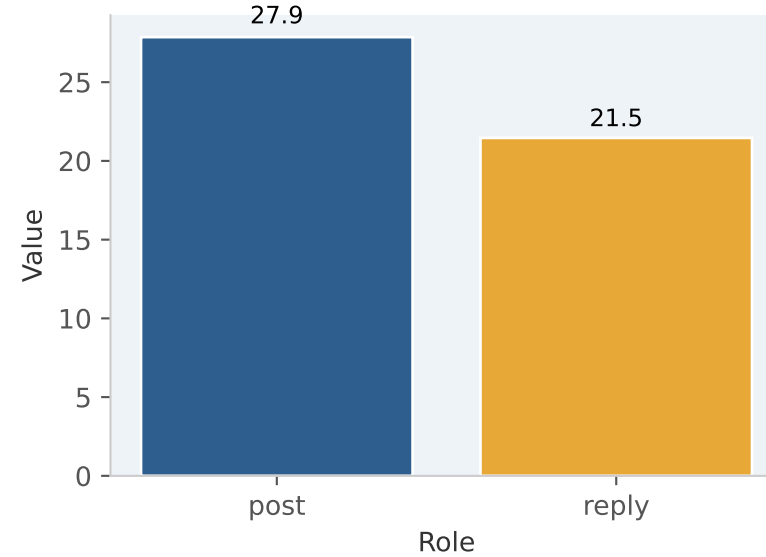
### Avg Chars per Sentence



### % Messages with '?'



### % Messages with '!'



### % Messages with '...'



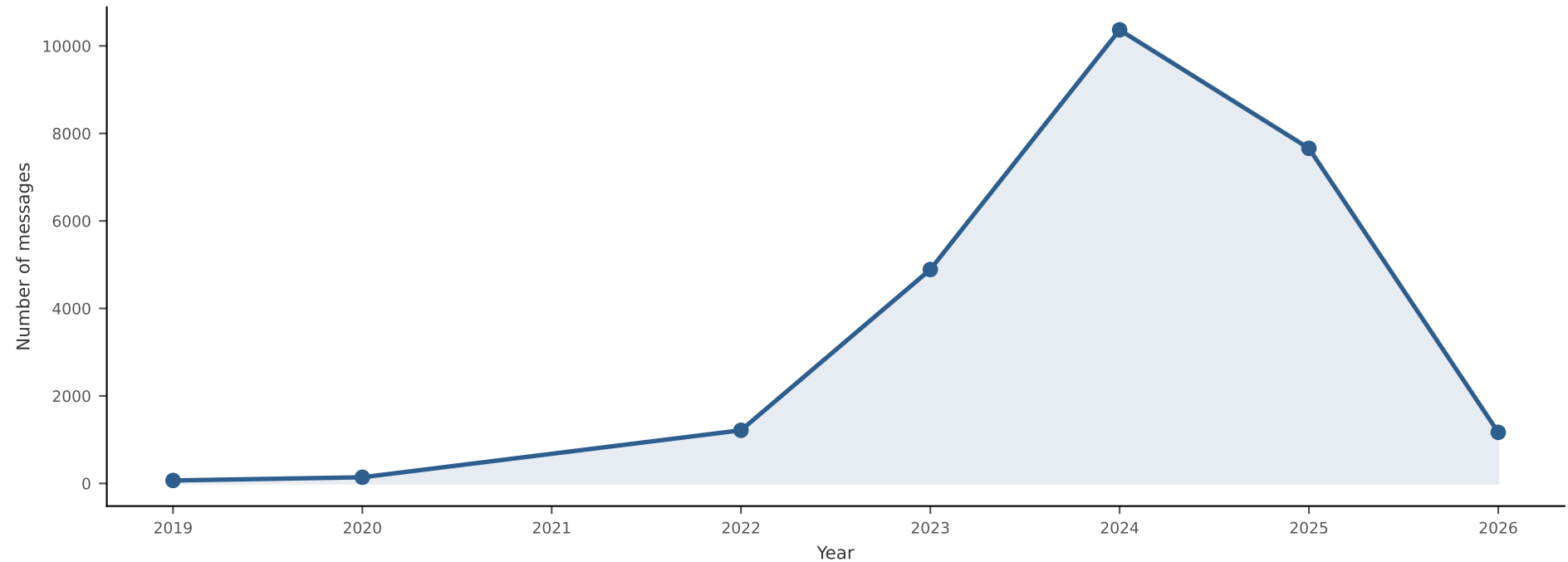
### **3. Time Series & CDS Prevalence**

# **Depression Connect Forum**

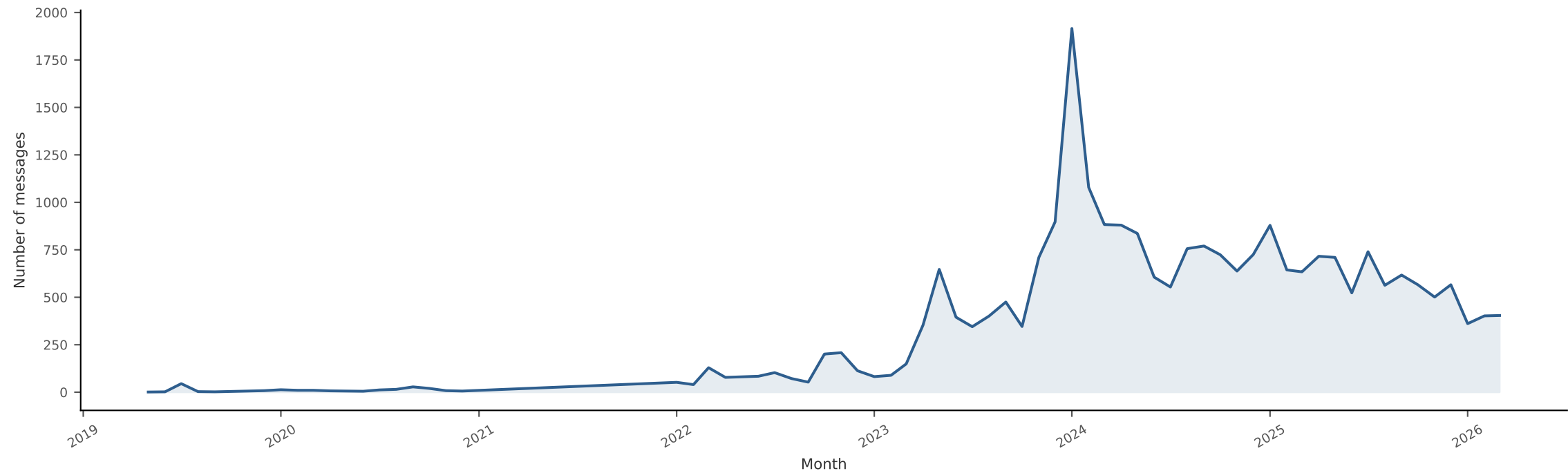
Exploratory Analysis — Time Series & CDS Prevalence

## **Section 1 — Forum Activity Over Time**

Messages per Year

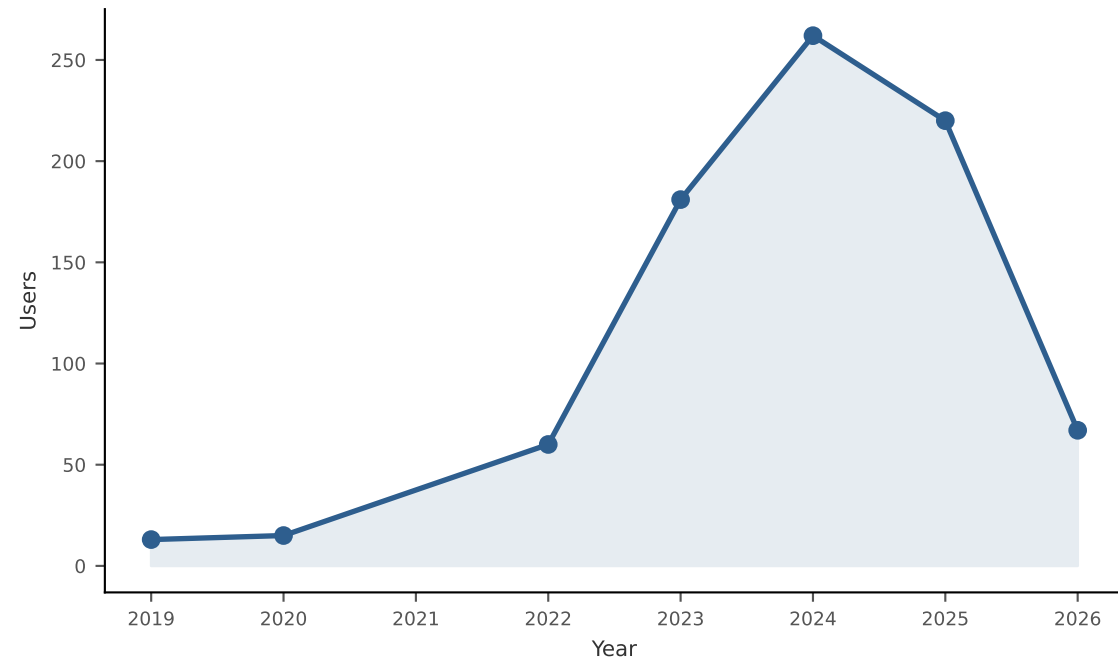


# Messages per Month

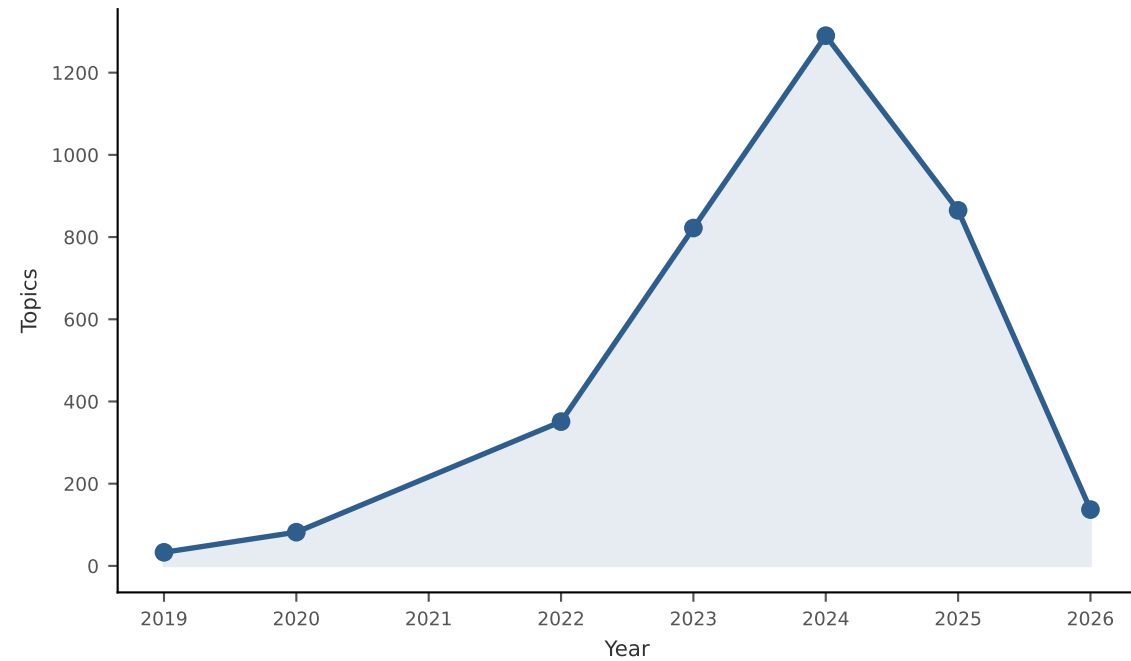


# Forum Activity — Year-level Metrics

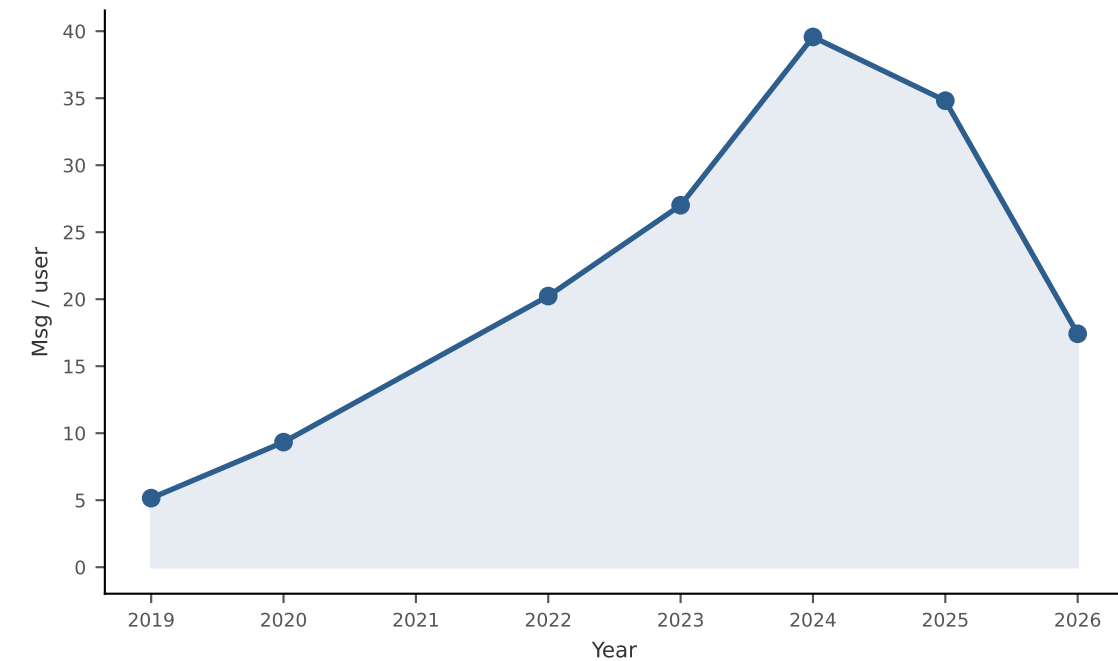
## Unique Users per Year



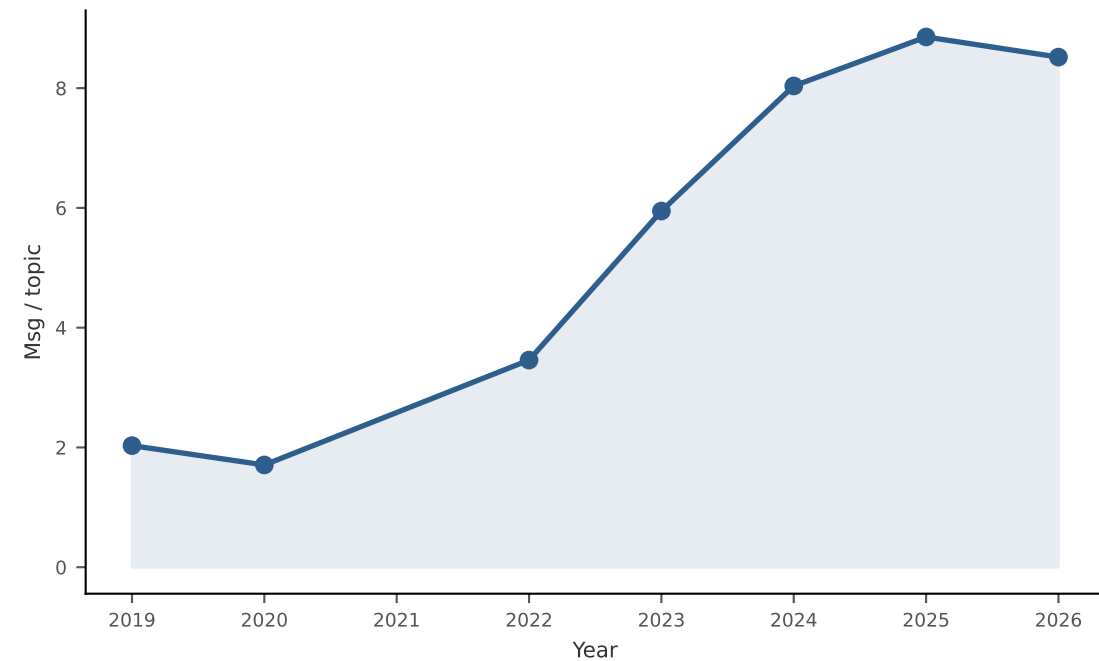
## Unique Topics per Year



## Messages per User per Year

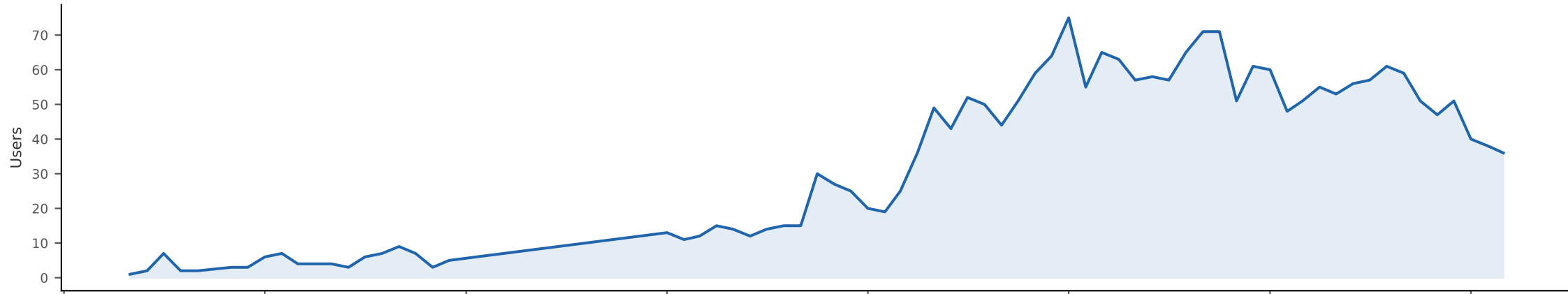


## Messages per Topic per Year

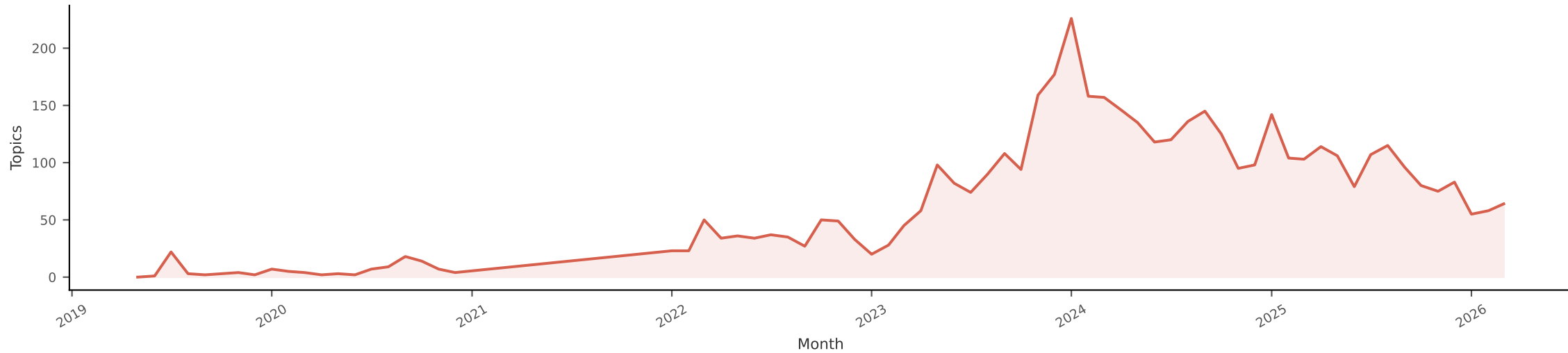


# Forum Activity — Month-level Metrics

## Unique Users per Month

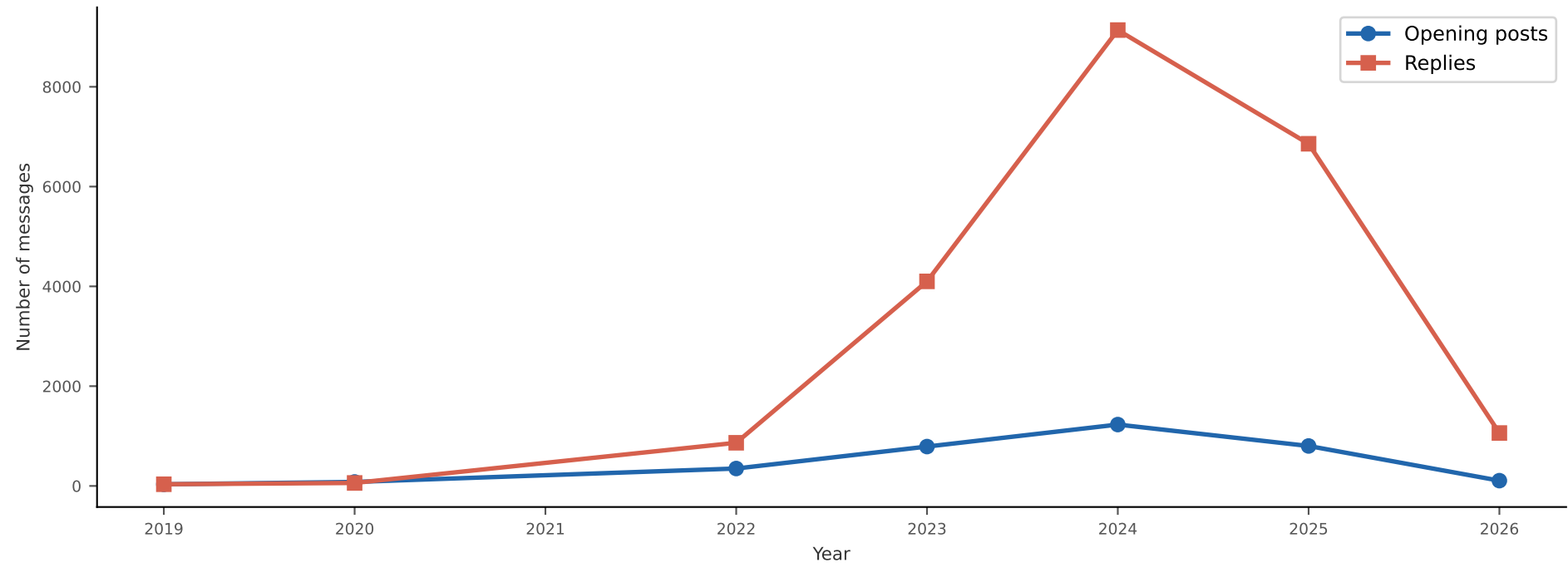


## Unique Topics per Month

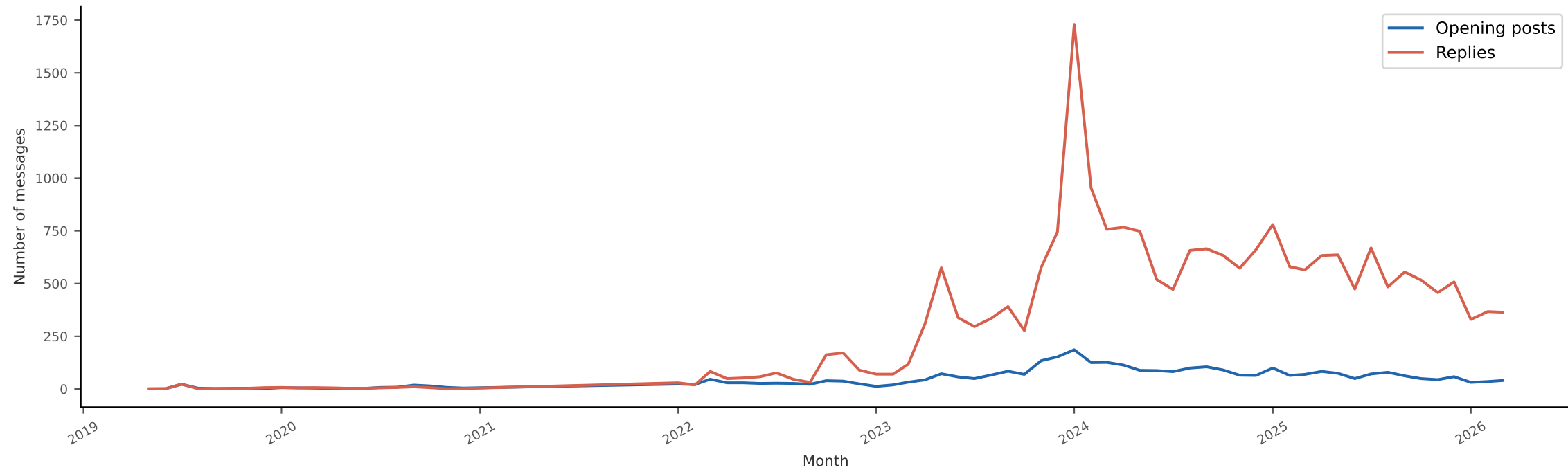




Message Volume by Role — Year Level



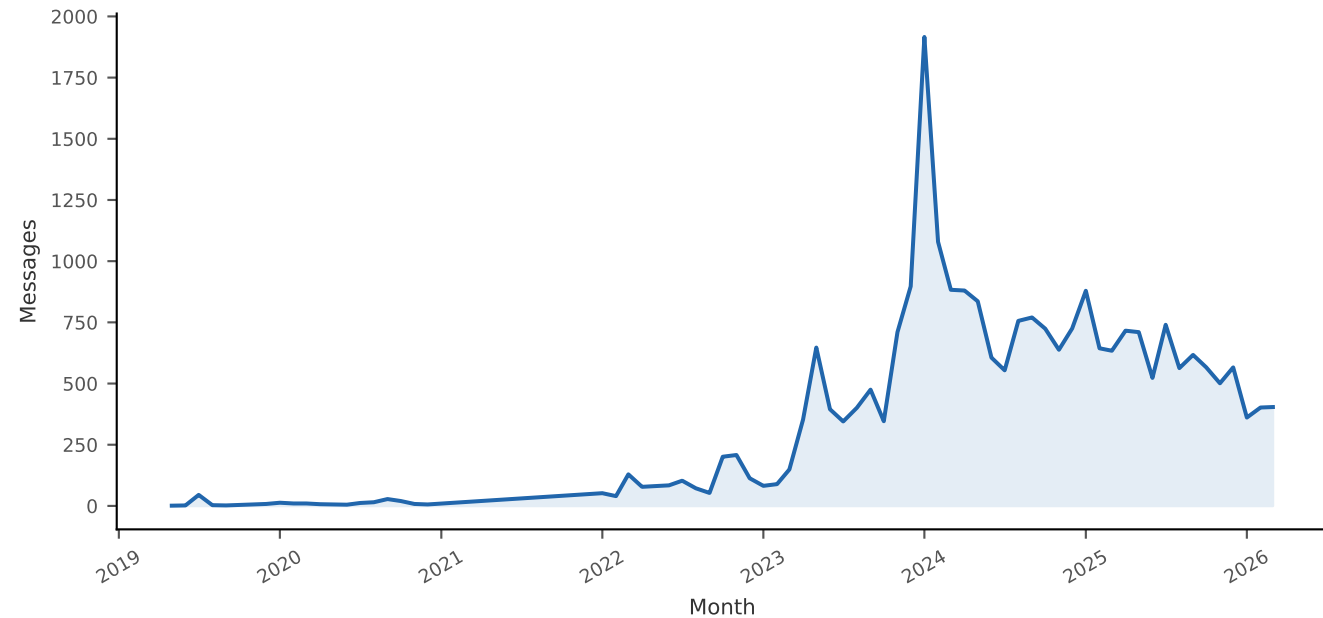
Message Volume by Role — Month Level



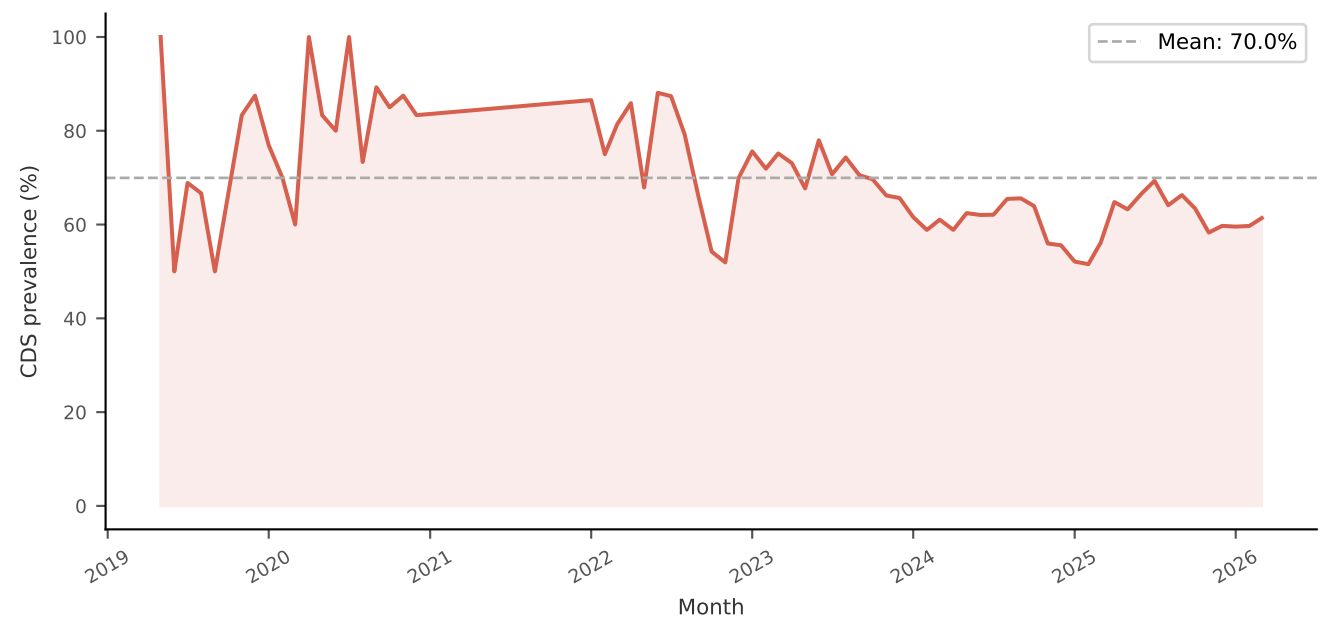
## **Section 2 — CDS Prevalence Over Time**

# Message Volume and CDS Prevalence Over Time (Monthly)

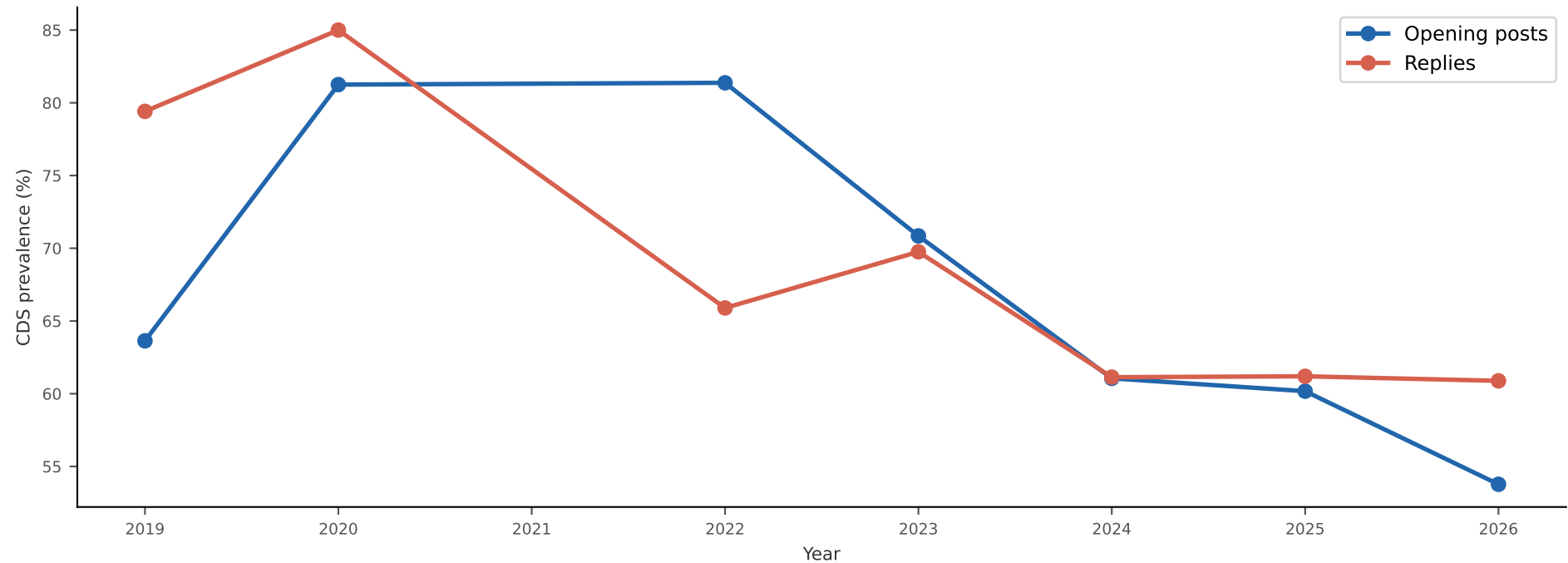
## Monthly Message Volume



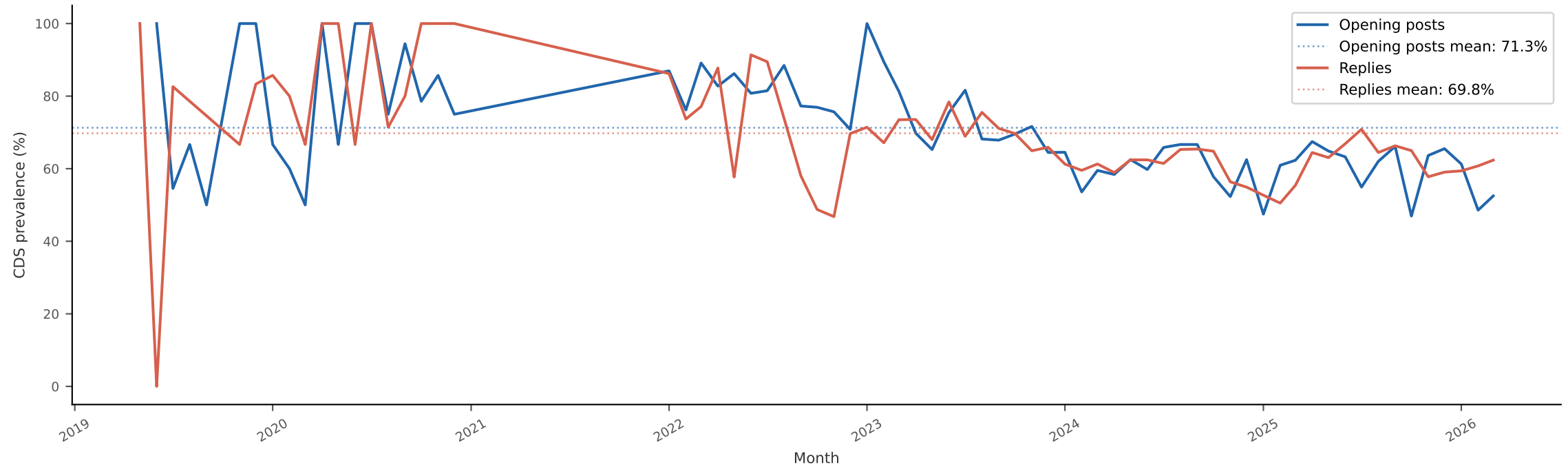
## Monthly CDS Prevalence (%)



CDS Prevalence by Role — Year Level

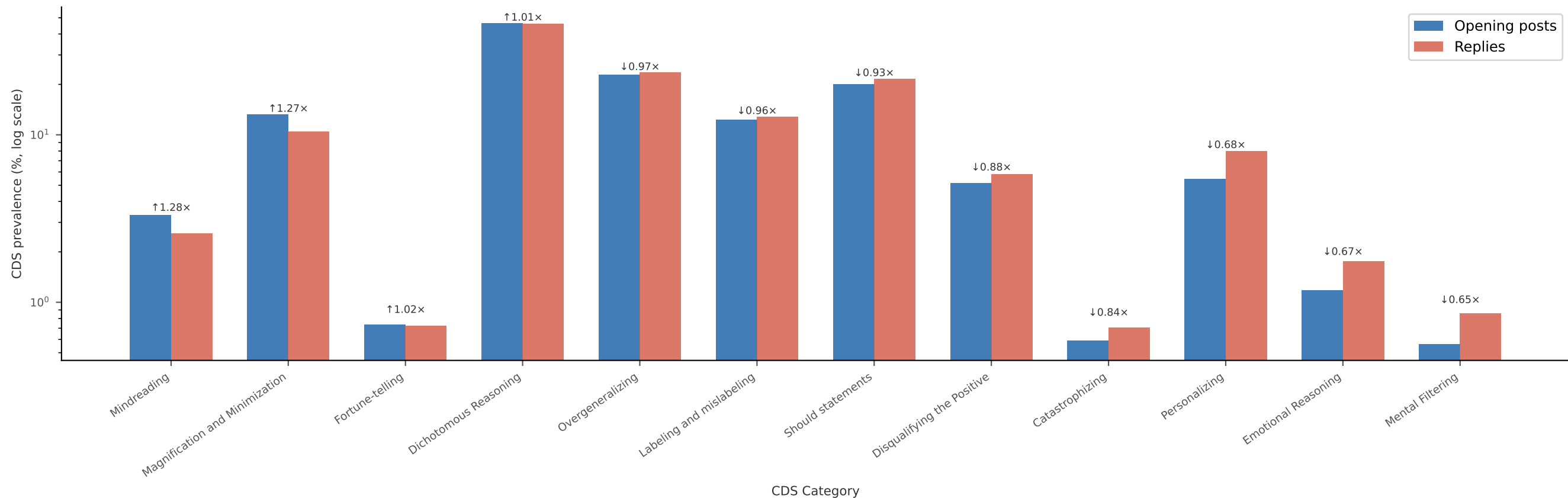


CDS Prevalence by Role — Month Level



## **Section 3 — CDS Prevalence by Category**

**CDS Prevalence by Category — Posts vs Replies**  
(sorted by prevalence ratio, log scale)

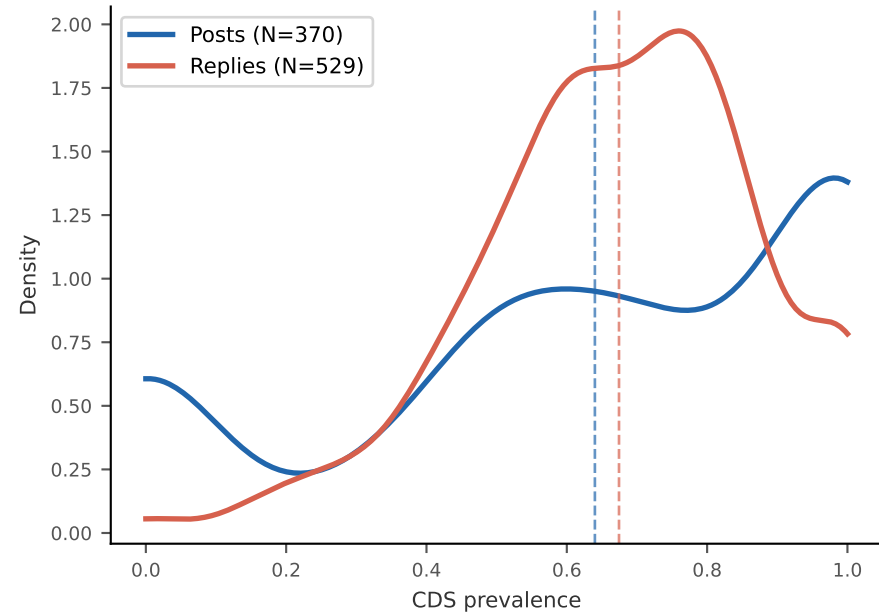




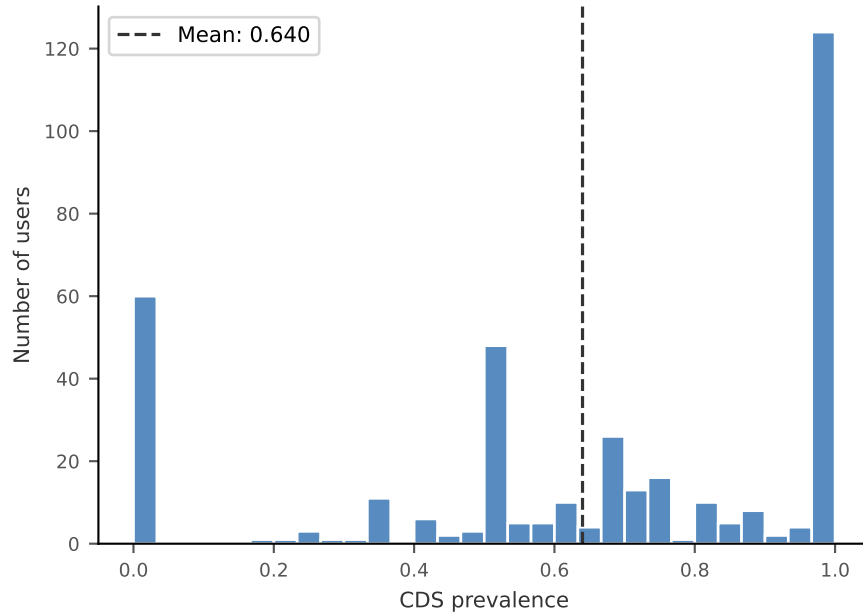
## **Section 4 — Within-User CDS Distribution**

# Within-User CDS Prevalence Distribution

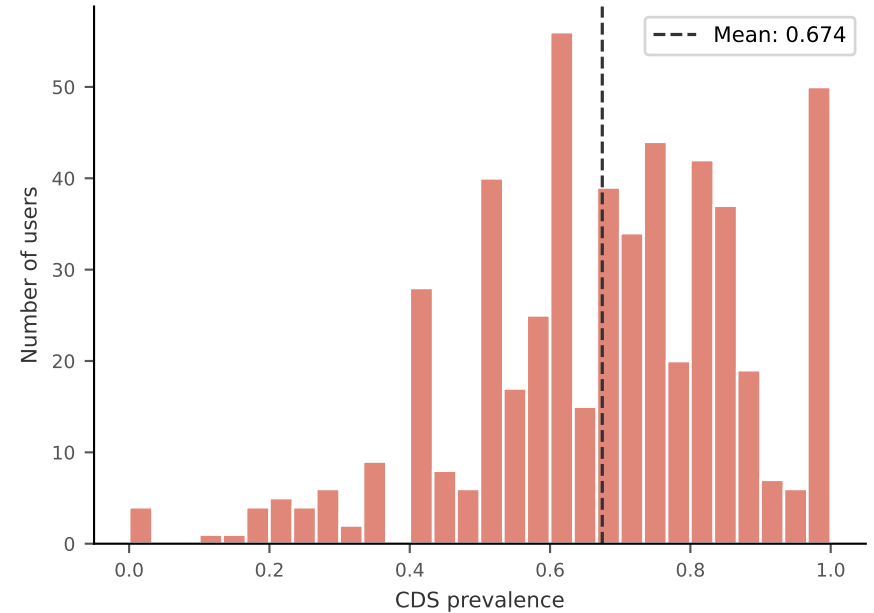
## KDE: Within-user CDS Prevalence



## Distribution: Posts



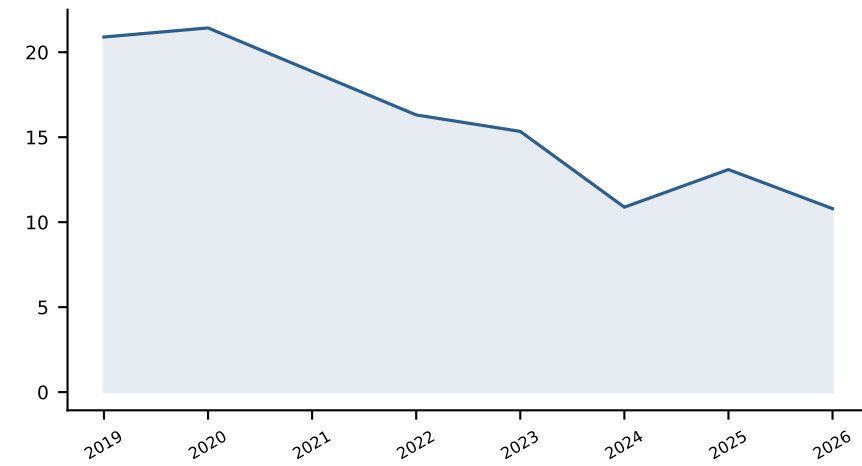
## Distribution: Replies



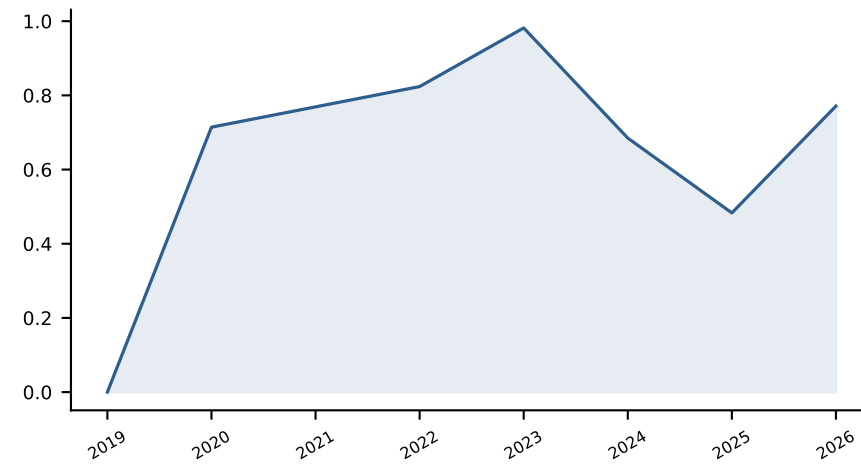
## **Section 5 — CDS per Category Over Time**

# CDS Prevalence per Category Over Time (Year Level)

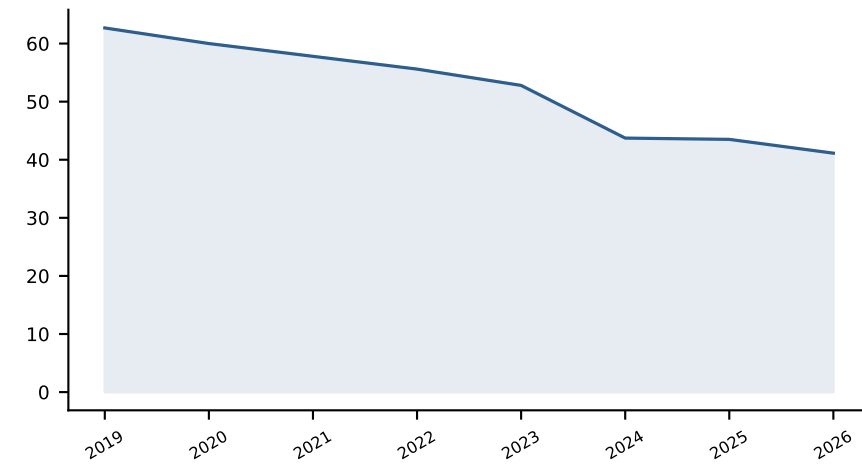
## Labeling And Mislabeling



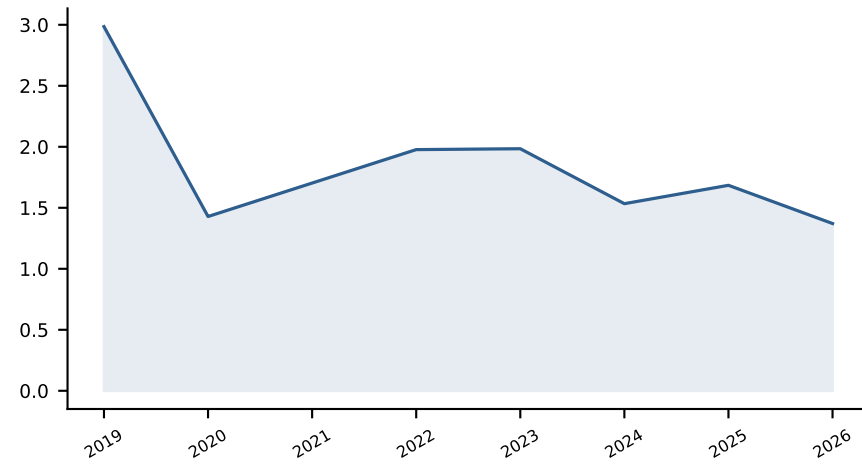
## Catastrophizing



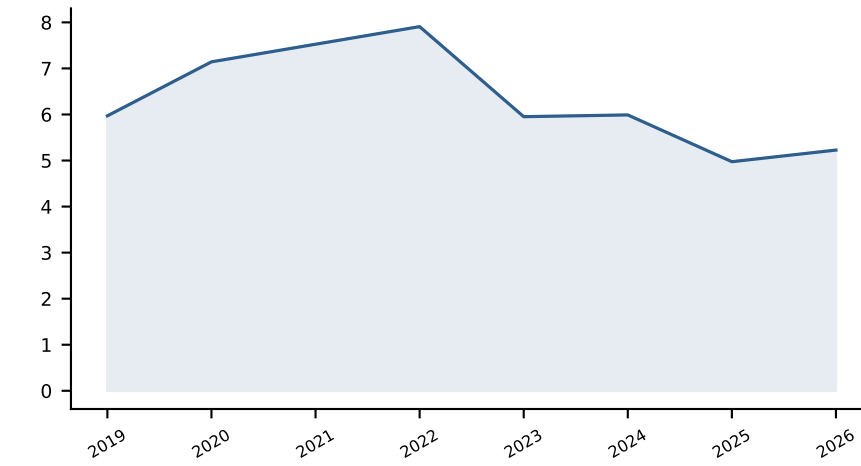
## Dichotomous Reasoning



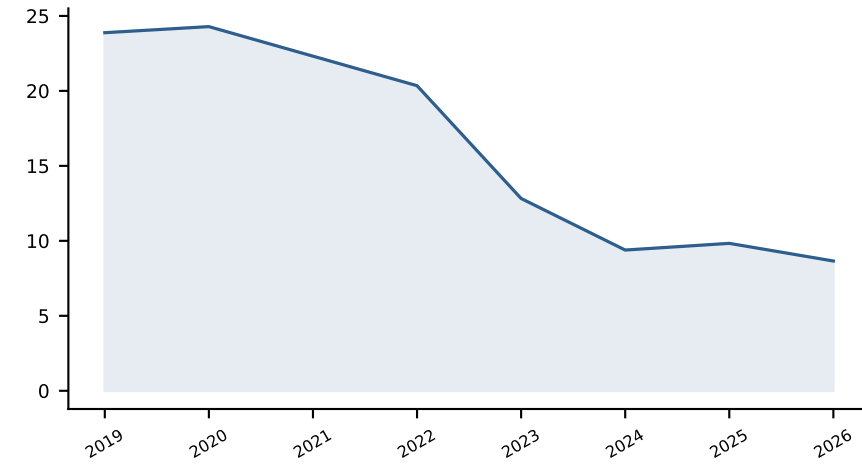
## Emotional Reasoning



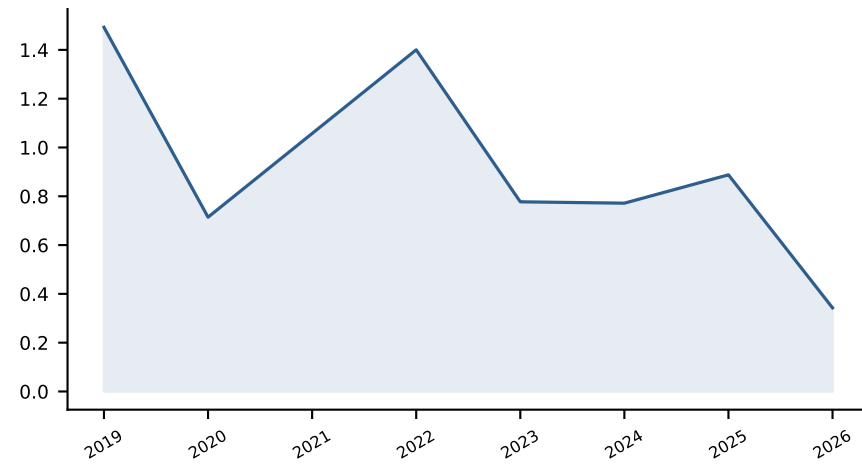
## Disqualifying The Positive



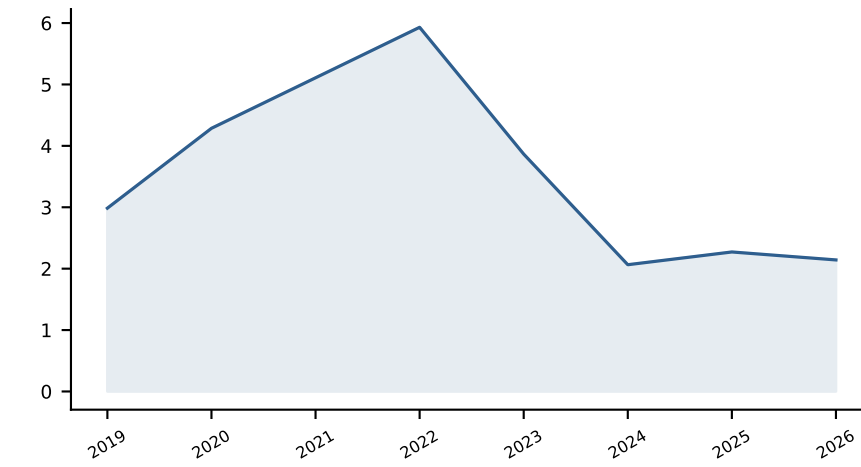
## Magnification And Minimization



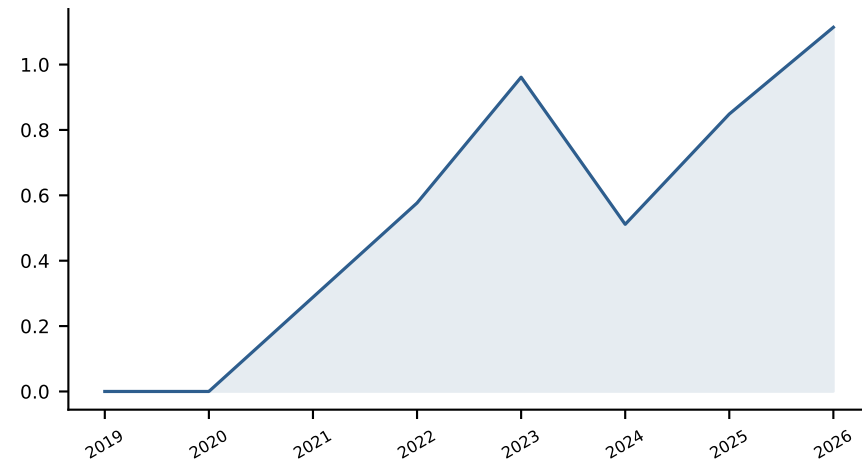
## Mental Filtering



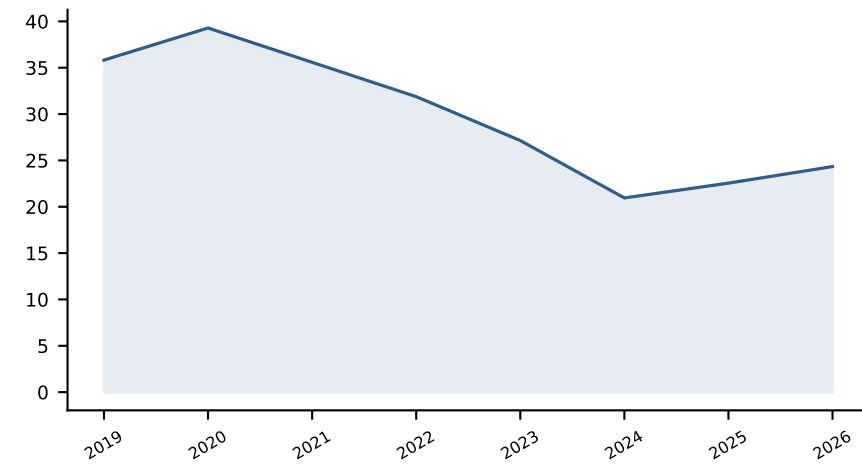
## Mindreading



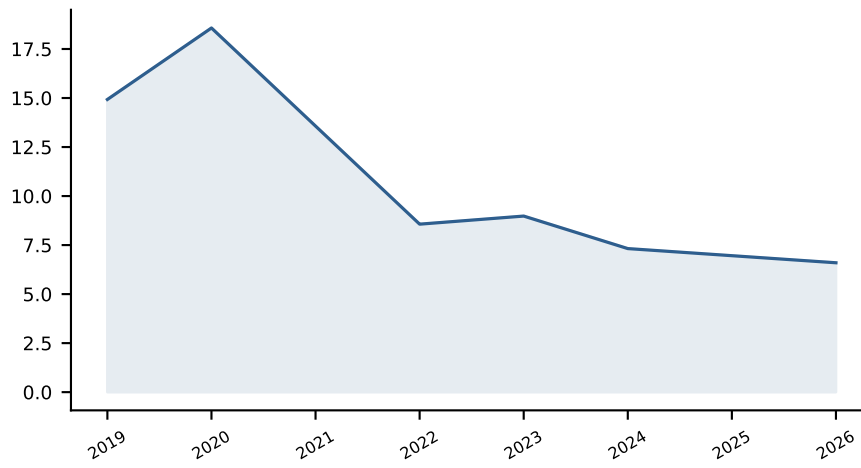
## Fortune-Telling



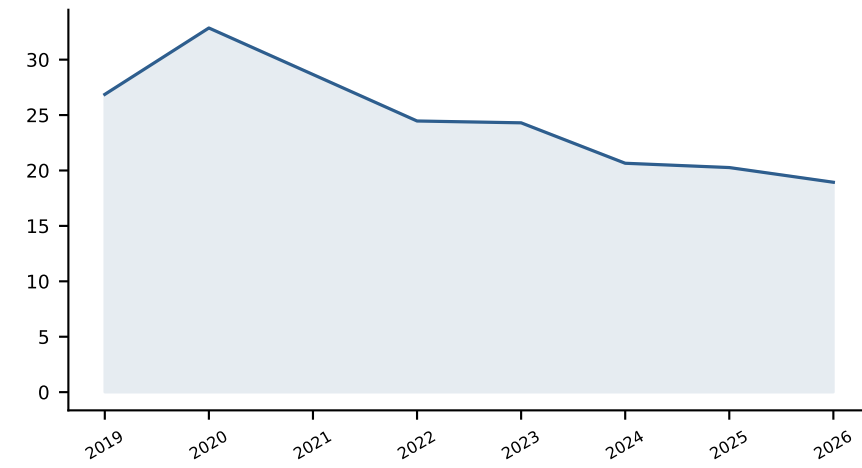
## Overgeneralizing



## Personalizing

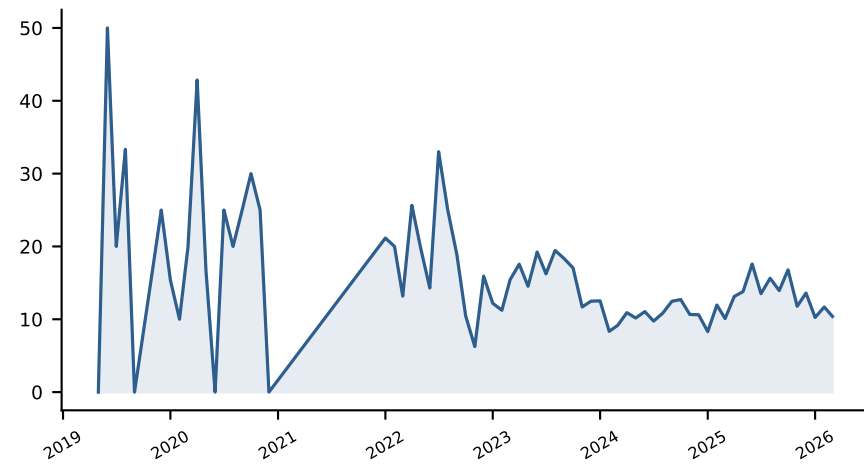


## Should Statements

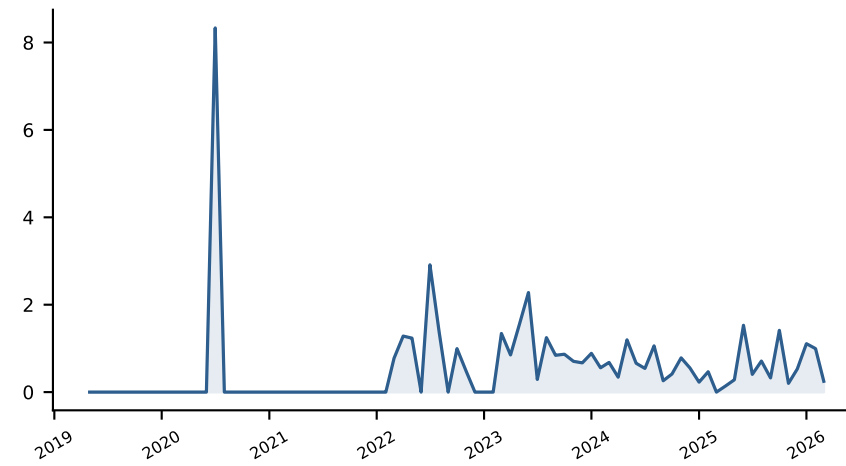


# CDS Prevalence per Category Over Time (Month Level)

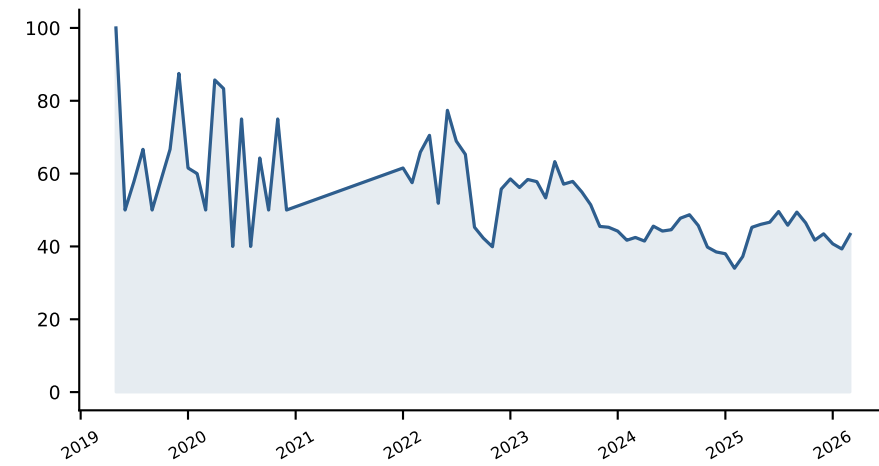
## Labeling And Mislabeling



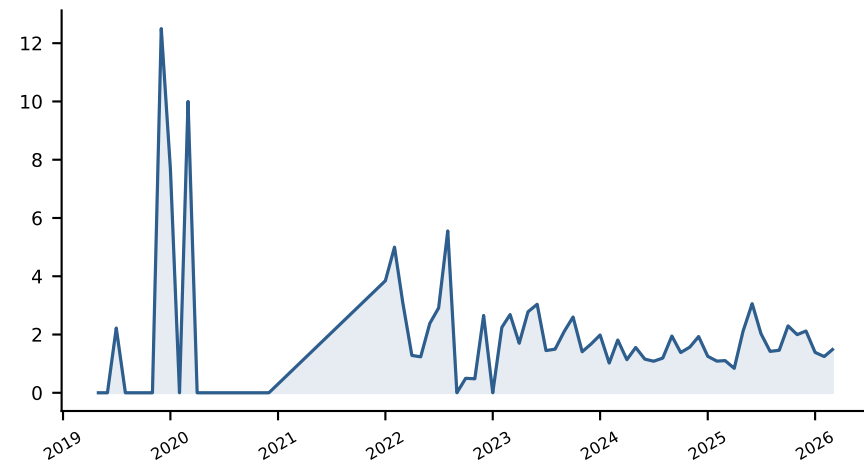
## Catastrophizing



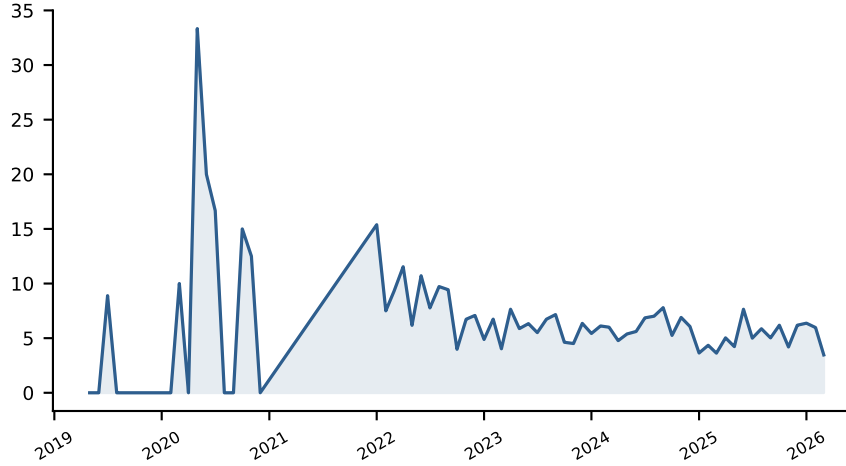
## Dichotomous Reasoning



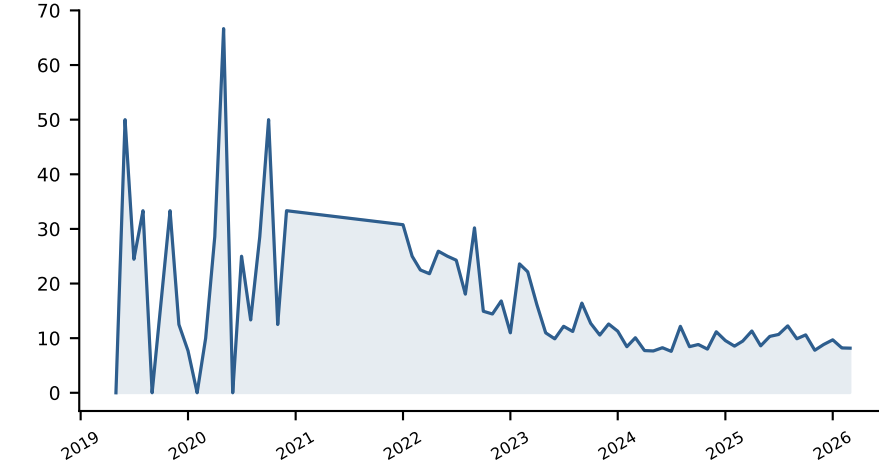
## Emotional Reasoning



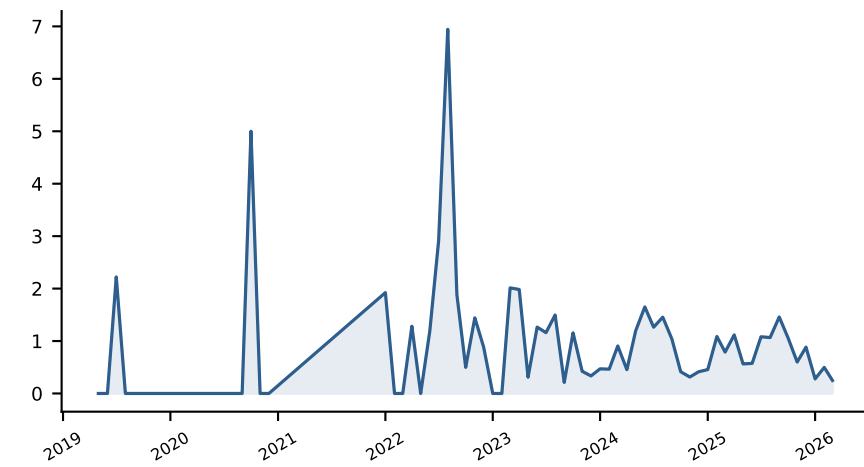
## Disqualifying The Positive



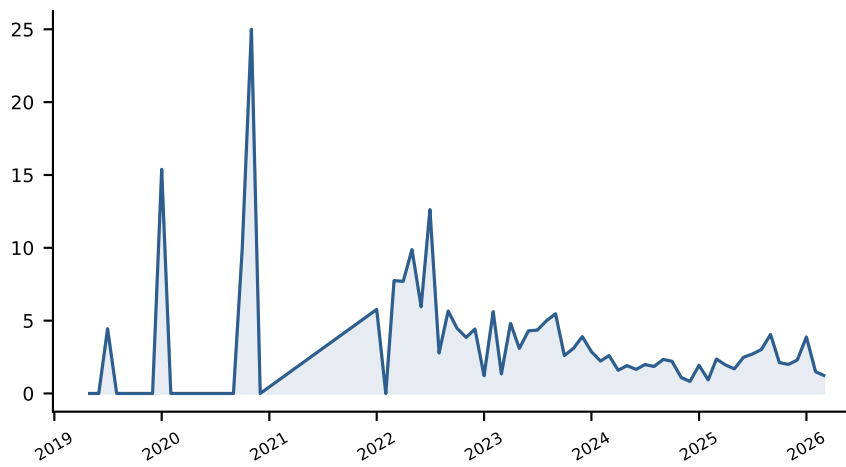
## Magnification And Minimization



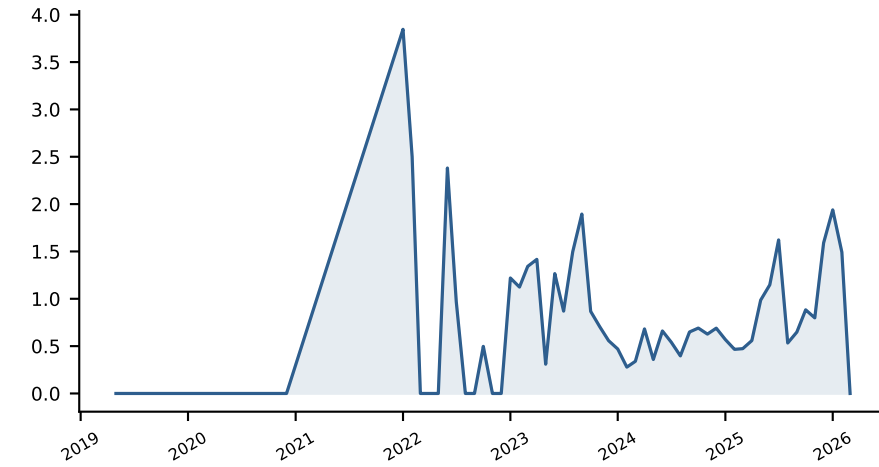
## Mental Filtering



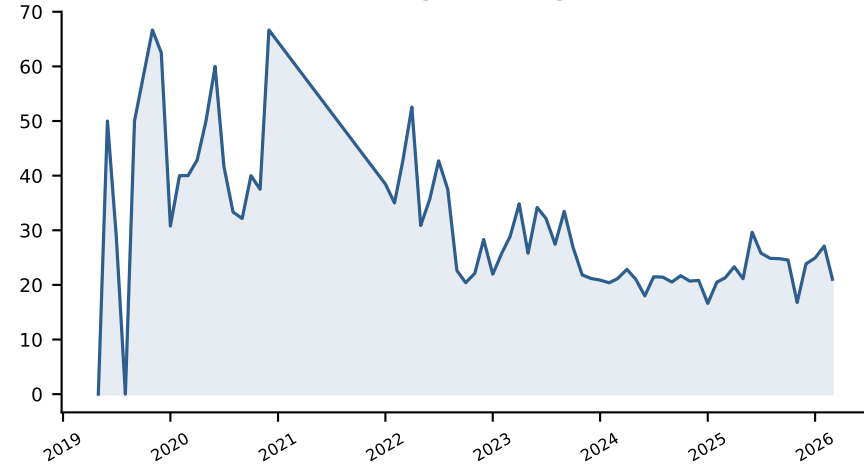
## Mindreading



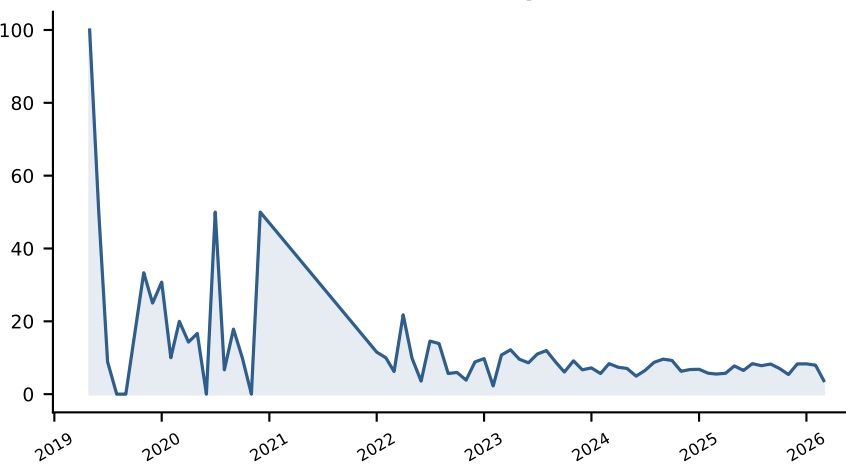
## Fortune-Telling



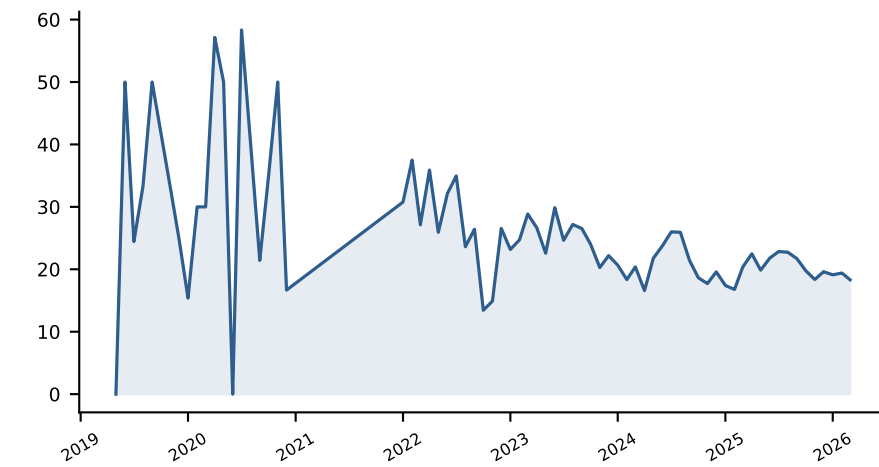
## Overgeneralizing



## Personalizing



## Should Statements



## **4. Cognitive Distortion (CDS) Prevalence**

# Depression Connect Forum

Most Common Cognitive Distortions (CDS Prevalence)

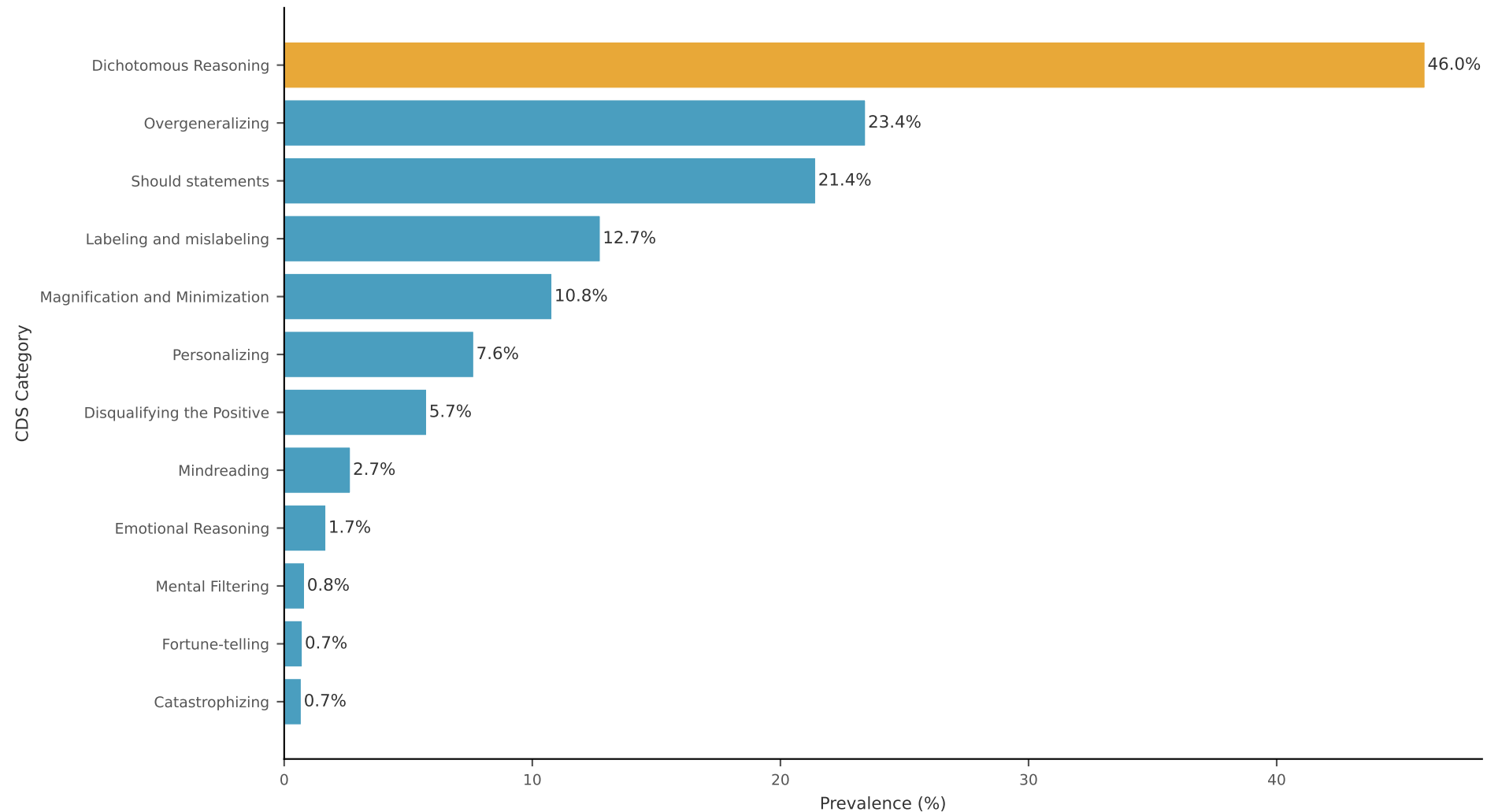
## **Section 1 — Category Ranking**



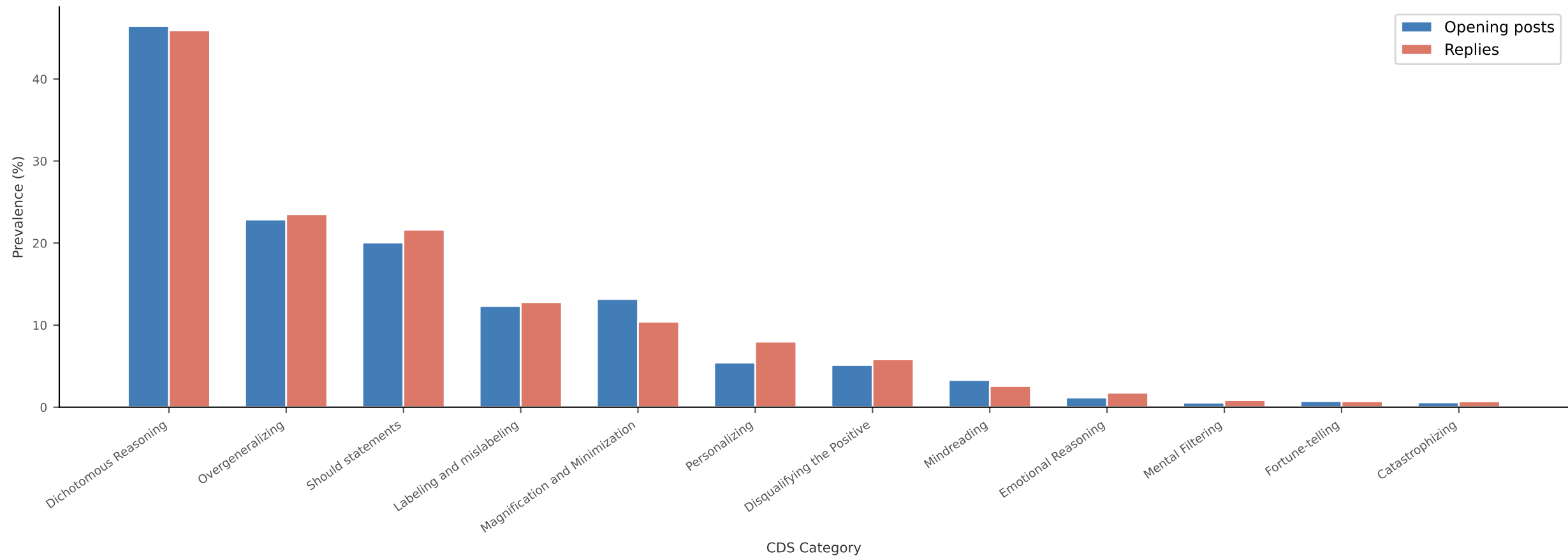
CDS Category Ranking — Mann-Whitney U (per-user rates) with BH correction  
(p-BH = Benjamini-Hochberg corrected; yellow rows significant after correction)

| Category                       | Overall % | Posts % | Replies % | Ratio P/R | N users (post) | N users (reply) | p-value (MWU) | p-BH   | Rank-biserial r |
|--------------------------------|-----------|---------|-----------|-----------|----------------|-----------------|---------------|--------|-----------------|
| Dichotomous Reasoning          | 45.98     | 46.46   | 45.91     | 1.012     | 370            | 529             | 0.0029        | 0.0029 | 0.1164          |
| Overgeneralizing               | 23.43     | 22.87   | 23.52     | 0.973     | 370            | 529             | 0.0           | 0.0    | 0.2065          |
| Should statements              | 21.42     | 20.07   | 21.63     | 0.928     | 370            | 529             | 0.0           | 0.0    | 0.2392          |
| Labeling and mislabeling       | 12.74     | 12.34   | 12.8      | 0.964     | 370            | 529             | 0.0           | 0.0    | 0.2881          |
| Magnification and Minimization | 10.79     | 13.19   | 10.42     | 1.266     | 370            | 529             | 0.0           | 0.0    | 0.2524          |
| Personalizing                  | 7.64      | 5.43    | 7.98      | 0.681     | 370            | 529             | 0.0           | 0.0    | 0.3756          |
| Disqualifying the Positive     | 5.74      | 5.14    | 5.83      | 0.88      | 370            | 529             | 0.0           | 0.0    | 0.2159          |
| Mindreading                    | 2.67      | 3.31    | 2.58      | 1.283     | 370            | 529             | 0.0           | 0.0    | 0.1229          |
| Emotional Reasoning            | 1.68      | 1.18    | 1.76      | 0.671     | 370            | 529             | 0.0           | 0.0    | 0.1853          |
| Mental Filtering               | 0.82      | 0.56    | 0.86      | 0.653     | 370            | 529             | 0.0           | 0.0    | 0.1146          |
| Fortune-telling                | 0.73      | 0.74    | 0.72      | 1.02      | 370            | 529             | 0.0001        | 0.0001 | 0.0775          |
| Catastrophizing                | 0.69      | 0.59    | 0.71      | 0.837     | 370            | 529             | 0.0           | 0.0    | 0.0995          |

## CDS Category Prevalence Ranking (% of messages containing at least one phrase from this category)



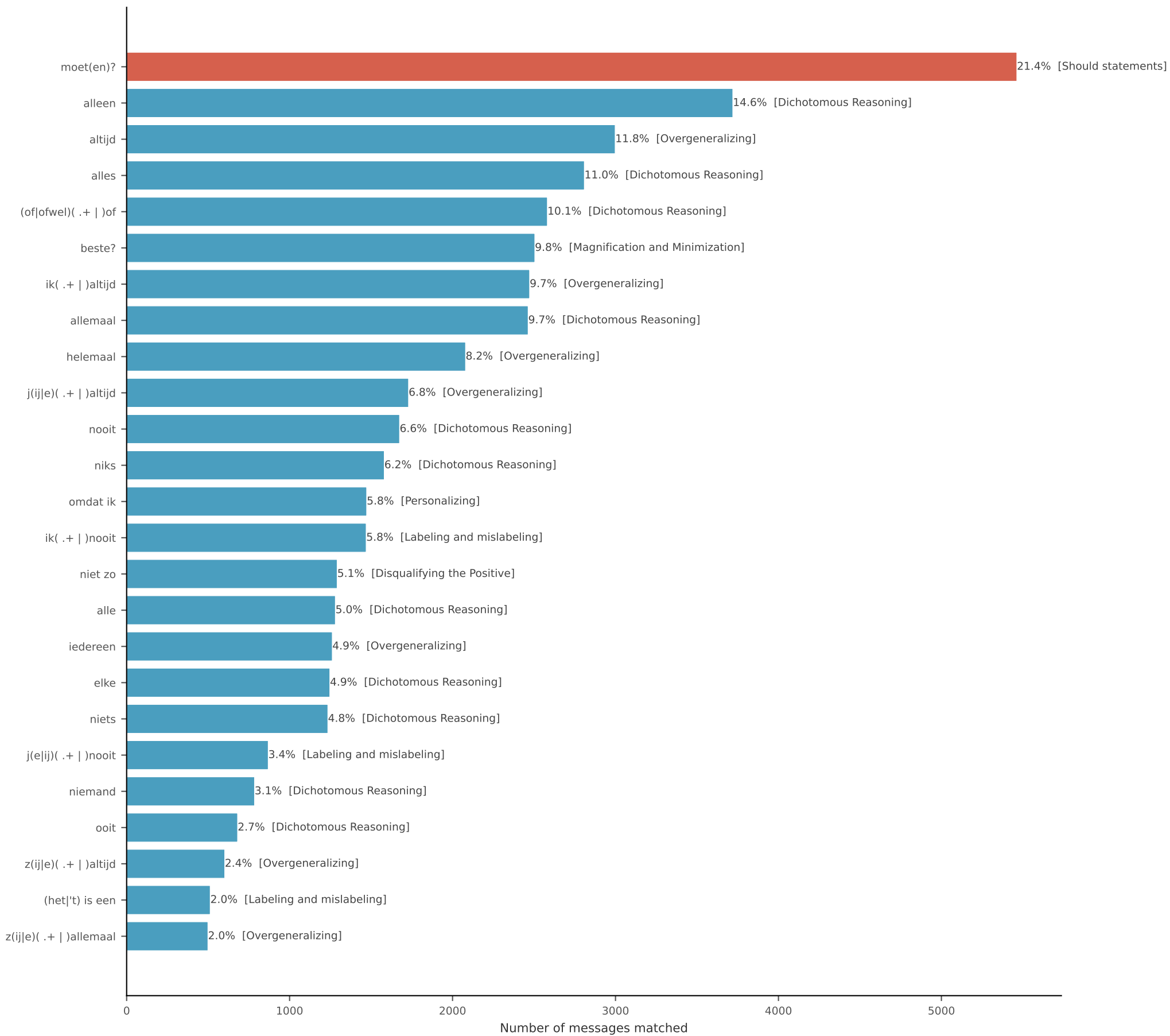
CDS Prevalence by Category — Opening Posts vs Replies



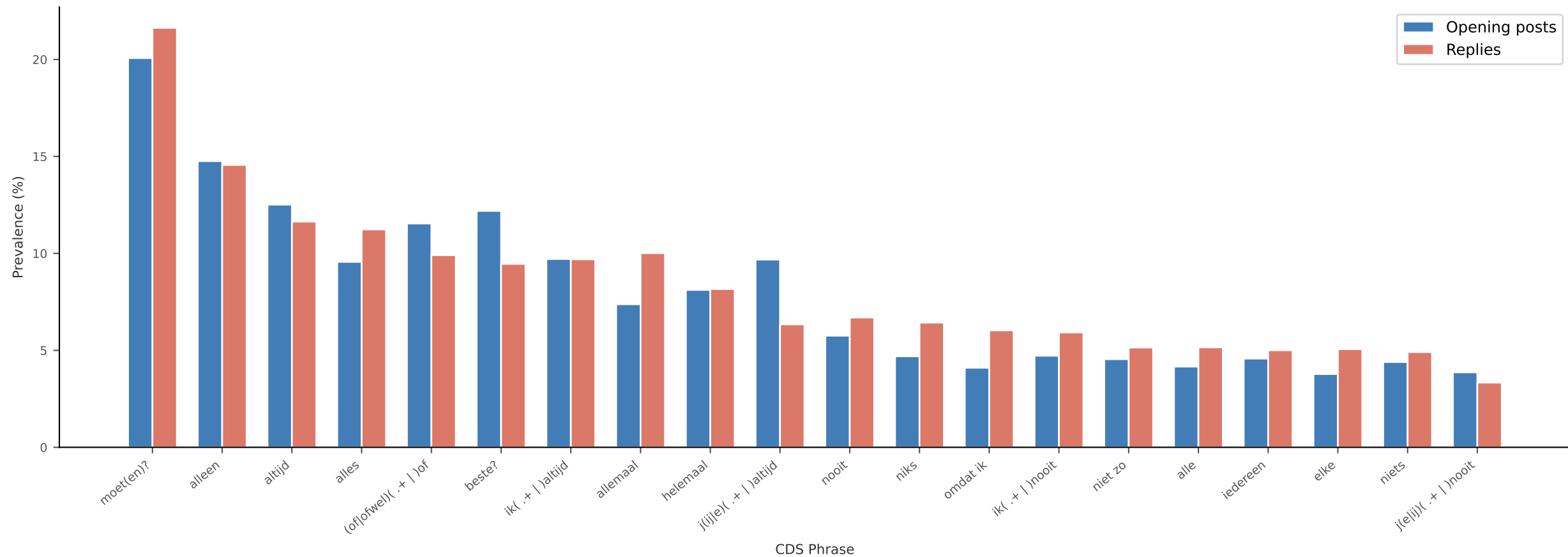
## **Section 2 — Individual CDS Phrases**

## Top 25 Most Frequent CDS Phrases (label = prevalence % | category)

CDS Phrase



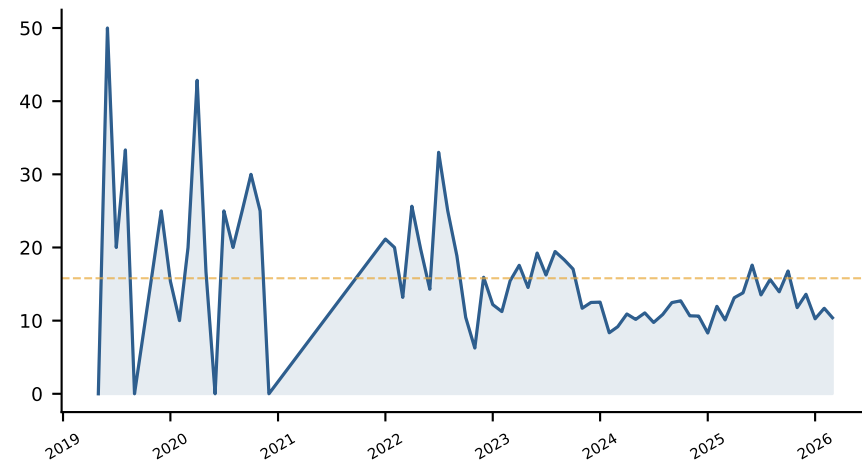
Top 20 CDS Phrases — Opening Posts vs Replies (prevalence %)



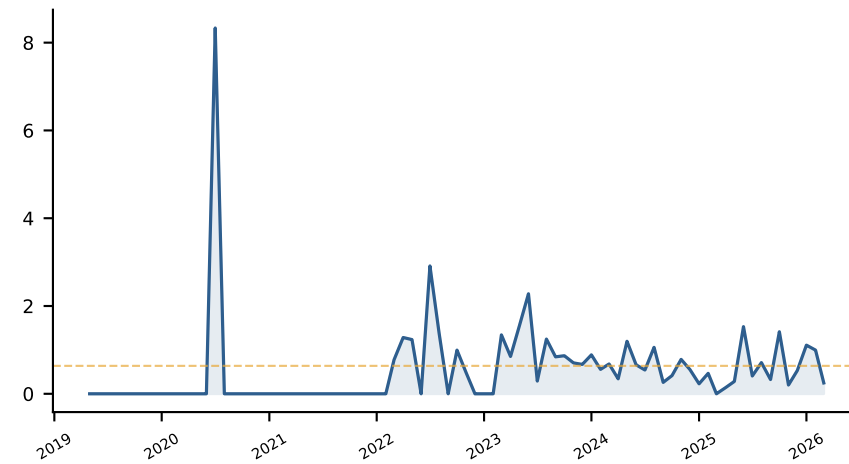
## **Section 3 — Category Trends Over Time**

# CDS Category Prevalence Over Time (Monthly, dashed = mean)

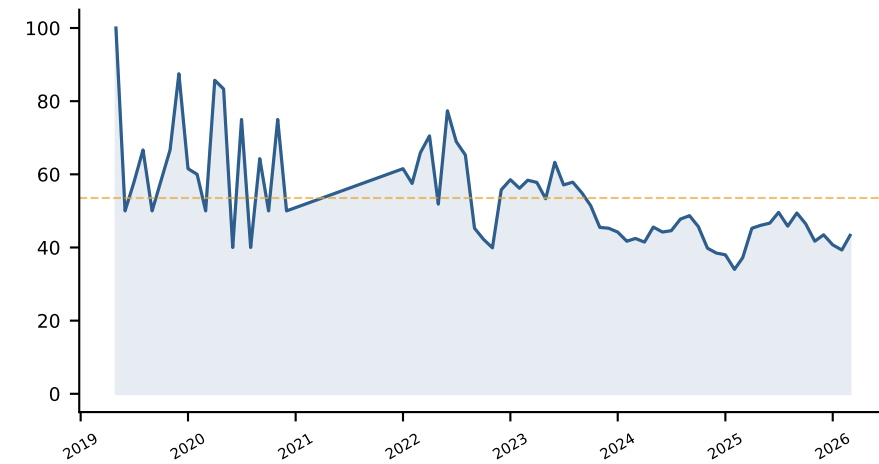
## Labeling and mislabeling



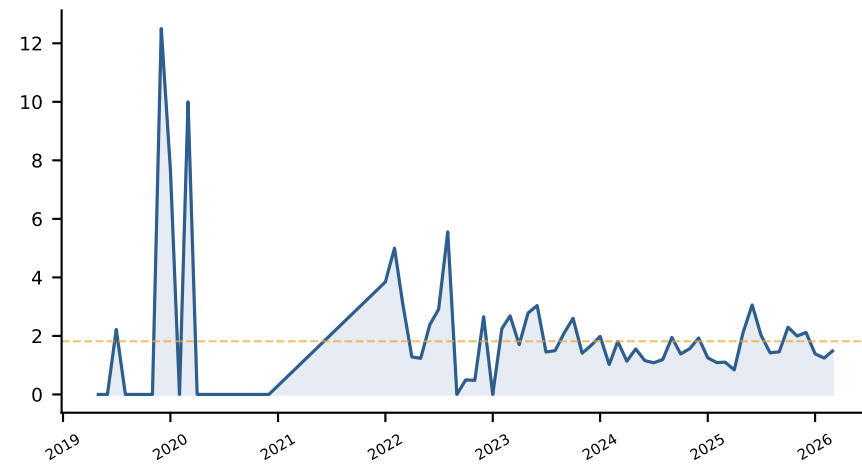
## Catastrophizing



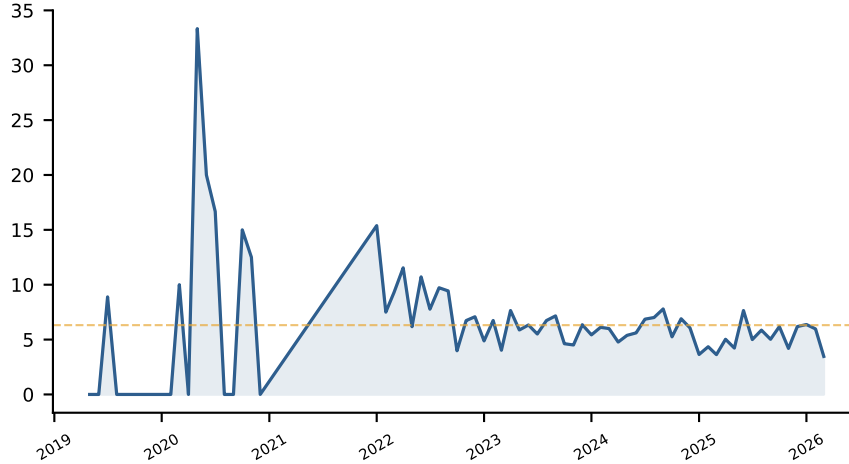
## Dichotomous Reasoning



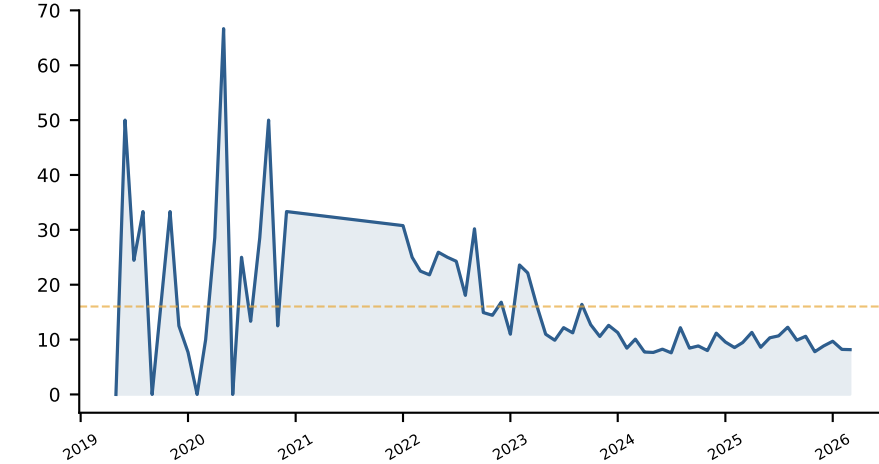
## Emotional Reasoning



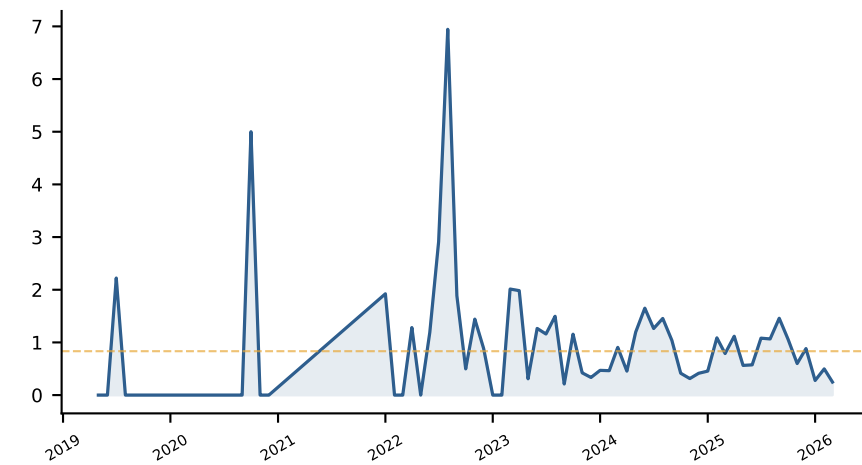
## Disqualifying the Positive



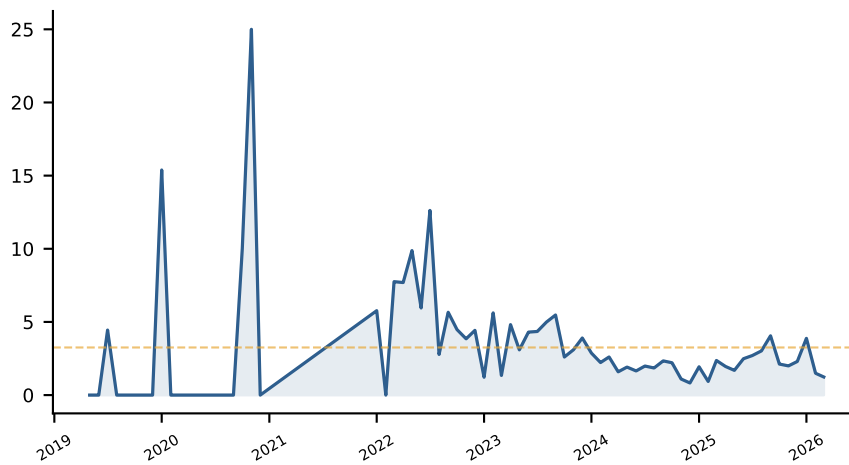
## Magnification and Minimization



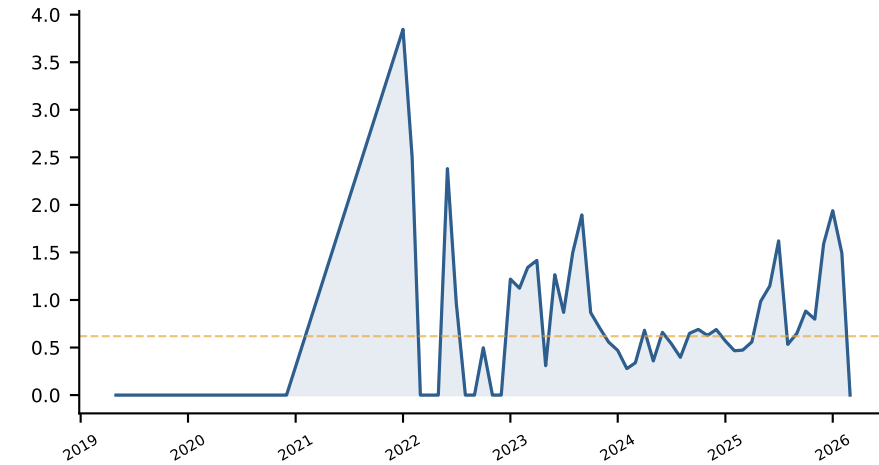
## Mental Filtering



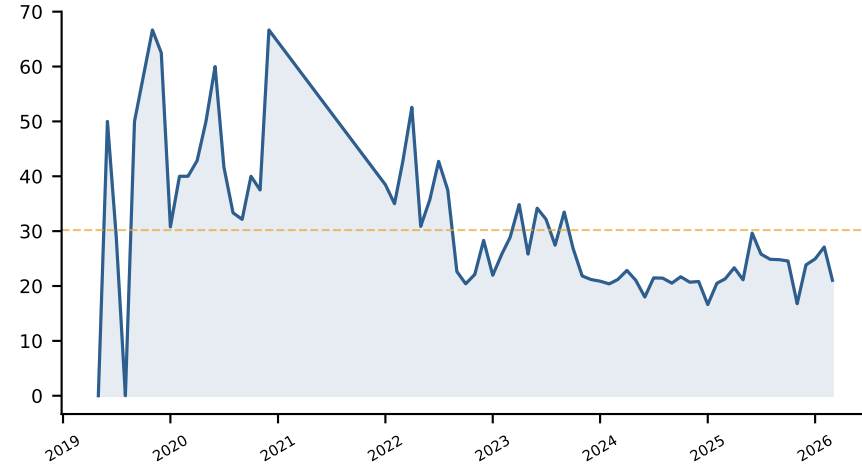
## Mindreading



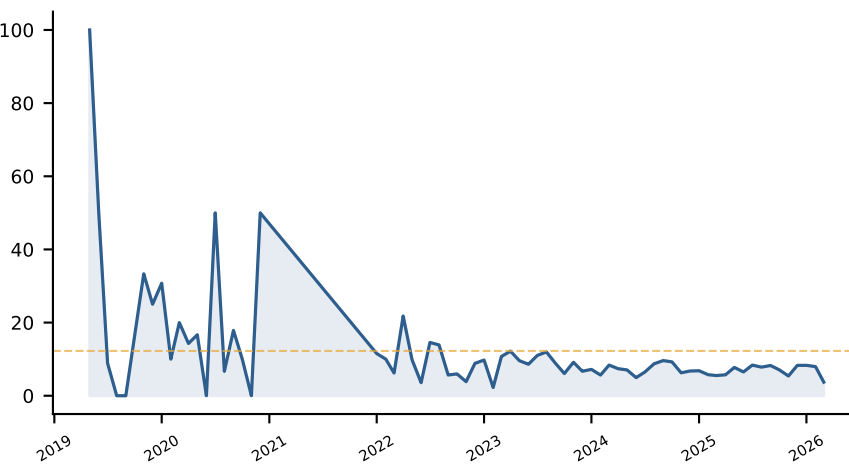
## Fortune-telling



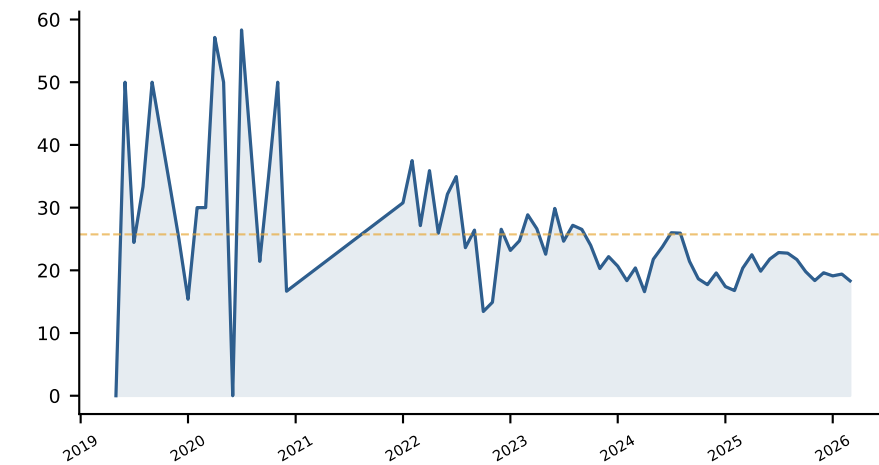
## Overgeneralizing



## Personalizing



## Should statements





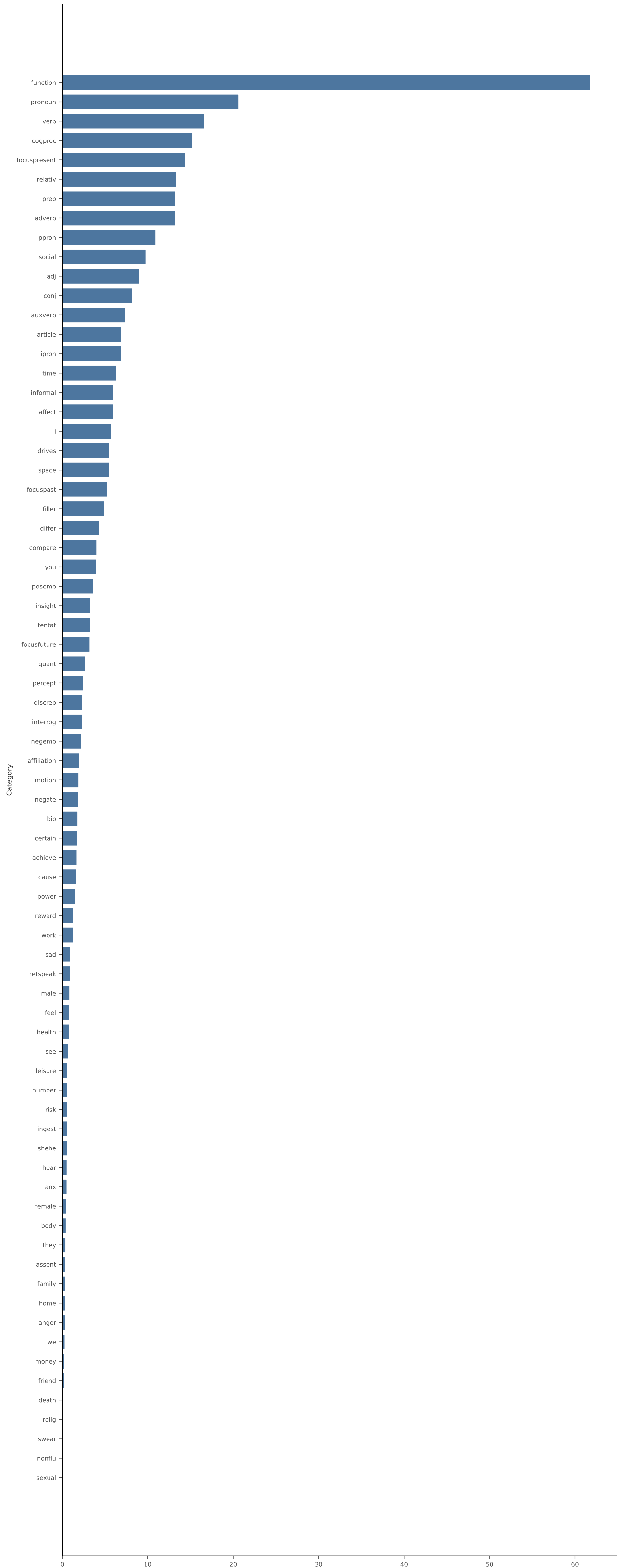
## **5. LIWC Psycholinguistic Analysis**

# Depression Connect Forum

LIWC Psycholinguistic Feature Analysis

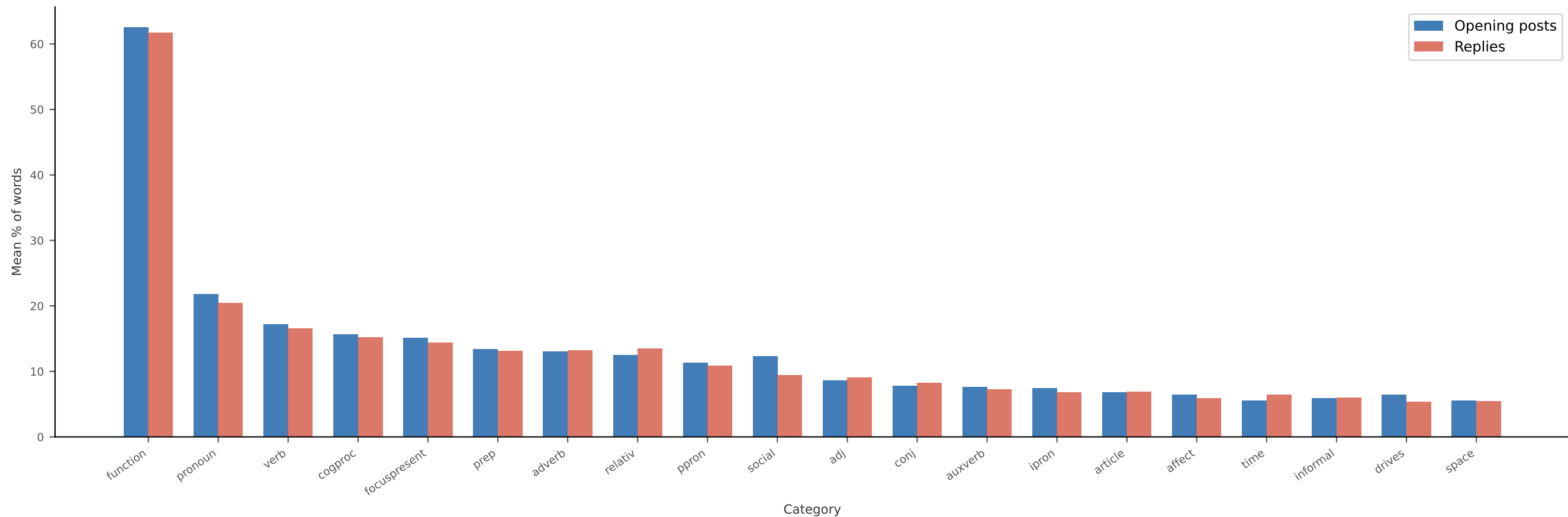
## **Section 1 — Overall Category Prevalence**

Overall LIWC Category Prevalence (mean % of words)



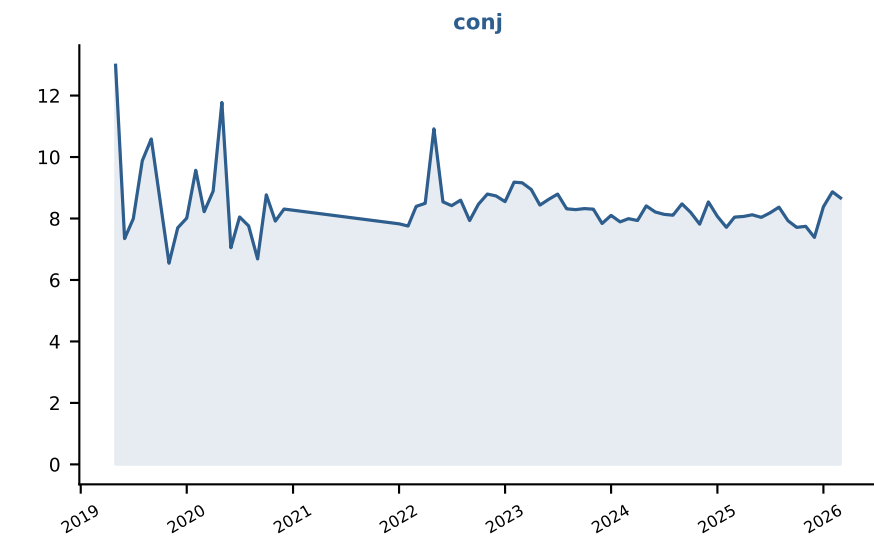
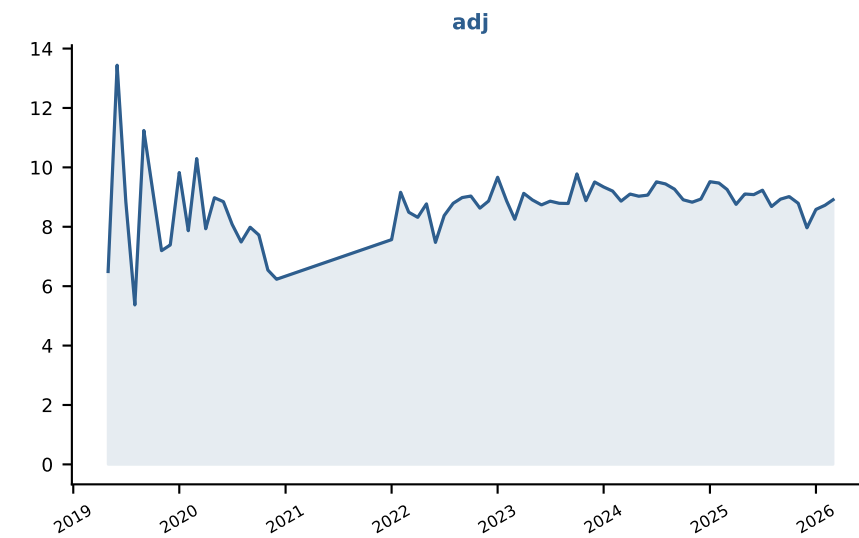
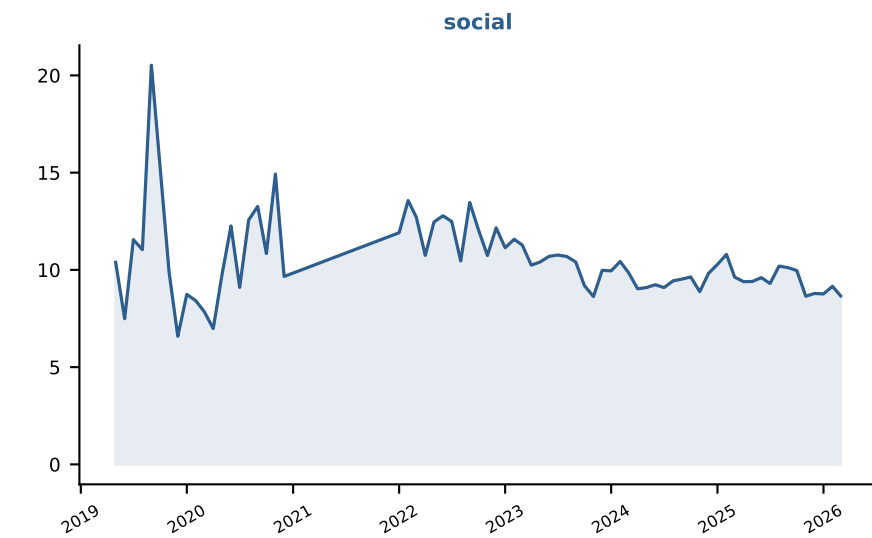
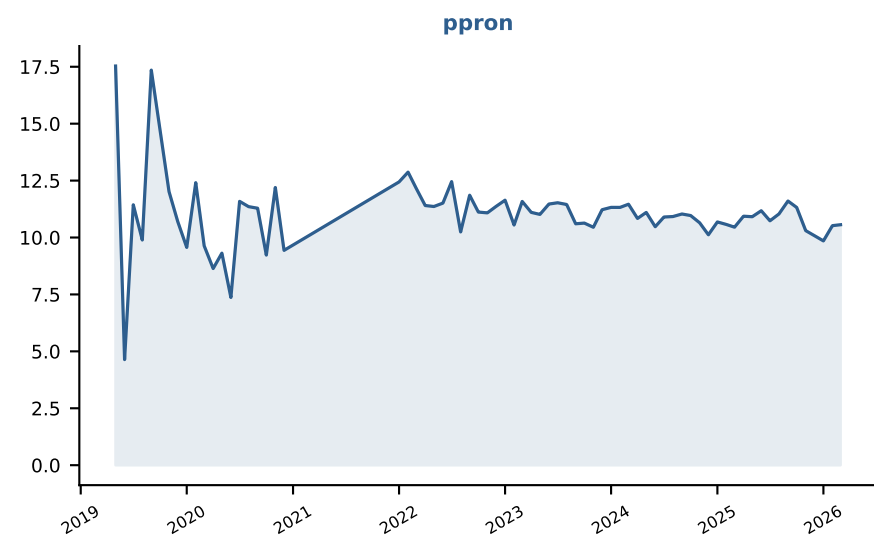
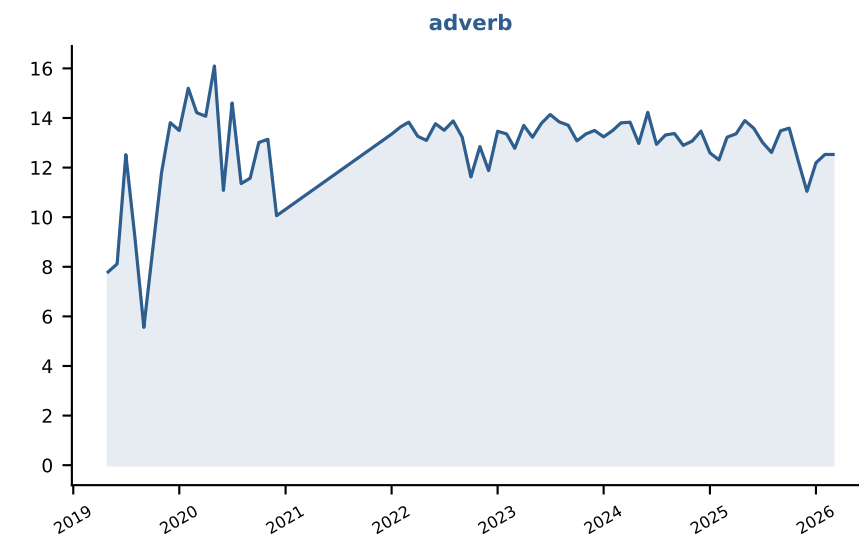
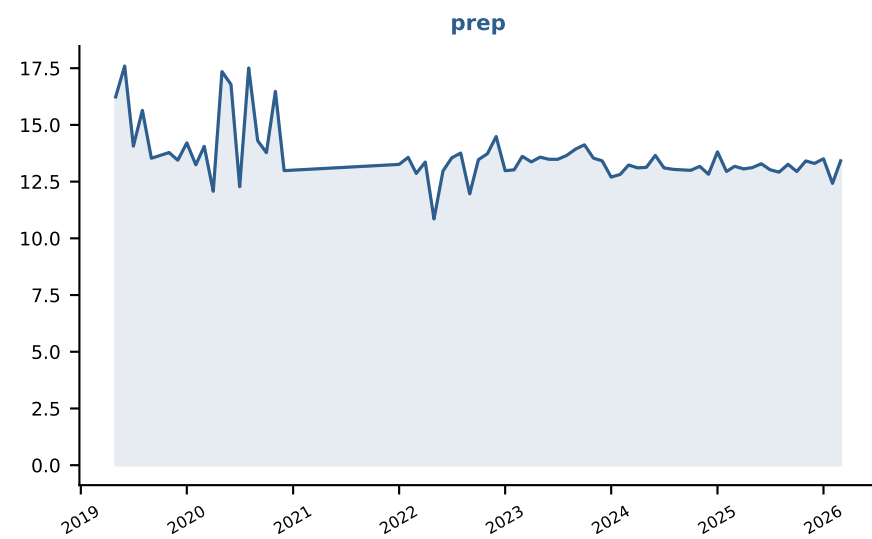
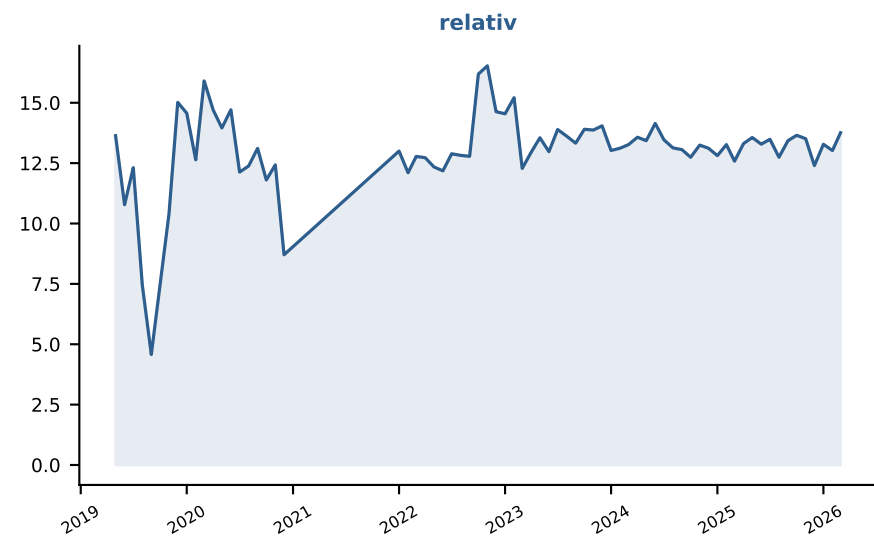
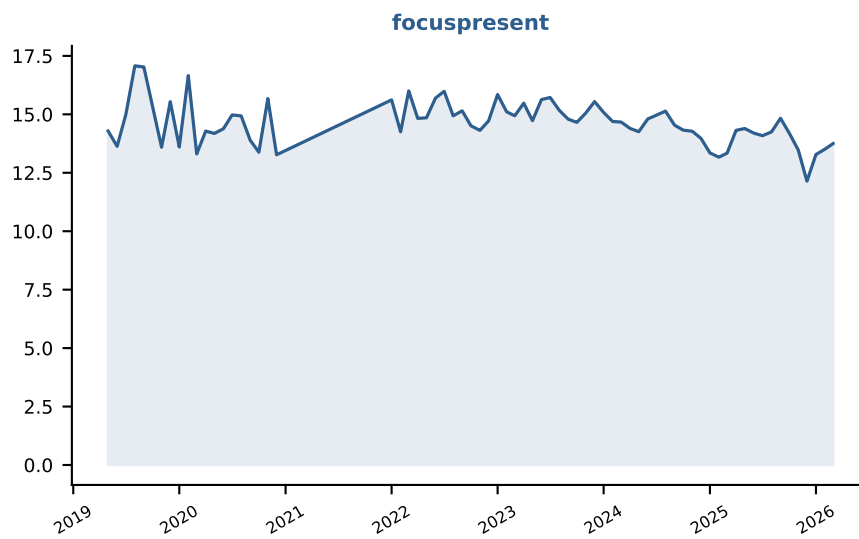
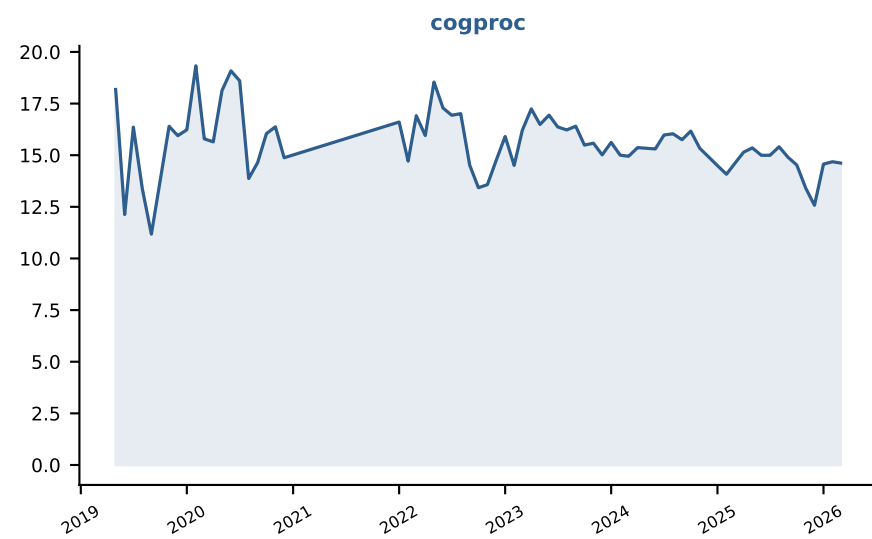
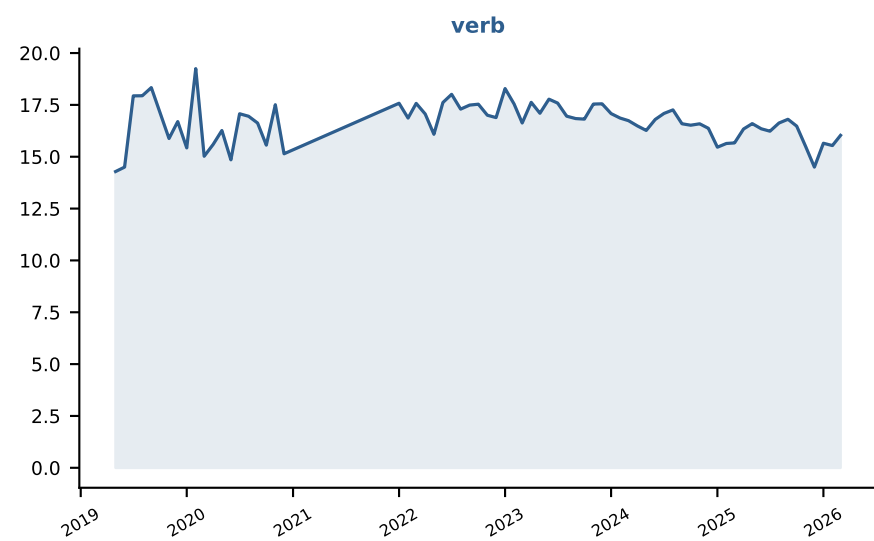
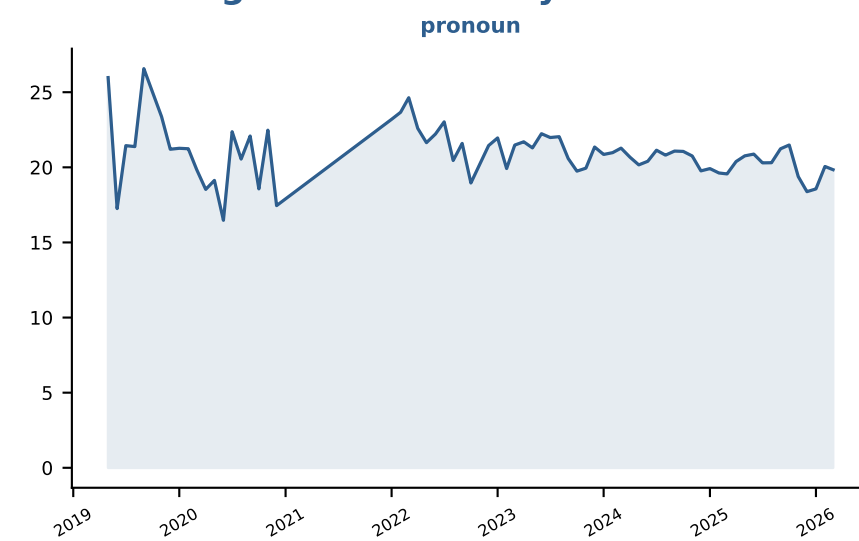
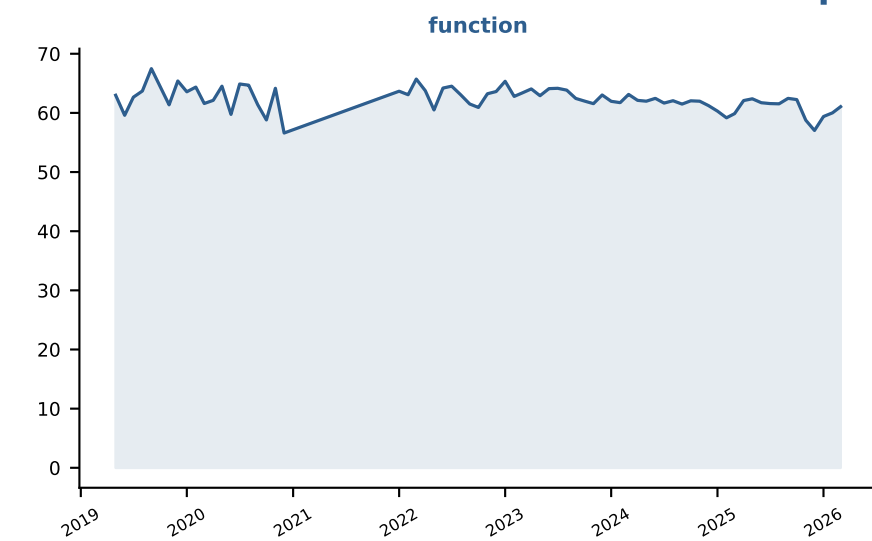
## **Section 2 — Posts vs Replies**

Top 20 LIWC Categories — Posts vs Replies (mean % of words)



## Section 3 — Top Categories Over Time

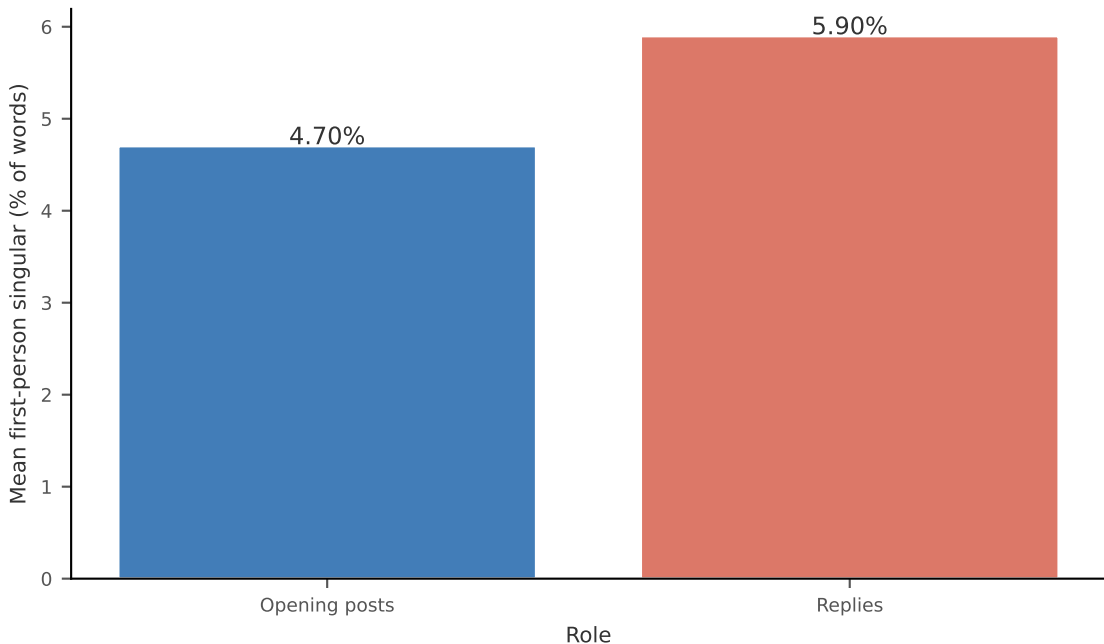
# Top 12 LIWC Categories — Monthly Prevalence Over Time





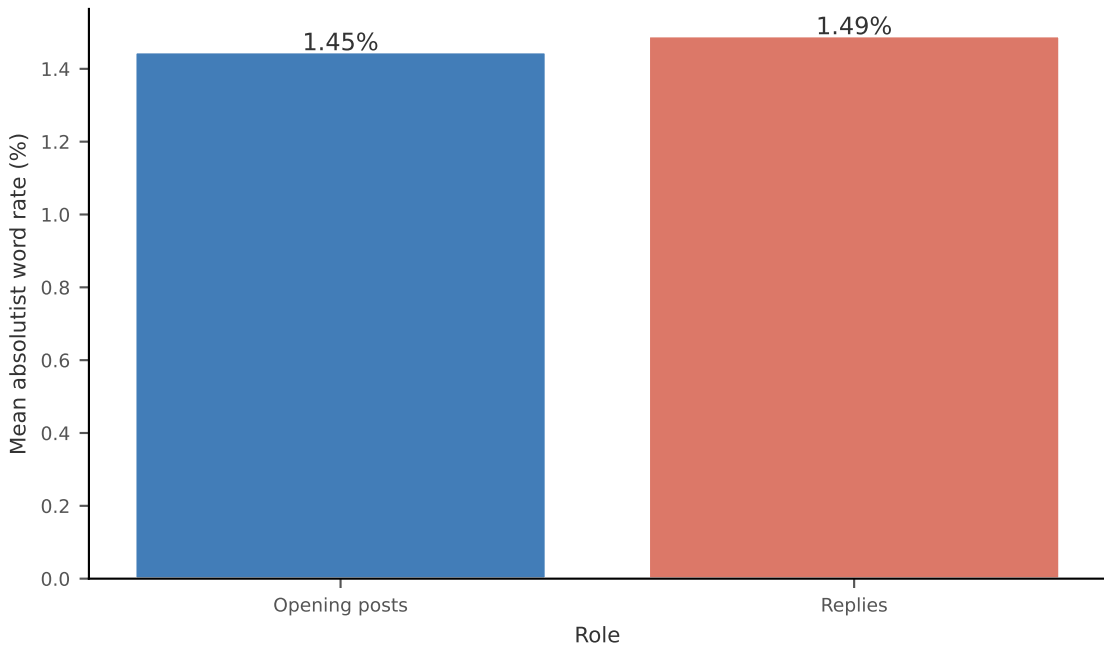
## **Section 4 — First-Person Singular by Role**

# First-Person Singular Rate by Role (self-focus marker; Smirnova 2018, Eichstaedt et al. 2018)



## **Section 5 — Absolutist Words**

## Absolutist Word Rate by Role (Dutch function-word list; Al-Mosaiwi & Johnstone 2018)



## **6. Per-User Longitudinal Trends**

# Depression Connect Forum

Longitudinal Analysis — Top 5 Most Active Posters

# Selected Users — Top by Raw Post Volume

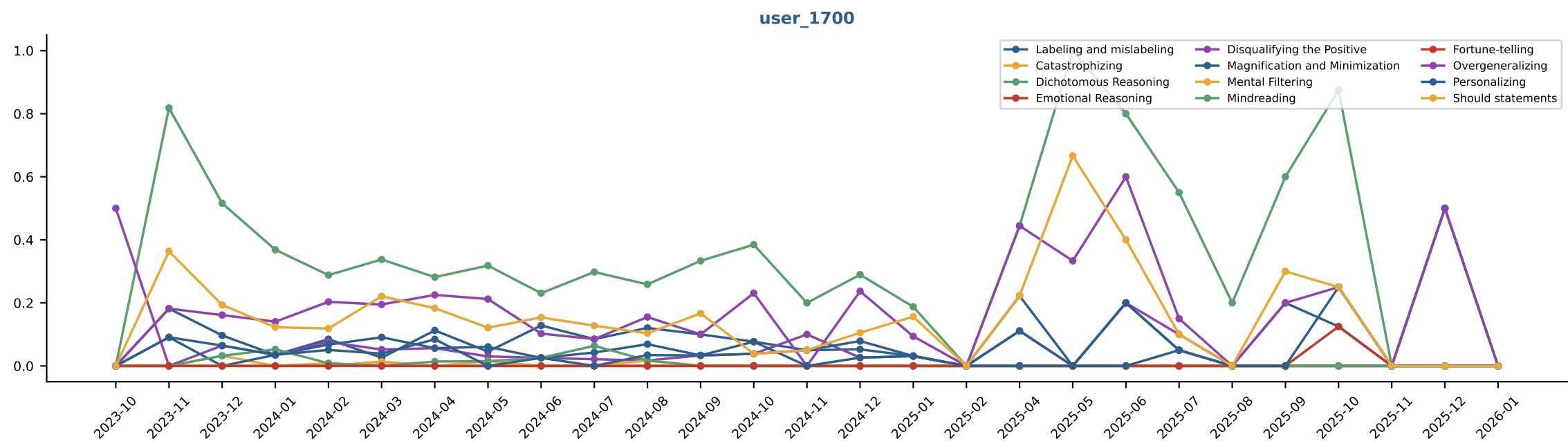
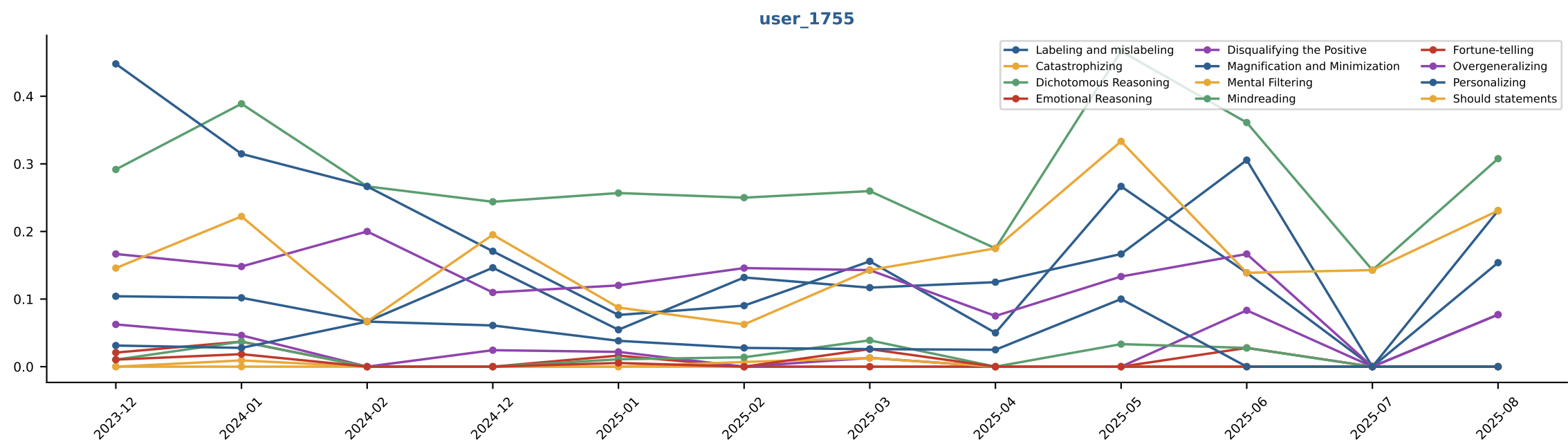
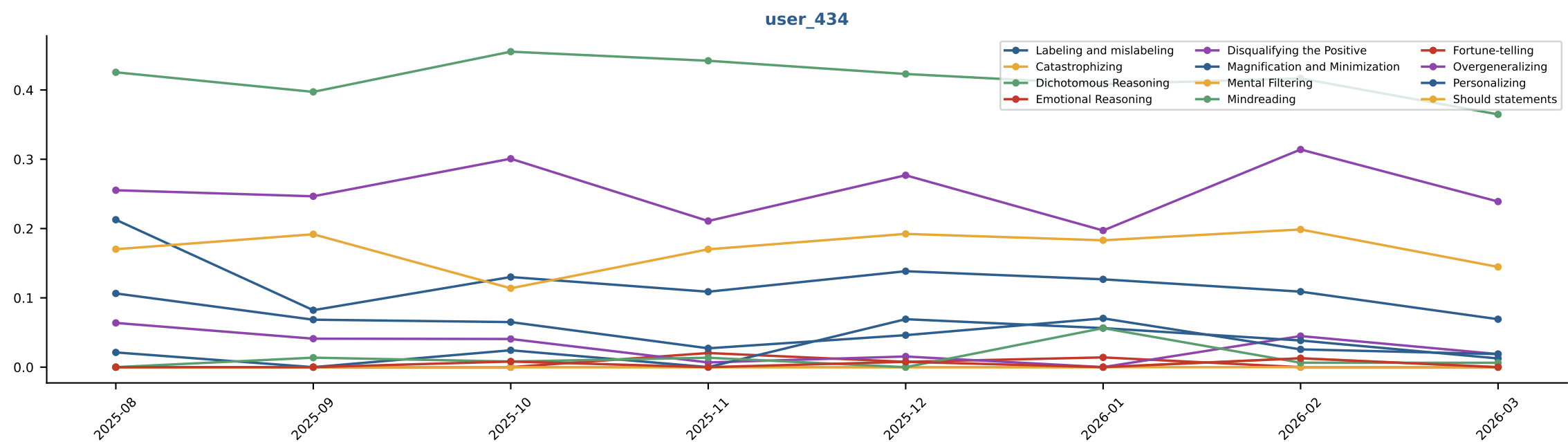
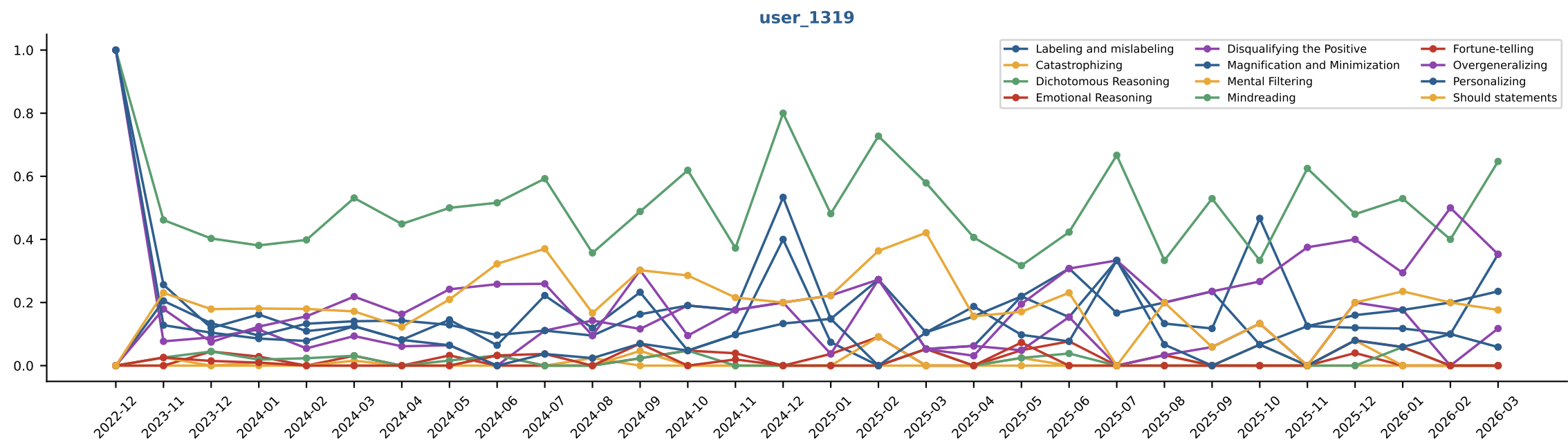
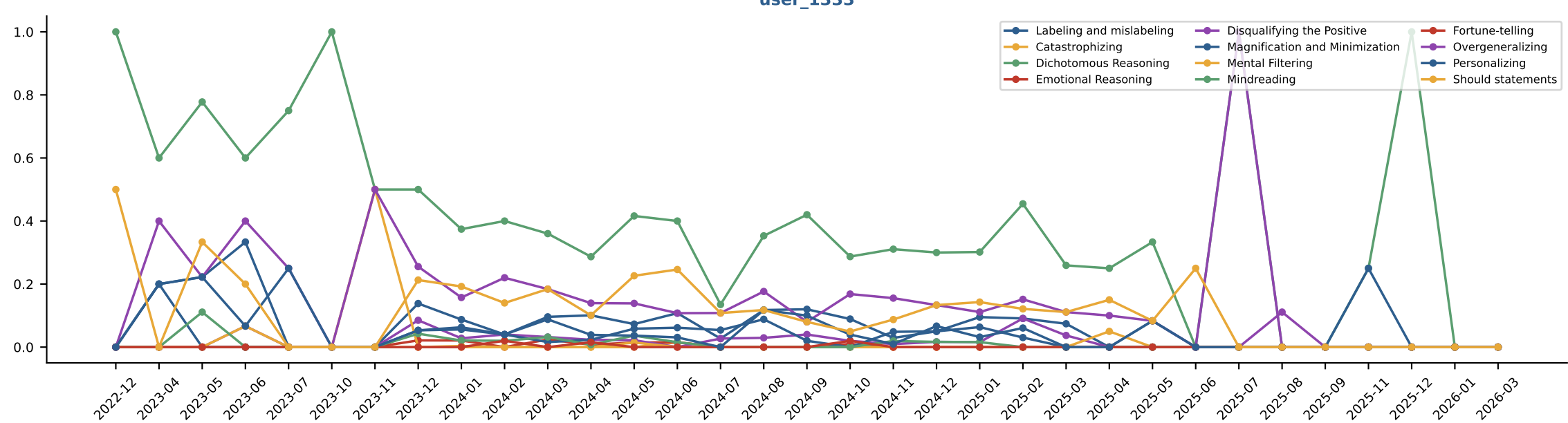
| User      | Posts | Active days | Posts / active month |
|-----------|-------|-------------|----------------------|
| user_1333 | 1486  | 1160        | 38.99                |
| user_1319 | 1046  | 1181        | 26.96                |
| user_434  | 906   | 229         | 120.43               |
| user_1755 | 831   | 630         | 40.15                |
| user_1700 | 799   | 800         | 30.4                 |

Selection is by total post count only. Note this over-weights high-intensity users and can miss long-active, moderate-volume users (see --select sustained).

## CDS Category Scores Per Month

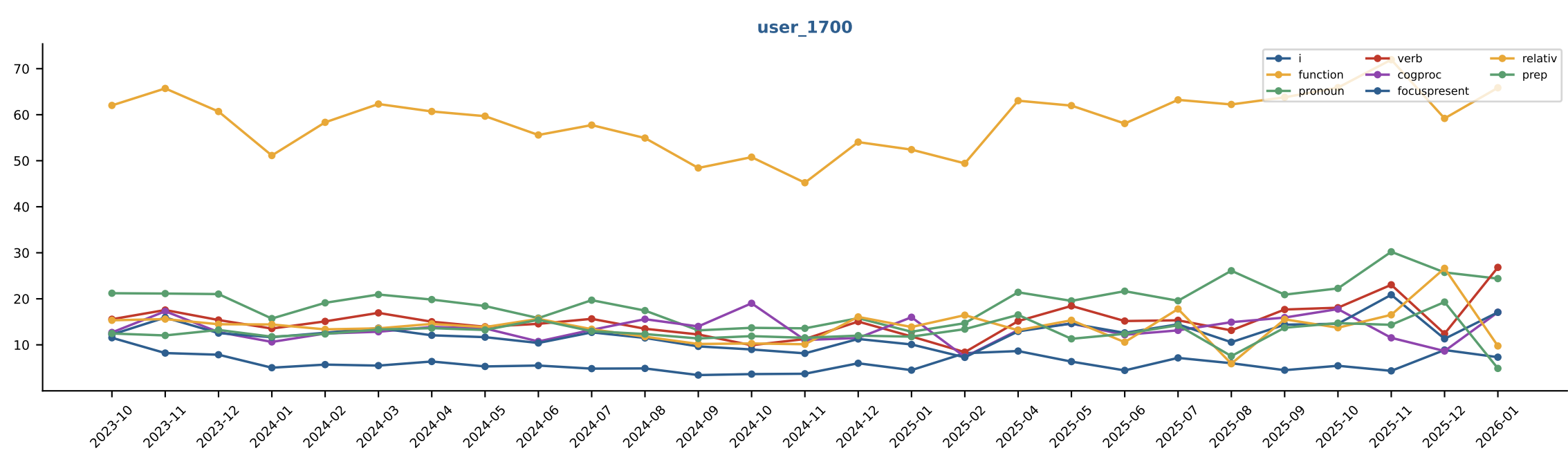
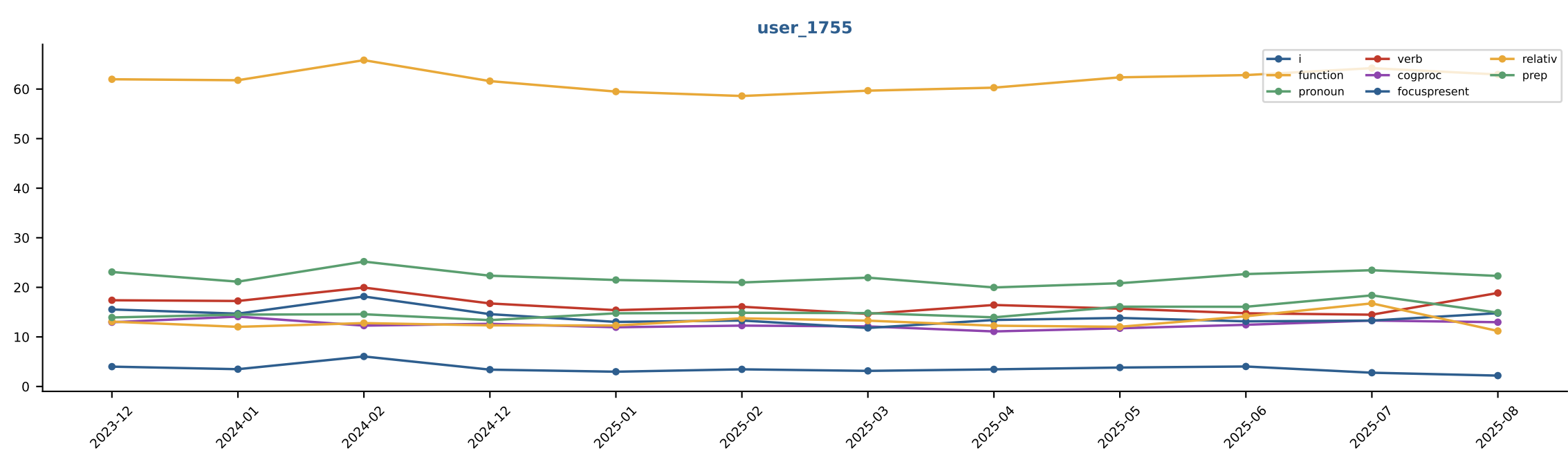
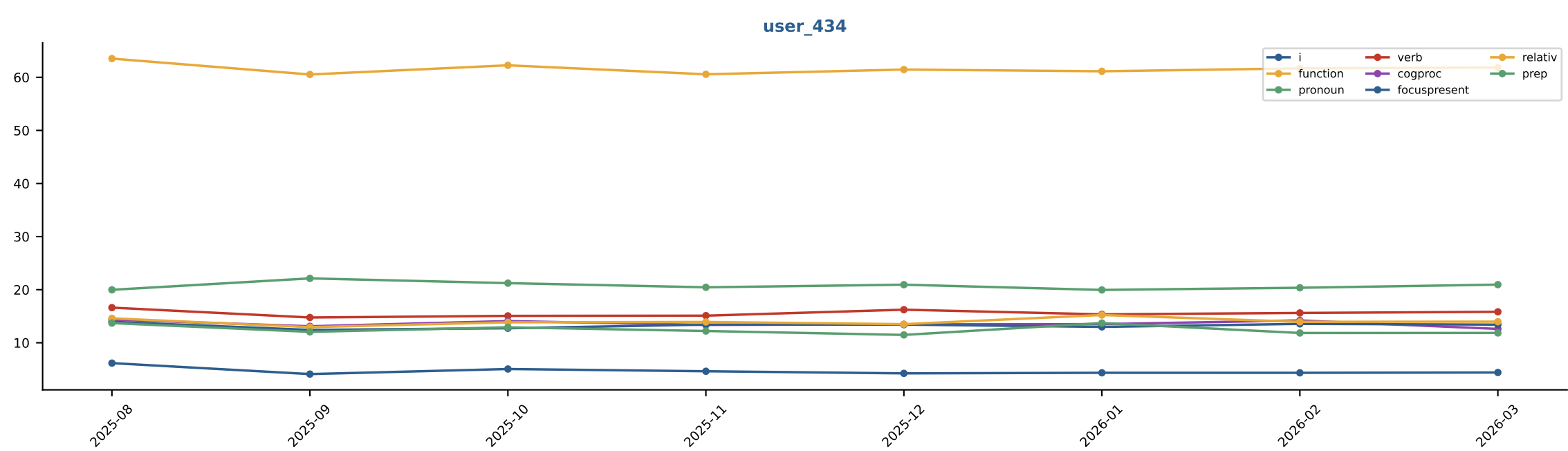
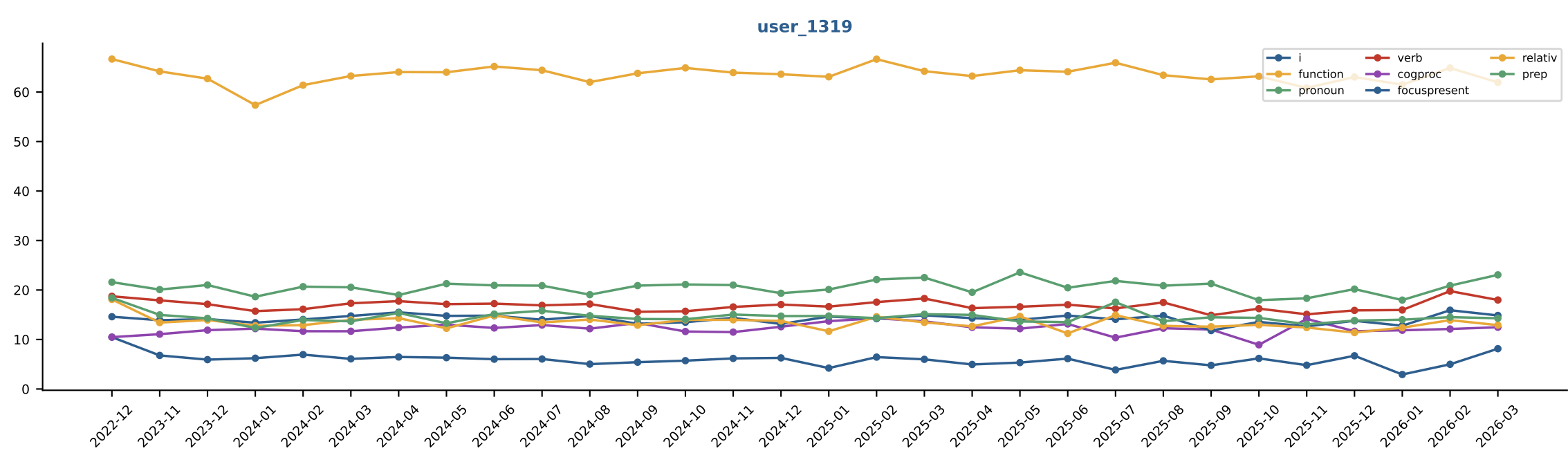
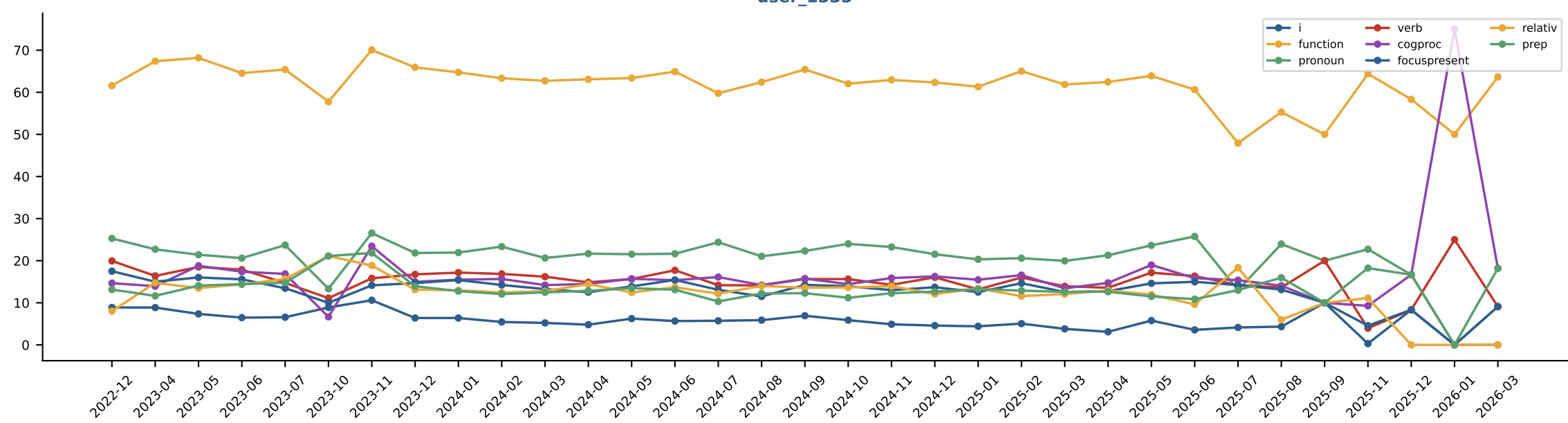


CDS Categories Over Time (mean per month)



## LIWC Category Scores Per Month

### Top LIWC Categories Over Time (mean % of words per month)



## **7. LIWC-22 Validation**

# LIWC-22 Validation Report

Custom scorer vs LIWC-22 CLI (Dutch LIWC-2015 dictionary)

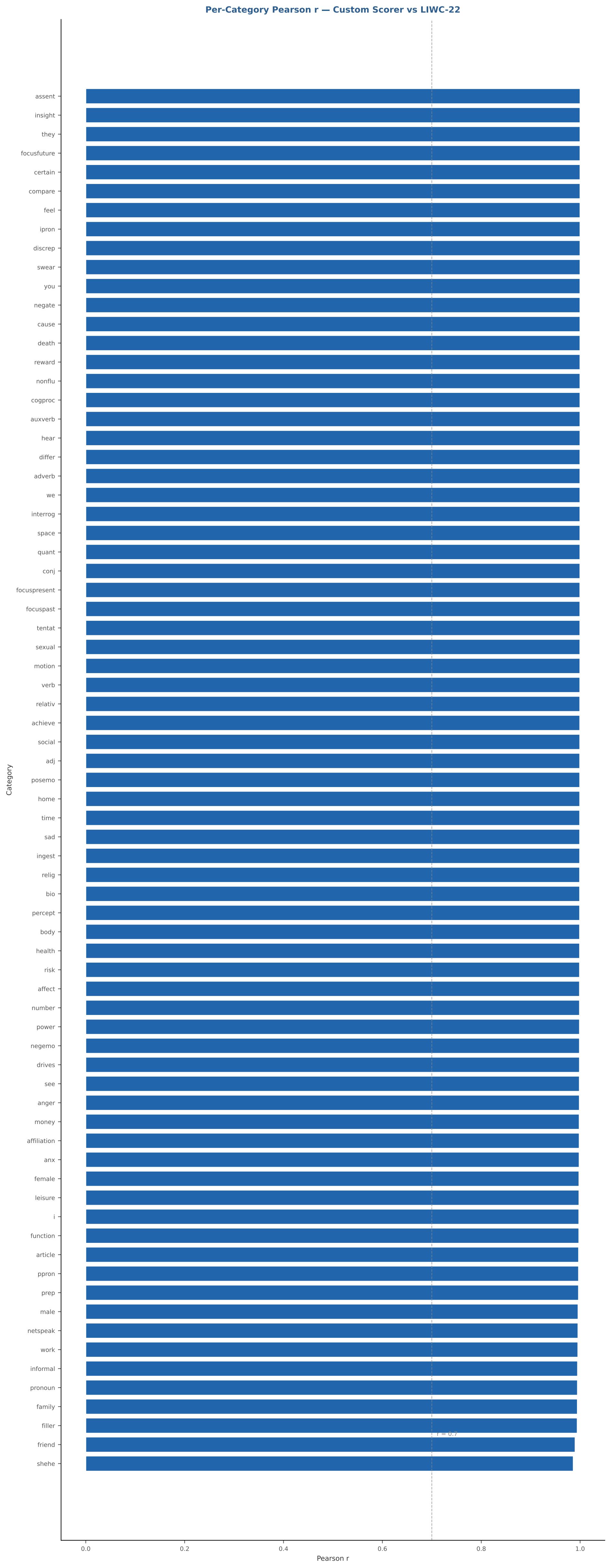
Dataset: New Data Only

# Section 1 — Scoring Agreement

Per-Category Scoring Agreement (sorted by MAE descending)  
MAE = mean absolute difference in percentage points; yellow = MAE > 2 pp

| Category     | Pearson r | p-value | MAE (pp) | Mean (custom) | Mean (LIWC-22) |
|--------------|-----------|---------|----------|---------------|----------------|
| function     | 0.9975    | 0.0     | 0.1682   | 61.8103       | 61.9654        |
| pronoun      | 0.9947    | 0.0     | 0.1538   | 20.6443       | 20.7727        |
| informal     | 0.9949    | 0.0     | 0.0891   | 6.0147        | 5.9415         |
| filler       | 0.9942    | 0.0     | 0.0821   | 4.9521        | 4.8836         |
| prep         | 0.9968    | 0.0     | 0.0747   | 13.204        | 13.1644        |
| ppron        | 0.9969    | 0.0     | 0.0639   | 10.9445       | 11.0042        |
| i            | 0.9975    | 0.0     | 0.0445   | 5.74          | 5.7825         |
| verb         | 0.9995    | 0.0     | 0.0367   | 16.62         | 16.6466        |
| relativ      | 0.9994    | 0.0     | 0.0345   | 13.3323       | 13.3505        |
| focuspresent | 0.9996    | 0.0     | 0.0317   | 14.4647       | 14.4887        |
| cogproc      | 0.9998    | 0.0     | 0.0317   | 15.2702       | 15.2957        |
| article      | 0.997     | 0.0     | 0.0315   | 6.9043        | 6.9333         |
| adverb       | 0.9997    | 0.0     | 0.0297   | 13.2029       | 13.2282        |
| social       | 0.9994    | 0.0     | 0.0267   | 9.8125        | 9.8206         |
| drives       | 0.9987    | 0.0     | 0.0261   | 5.5124        | 5.5081         |
| adj          | 0.9993    | 0.0     | 0.023    | 9.035         | 9.0462         |
| affect       | 0.9988    | 0.0     | 0.0227   | 5.9599        | 5.9607         |
| time         | 0.9992    | 0.0     | 0.0191   | 6.316         | 6.3239         |
| conj         | 0.9997    | 0.0     | 0.0189   | 8.1804        | 8.195          |
| auxverb      | 0.9998    | 0.0     | 0.0172   | 7.3418        | 7.3552         |
| work         | 0.9955    | 0.0     | 0.0161   | 1.2968        | 1.2869         |
| affiliation  | 0.9982    | 0.0     | 0.0151   | 1.9961        | 1.989          |
| space        | 0.9997    | 0.0     | 0.0149   | 5.501         | 5.51           |
| ipron        | 0.9999    | 0.0     | 0.0148   | 6.9022        | 6.9139         |
| focuspast    | 0.9996    | 0.0     | 0.0138   | 5.2866        | 5.2959         |
| negemo       | 0.9987    | 0.0     | 0.0134   | 2.2596        | 2.2561         |
| differ       | 0.9997    | 0.0     | 0.0123   | 4.3389        | 4.3446         |
| shehe        | 0.9862    | 0.0     | 0.0112   | 0.5737        | 0.5842         |
| percept      | 0.9991    | 0.0     | 0.011    | 2.4638        | 2.4644         |
| posemo       | 0.9993    | 0.0     | 0.0107   | 3.6492        | 3.6537         |
| tentat       | 0.9996    | 0.0     | 0.0103   | 3.283         | 3.2905         |
| male         | 0.9959    | 0.0     | 0.0101   | 0.8976        | 0.8915         |
| bio          | 0.9991    | 0.0     | 0.0099   | 1.8169        | 1.8165         |
| netspeak     | 0.9957    | 0.0     | 0.0091   | 0.9805        | 0.9757         |
| compare      | 0.9999    | 0.0     | 0.0091   | 4.0468        | 4.0533         |
| you          | 0.9998    | 0.0     | 0.0085   | 3.9906        | 3.9965         |
| insight      | 0.9999    | 0.0     | 0.0084   | 3.2922        | 3.2976         |
| quant        | 0.9997    | 0.0     | 0.0081   | 2.716         | 2.7195         |
| leisure      | 0.9977    | 0.0     | 0.0076   | 0.624         | 0.6199         |
| focusfuture  | 0.9999    | 0.0     | 0.0074   | 3.2442        | 3.2501         |
| power        | 0.9988    | 0.0     | 0.0069   | 1.5593        | 1.5605         |
| interrog     | 0.9997    | 0.0     | 0.0065   | 2.3332        | 2.3377         |
| see          | 0.9985    | 0.0     | 0.006    | 0.7231        | 0.7209         |
| motion       | 0.9996    | 0.0     | 0.0058   | 1.932         | 1.9343         |
| discrep      | 0.9999    | 0.0     | 0.0057   | 2.3763        | 2.3803         |
| achieve      | 0.9994    | 0.0     | 0.0057   | 1.715         | 1.7176         |
| negate       | 0.9998    | 0.0     | 0.0055   | 1.8803        | 1.8828         |
| health       | 0.999     | 0.0     | 0.005    | 0.8211        | 0.8197         |
| cause        | 0.9998    | 0.0     | 0.0048   | 1.6206        | 1.6229         |
| certain      | 0.9999    | 0.0     | 0.0047   | 1.7475        | 1.7507         |
| reward       | 0.9998    | 0.0     | 0.0041   | 1.3117        | 1.3134         |
| sad          | 0.9992    | 0.0     | 0.0037   | 0.9859        | 0.9869         |
| anx          | 0.9981    | 0.0     | 0.0036   | 0.5294        | 0.5283         |
| family       | 0.9945    | 0.0     | 0.0036   | 0.3517        | 0.3498         |
| risk         | 0.9989    | 0.0     | 0.0036   | 0.5893        | 0.5887         |
| ingest       | 0.9992    | 0.0     | 0.0034   | 0.5829        | 0.5834         |
| friend       | 0.9899    | 0.0     | 0.0026   | 0.2489        | 0.2476         |
| feel         | 0.9999    | 0.0     | 0.0026   | 0.884         | 0.8856         |
| number       | 0.9988    | 0.0     | 0.0026   | 0.6021        | 0.6026         |
| female       | 0.9977    | 0.0     | 0.0023   | 0.508         | 0.508          |
| body         | 0.999     | 0.0     | 0.0021   | 0.4278        | 0.4284         |
| hear         | 0.9998    | 0.0     | 0.002    | 0.537         | 0.538          |
| anger        | 0.9985    | 0.0     | 0.002    | 0.3217        | 0.3216         |
| home         | 0.9993    | 0.0     | 0.0018   | 0.3303        | 0.3302         |
| money        | 0.9982    | 0.0     | 0.0016   | 0.2538        | 0.2538         |
| they         | 0.9999    | 0.0     | 0.0012   | 0.3954        | 0.3961         |
| we           | 0.9997    | 0.0     | 0.0012   | 0.2974        | 0.2976         |
| assent       | 1.0       | 0.0     | 0.0009   | 0.3568        | 0.3573         |
| relig        | 0.9991    | 0.0     | 0.0006   | 0.0674        | 0.0673         |
| death        | 0.9998    | 0.0     | 0.0004   | 0.0912        | 0.0913         |
| swear        | 0.9999    | 0.0     | 0.0003   | 0.0498        | 0.0499         |
| nonflu       | 0.9998    | 0.0     | 0.0002   | 0.0454        | 0.0454         |
| sexual       | 0.9996    | 0.0     | 0.0001   | 0.0146        | 0.0146         |

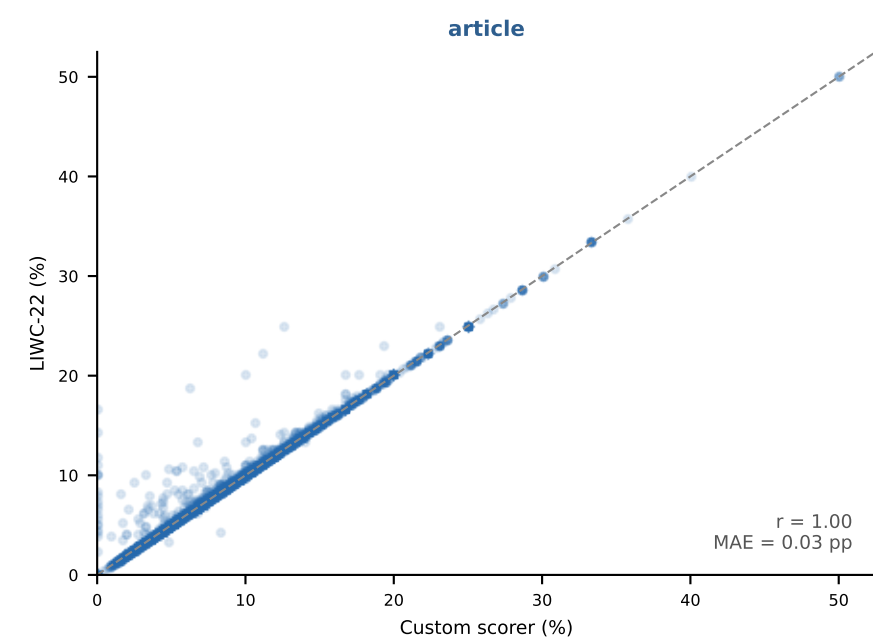
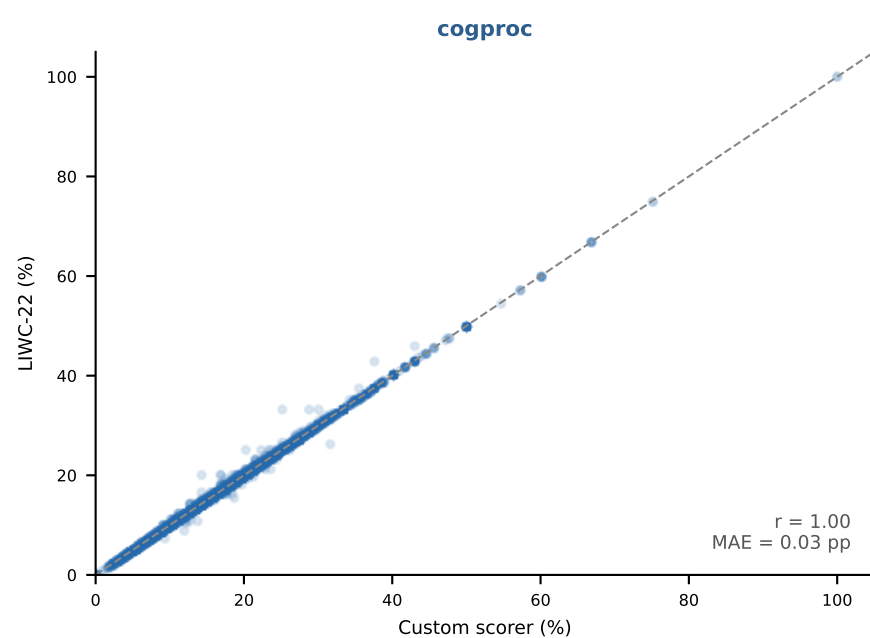
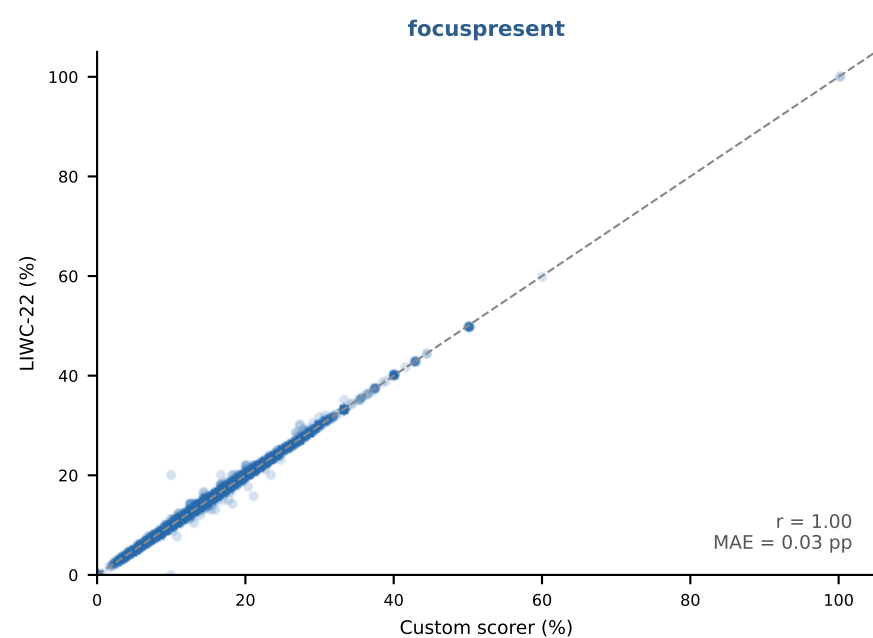
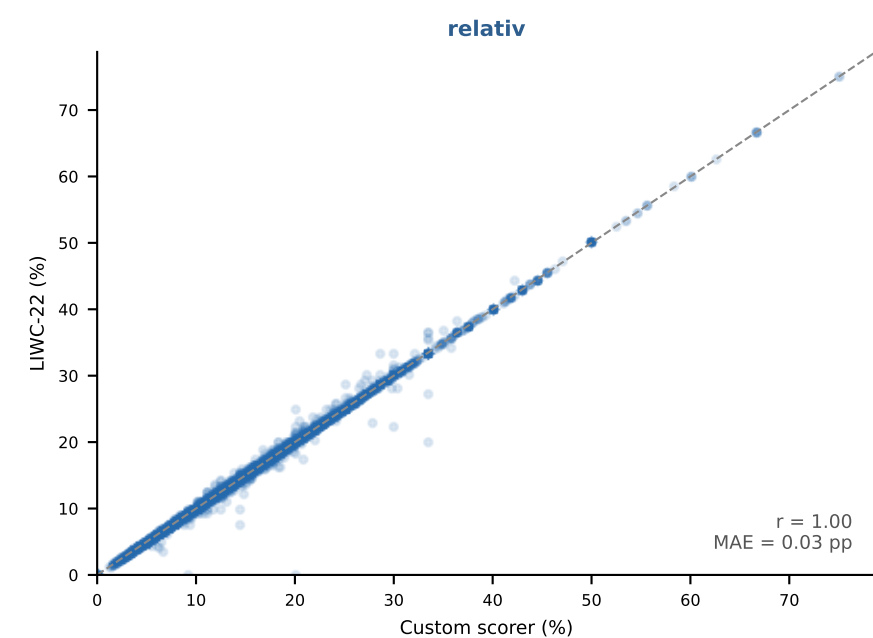
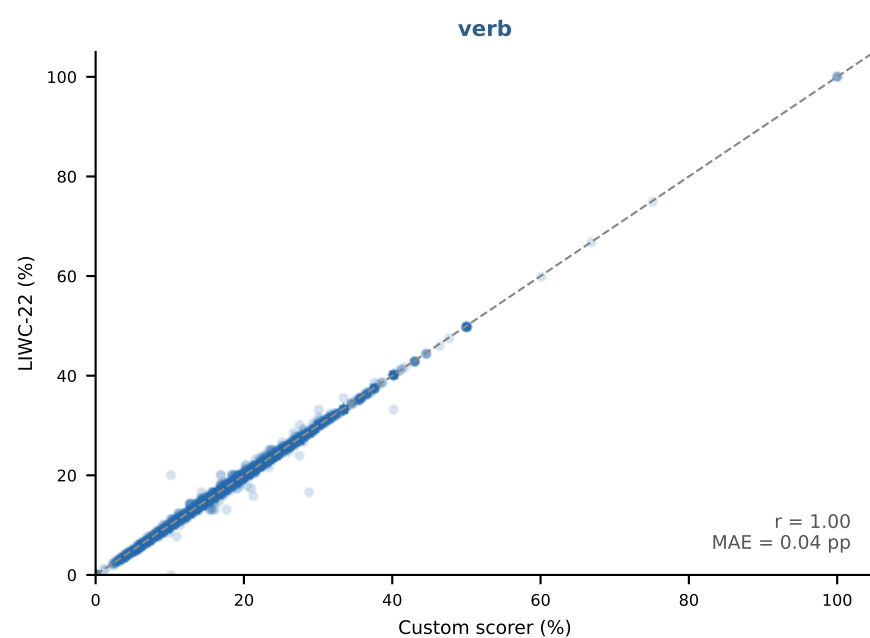
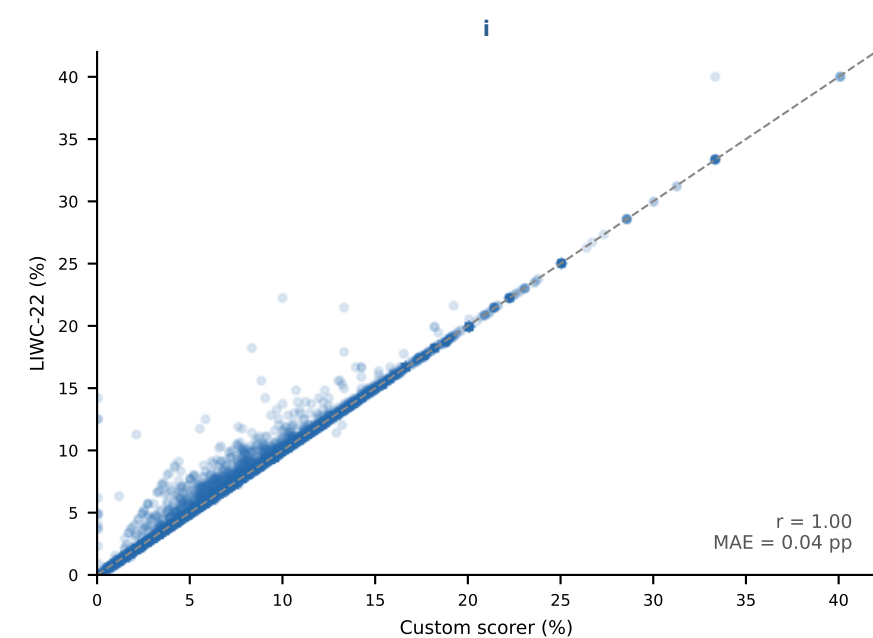
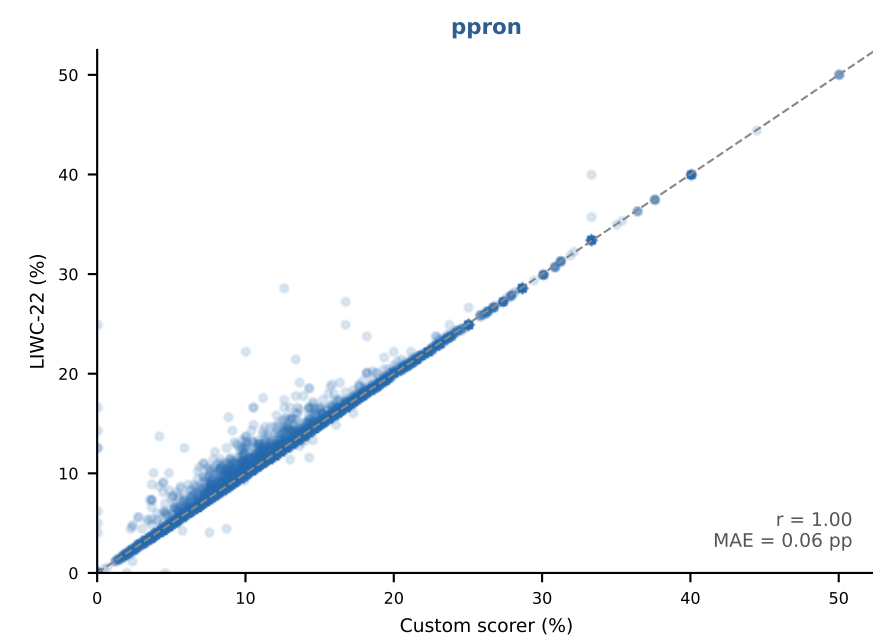
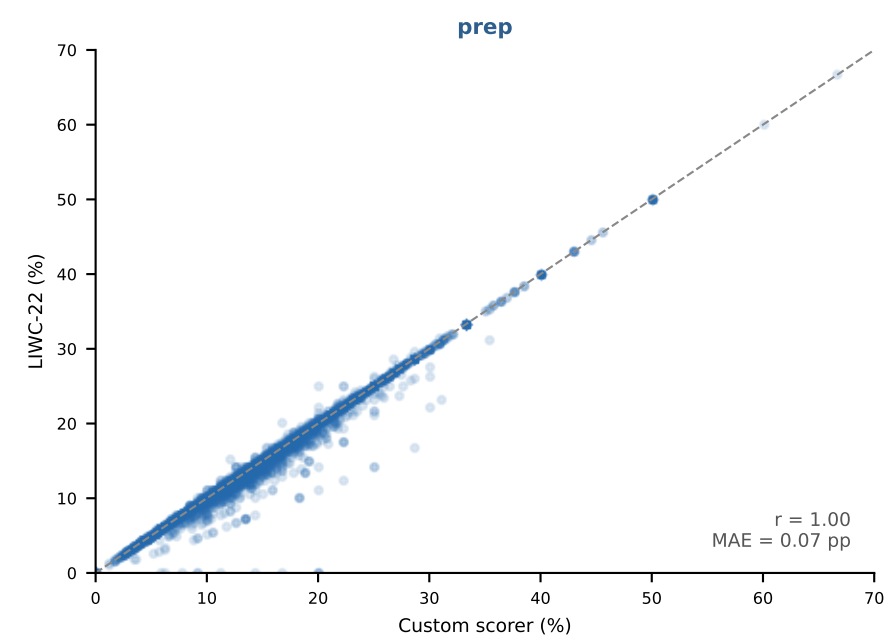
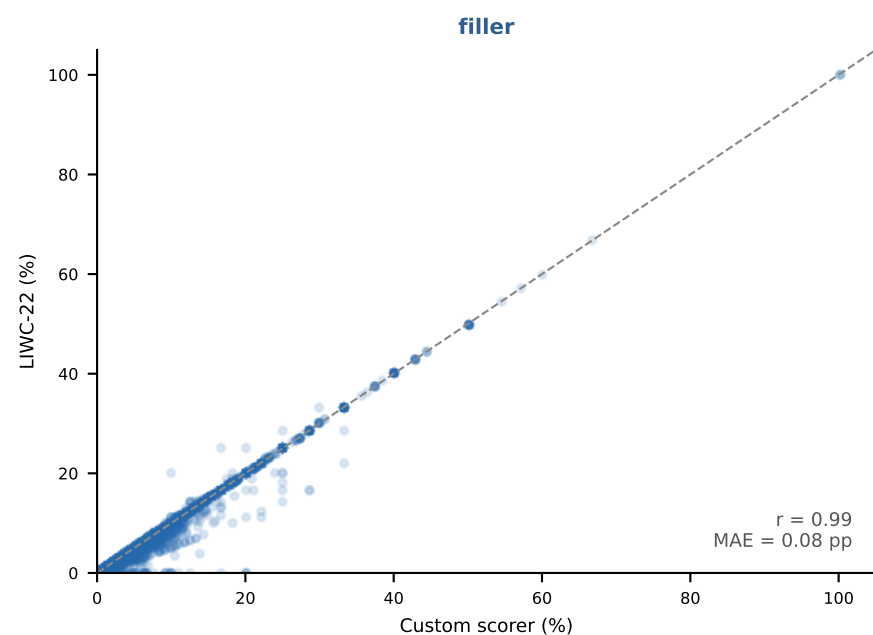
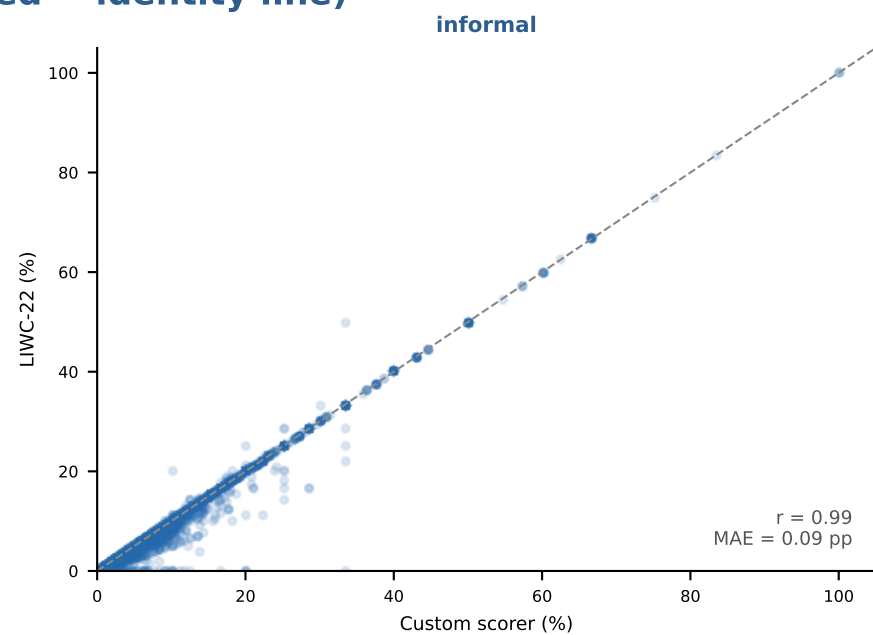
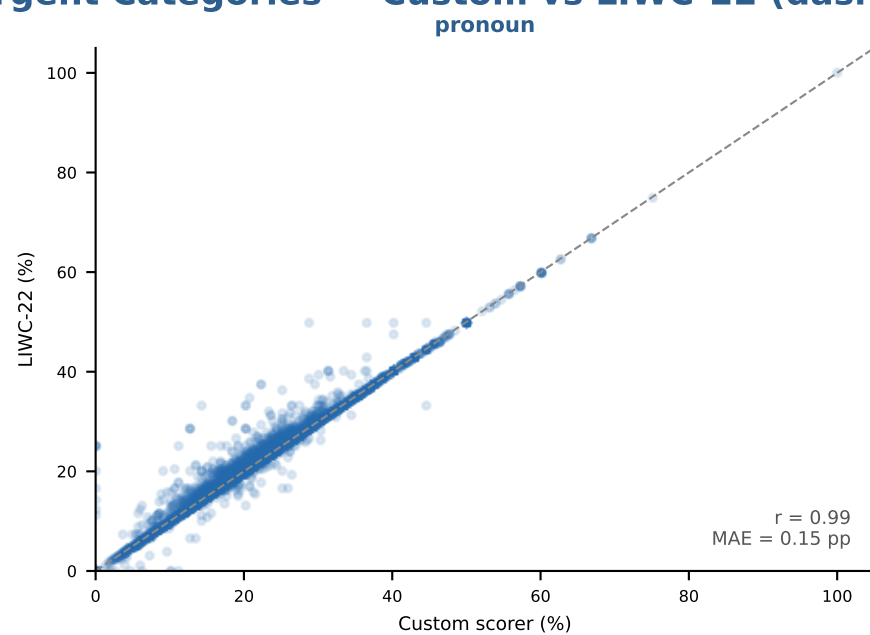
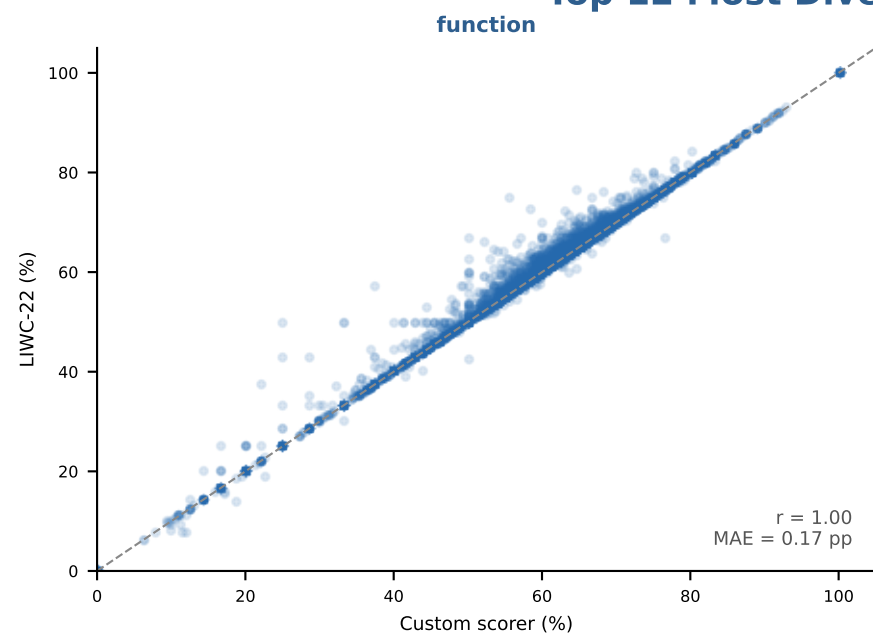
Per-Category Pearson r — Custom Scorer vs LIWC-22





## **Section 2 — Divergence Analysis**

# Top 12 Most Divergent Categories — Custom vs LIWC-22 (dashed = identity line)



## **Section 3 — LIWC-22 Summary Variables**

## LIWC-22 Summary Variables — Not Available with External Dictionary

The four LIWC-22 summary variables are not available in this run:

Analytic · Clout · Authentic · Tone

These scores are computed by LIWC-22 from a proprietary regression model trained on the built-in English dictionary. They do not generalise to external dictionary files, so LIWC-22 omits them when a .dicx file is supplied (as here, with the Dutch LIWC-2015 dictionary).

To obtain these variables you would need to run LIWC-22 with its built-in English dictionary (-d LIWC22), which is not appropriate for Dutch text.

See docs/statistical\_decisions.md §5 for context.

## **Section 4 — Category Coverage Differences**

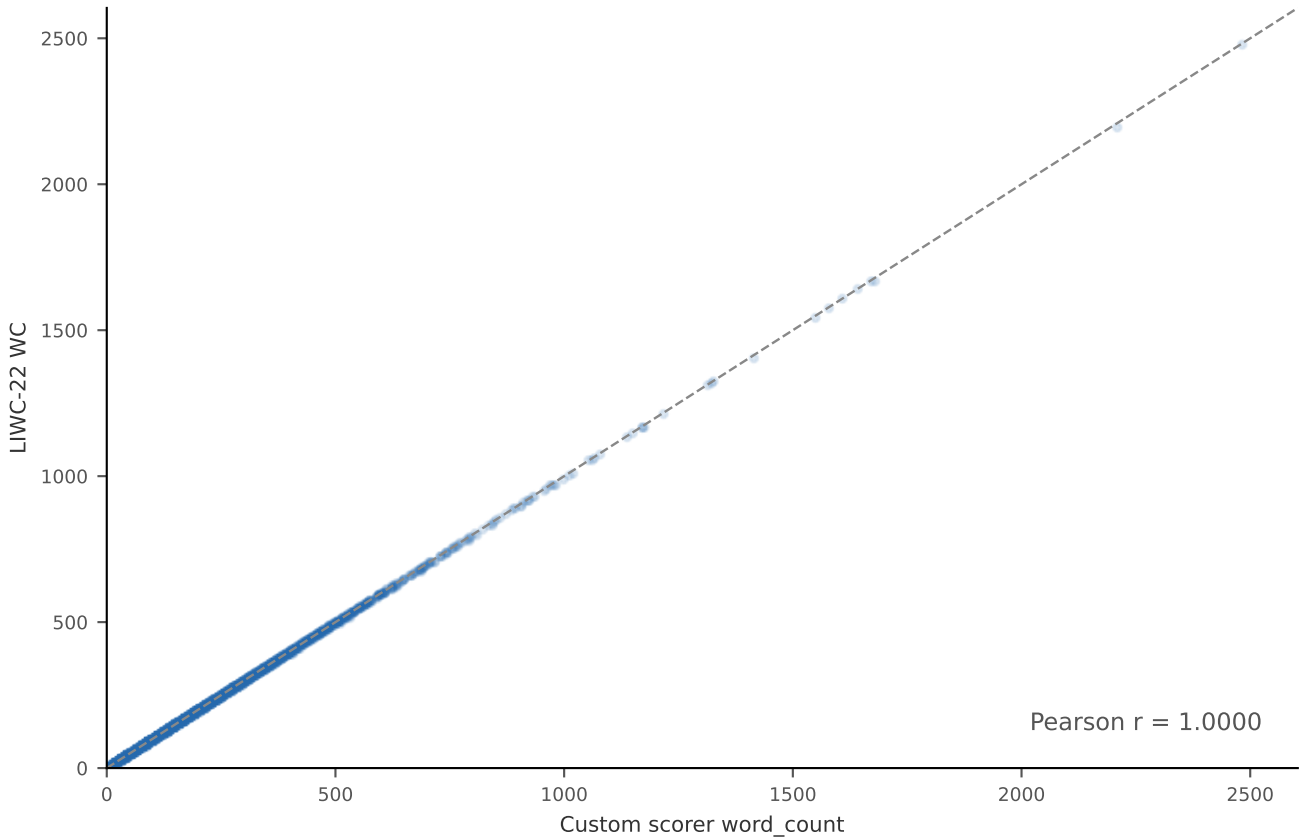
**Category Coverage Differences**

**(custom-only includes fps\_dutch fallback; LIWC-22-only includes structural metrics like WC, Emoji, AllPunc)**

| Custom scorer only | LIWC-22 only |
|--------------------|--------------|
| —                  | —            |

## **Section 5 — Diagnostics**

# Tokenisation Sanity Check — Word Count Agreement (custom scorer re.findall vs LIWC-22 WC; dashed = identity)





## **8. Pandemic-Period Comparison**

# Pandemic-Period Comparison

Psycholinguistic markers pre / during / post pandemic  
after Yahya & Abdul Rahim (2023), with per-user rank-based statistics

Dataset: New Data Only

# Section 1 — Diagnostics

## Period Definitions

```
pre      : PostDate < 2020-03-11
during   : 2020-03-11 ≤ PostDate < 2022-03-23
post     : PostDate ≥ 2022-03-23
```

Cutoff = WHO pandemic declaration (2020-03-11).

End = end of most Dutch COVID restrictions (2022-03-23).

**Dataset Variant × Pandemic Period — messages and unique users**  
⚠ **Period effects are confounded with dataset variant (old export ≈ pre+during, new export ≈ post); yellow = near-empty cell**

| variant  | period | messages | users |
|----------|--------|----------|-------|
| old      | pre    | 4505     | 169   |
| old      | during | 11102    | 291   |
| old      | post   | 2943     | 96    |
| new_only | pre    | 94       | 15    |
| new_only | during | 318      | 23    |
| new_only | post   | 25092    | 520   |
| combined | pre    | 4865     | 177   |
| combined | during | 11960    | 307   |
| combined | post   | 6592     | 247   |

**Users by Number of Periods Appeared In**  
**Users in >1 period violate between-group independence — see the single-period sensitivity analysis in Section 4**

| periods_appeared_in | n_users | pct_users |
|---------------------|---------|-----------|
| 1                   | 508     | 95.8      |
| 2                   | 16      | 3.0       |
| 3                   | 6       | 1.1       |

## **Section 2 — Dictionary Introspection**

**Yahya & Abdul Rahim (2023) §2.3 Categories vs Available Dictionaries**  
**Only available categories are analyzed; '—' = genuinely absent from that source (omission, not a bug)**

| yahya_group    | liwc2015                                                      | liwc22                                                        | dic_crosscheck                                                |
|----------------|---------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|
| pronouns       | i, ipron, ppron, pronoun, shehe, they, we, you                | i, ipron, ppron, pronoun, shehe, they, we, you                | i, ipron, ppron, pronoun, shehe, they, we, you                |
| function_words | article, auxverb, conj, function, negate, number, prep, quant | article, auxverb, conj, function, negate, number, prep, quant | article, auxverb, conj, function, negate, number, prep, quant |
| other_grammar  | adj, adverb, compare, interrog, verb                          | adj, adverb, compare, interrog, verb                          | adj, adverb, compare, interrog, verb                          |
| tense          | focusfuture, focuspast, focuspresent                          | focusfuture, focuspast, focuspresent                          | focusfuture, focuspast, focuspresent                          |
| emotion        | affect, anger, anx, negemo, posemo, sad                       | affect, anger, anx, negemo, posemo, sad                       | affect, anger, anx, negemo, posemo, sad                       |
| informal       | assent, filler, informal, netspeak, nonflu, swear             | assent, filler, informal, netspeak, nonflu, swear             | assent, filler, informal, netspeak, nonflu, swear             |
| summary        | —                                                             | —                                                             | —                                                             |
| absolutist     | absolutist                                                    | —                                                             | —                                                             |



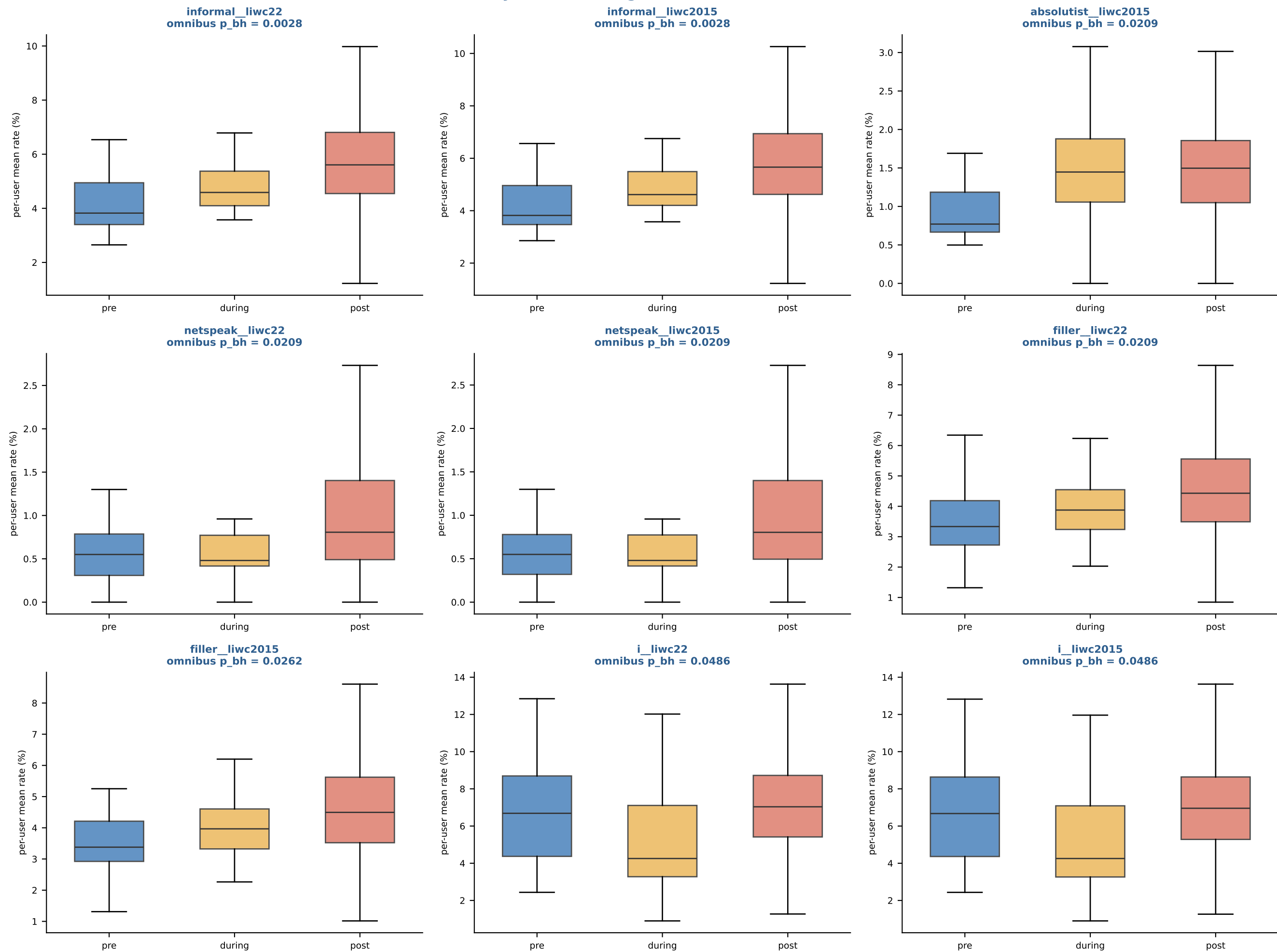
## **Section 3 — Main Results (all users)**

| Kruskal-Wallis Omnibus per Feature (per-user rates)<br>effect_size = epsilon-squared; p_bh = BH over pooled omnibus + post-hoc family; yellow = significant |       |          |        |            |           |          |             |      |          |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------|--------|------------|-----------|----------|-------------|------|----------|-------------|
| feature                                                                                                                                                     | n_pre | n_during | n_post | comparison | statistic | p_raw    | effect_size | note | p_bh     | significant |
| informal_liwc22                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 20.198    | 4.1e-05  | 0.0363      |      | 0.002812 | True        |
| informal_liwc2015                                                                                                                                           | 15.0  | 23.0     | 520.0  | omnibus    | 19.4482   | 6e-05    | 0.0349      |      | 0.002812 | True        |
| absolutist_liwc2015                                                                                                                                         | 15.0  | 23.0     | 520.0  | omnibus    | 11.9202   | 0.00258  | 0.0214      |      | 0.020863 | True        |
| netspeak_liwc22                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 12.6004   | 0.001836 | 0.0226      |      | 0.020863 | True        |
| netspeak_liwc2015                                                                                                                                           | 15.0  | 23.0     | 520.0  | omnibus    | 12.9307   | 0.001556 | 0.0232      |      | 0.020863 | True        |
| filler_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 11.9039   | 0.002601 | 0.0214      |      | 0.020863 | True        |
| filler_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 11.0912   | 0.003905 | 0.0199      |      | 0.026217 | True        |
| i_liwc22                                                                                                                                                    | 15.0  | 23.0     | 520.0  | omnibus    | 9.2881    | 0.009619 | 0.0167      |      | 0.048619 | True        |
| i_liwc2015                                                                                                                                                  | 15.0  | 23.0     | 520.0  | omnibus    | 9.2452    | 0.009827 | 0.0166      |      | 0.048619 | True        |
| they_liwc2015                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 6.413     | 0.040498 | 0.0115      |      | 0.16537  | False       |
| sad_liwc22                                                                                                                                                  | 15.0  | 23.0     | 520.0  | omnibus    | 6.0941    | 0.0475   | 0.0109      |      | 0.16537  | False       |
| sad_liwc2015                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 6.1533    | 0.046114 | 0.011       |      | 0.16537  | False       |
| they_liwc22                                                                                                                                                 | 15.0  | 23.0     | 520.0  | omnibus    | 6.4345    | 0.040065 | 0.0116      |      | 0.16537  | False       |
| negemo_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 6.0952    | 0.047474 | 0.0109      |      | 0.16537  | False       |
| negemo_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 5.9544    | 0.050936 | 0.0107      |      | 0.170999 | False       |
| you_liwc2015                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 5.0497    | 0.08007  | 0.0091      |      | 0.217197 | False       |
| you_liwc22                                                                                                                                                  | 15.0  | 23.0     | 520.0  | omnibus    | 5.1115    | 0.077634 | 0.0092      |      | 0.217197 | False       |
| prep_liwc2015                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 4.9876    | 0.082594 | 0.009       |      | 0.217197 | False       |
| ipron_liwc22                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 5.0023    | 0.081991 | 0.009       |      | 0.217197 | False       |
| prep_liwc22                                                                                                                                                 | 15.0  | 23.0     | 520.0  | omnibus    | 5.0906    | 0.07845  | 0.0091      |      | 0.217197 | False       |
| adj_liwc2015                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 4.9492    | 0.084198 | 0.0089      |      | 0.217197 | False       |
| ipron_liwc2015                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 4.9187    | 0.085492 | 0.0088      |      | 0.217197 | False       |
| nonflu_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 5.2843    | 0.071208 | 0.0095      |      | 0.217197 | False       |
| adj_liwc22                                                                                                                                                  | 15.0  | 23.0     | 520.0  | omnibus    | 4.7392    | 0.093516 | 0.0085      |      | 0.219763 | False       |
| we_liwc2015                                                                                                                                                 | 15.0  | 23.0     | 520.0  | omnibus    | 4.747     | 0.093152 | 0.0085      |      | 0.219763 | False       |
| we_liwc22                                                                                                                                                   | 15.0  | 23.0     | 520.0  | omnibus    | 4.7592    | 0.092589 | 0.0085      |      | 0.219763 | False       |
| pronoun_liwc2015                                                                                                                                            | 15.0  | 23.0     | 520.0  | omnibus    | 4.4844    | 0.106225 | 0.0081      |      | 0.243541 | False       |
| pronoun_liwc22                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 4.4327    | 0.109006 | 0.008       |      | 0.243966 | False       |
| shehe_liwc2015                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 3.8922    | 0.142827 | 0.007       |      | 0.303082 | False       |
| shehe_liwc22                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 3.8608    | 0.145092 | 0.0069      |      | 0.303082 | False       |
| focuspast_liwc2015                                                                                                                                          | 15.0  | 23.0     | 520.0  | omnibus    | 3.5761    | 0.167283 | 0.0064      |      | 0.328197 | False       |
| focuspast_liwc22                                                                                                                                            | 15.0  | 23.0     | 520.0  | omnibus    | 3.5312    | 0.171082 | 0.0063      |      | 0.328197 | False       |
| nonflu_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 3.2503    | 0.196885 | 0.0058      |      | 0.362886 | False       |
| assent_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 3.1495    | 0.20706  | 0.0057      |      | 0.374301 | False       |
| quant_liwc2015                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 3.0445    | 0.218224 | 0.0055      |      | 0.387039 | False       |
| quant_liwc22                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 2.9826    | 0.225082 | 0.0054      |      | 0.39181  | False       |
| adverb_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 2.816     | 0.244636 | 0.0051      |      | 0.410638 | False       |
| adverb_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 2.8235    | 0.243714 | 0.0051      |      | 0.410638 | False       |
| function_liwc22                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 2.7701    | 0.250314 | 0.005       |      | 0.412799 | False       |
| function_liwc2015                                                                                                                                           | 15.0  | 23.0     | 520.0  | omnibus    | 2.6858    | 0.261084 | 0.0048      |      | 0.423136 | False       |
| assent_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 2.5362    | 0.281363 | 0.0046      |      | 0.448273 | False       |
| posemo_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 2.4788    | 0.289561 | 0.0045      |      | 0.453645 | False       |
| posemo_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 2.4028    | 0.300779 | 0.0043      |      | 0.463495 | False       |
| article_liwc22                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 2.2727    | 0.32099  | 0.0041      |      | 0.486662 | False       |
| article_liwc2015                                                                                                                                            | 15.0  | 23.0     | 520.0  | omnibus    | 2.1718    | 0.337599 | 0.0039      |      | 0.50372  | False       |
| anger_liwc22                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 2.0541    | 0.358056 | 0.0037      |      | 0.517805 | False       |
| anger_liwc2015                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 2.076     | 0.354158 | 0.0037      |      | 0.517805 | False       |
| focuspresent_liwc2015                                                                                                                                       | 15.0  | 23.0     | 520.0  | omnibus    | 1.5778    | 0.454352 | 0.0028      |      | 0.639541 | False       |
| focuspresent_liwc22                                                                                                                                         | 15.0  | 23.0     | 520.0  | omnibus    | 1.5712    | 0.455843 | 0.0028      |      | 0.639541 | False       |
| swear_liwc2015                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 1.4252    | 0.490372 | 0.0026      |      | 0.668043 | False       |
| swear_liwc22                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 1.4527    | 0.483674 | 0.0026      |      | 0.668043 | False       |
| focusfuture_liwc2015                                                                                                                                        | 15.0  | 23.0     | 520.0  | omnibus    | 1.1593    | 0.560087 | 0.0021      |      | 0.746401 | False       |
| focusfuture_liwc22                                                                                                                                          | 15.0  | 23.0     | 520.0  | omnibus    | 1.1113    | 0.573701 | 0.002       |      | 0.746401 | False       |
| affect_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 1.1336    | 0.567332 | 0.002       |      | 0.746401 | False       |
| verb_liwc22                                                                                                                                                 | 15.0  | 23.0     | 520.0  | omnibus    | 1.0907    | 0.579652 | 0.002       |      | 0.746401 | False       |
| verb_liwc2015                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 1.0577    | 0.58927  | 0.0019      |      | 0.748533 | False       |
| affect_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 1.0036    | 0.605446 | 0.0018      |      | 0.758826 | False       |
| anx_liwc2015                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 0.6479    | 0.7233   | 0.0012      |      | 0.894608 | False       |
| anx_liwc22                                                                                                                                                  | 15.0  | 23.0     | 520.0  | omnibus    | 0.5776    | 0.749155 | 0.001       |      | 0.908324 | False       |
| compare_liwc2015                                                                                                                                            | 15.0  | 23.0     | 520.0  | omnibus    | 0.3735    | 0.829662 | 0.0007      |      | 0.932345 | False       |
| auxverb_liwc2015                                                                                                                                            | 15.0  | 23.0     | 520.0  | omnibus    | 0.3873    | 0.82394  | 0.0007      |      | 0.932345 | False       |
| number_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 0.3714    | 0.830534 | 0.0007      |      | 0.932345 | False       |
| compare_liwc22                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 0.3671    | 0.8323   | 0.0007      |      | 0.932345 | False       |
| auxverb_liwc22                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 0.3414    | 0.843077 | 0.0006      |      | 0.932345 | False       |
| number_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 0.3555    | 0.837145 | 0.0006      |      | 0.932345 | False       |
| conj_liwc2015                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 0.2706    | 0.873461 | 0.0005      |      | 0.94374  | False       |
| conj_liwc22                                                                                                                                                 | 15.0  | 23.0     | 520.0  | omnibus    | 0.2813    | 0.868808 | 0.0005      |      | 0.94374  | False       |
| negate_liwc22                                                                                                                                               | 15.0  | 23.0     | 520.0  | omnibus    | 0.1516    | 0.926989 | 0.0003      |      | 0.970638 | False       |
| negate_liwc2015                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 0.1396    | 0.932601 | 0.0003      |      | 0.970638 | False       |
| ppron_liwc2015                                                                                                                                              | 15.0  | 23.0     | 520.0  | omnibus    | 0.1521    | 0.926766 | 0.0003      |      | 0.970638 | False       |
| ppron_liwc22                                                                                                                                                | 15.0  | 23.0     | 520.0  | omnibus    | 0.1245    | 0.93966  | 0.0002      |      | 0.970638 | False       |
| interrog_liwc2015                                                                                                                                           | 15.0  | 23.0     | 520.0  | omnibus    | 0.013     | 0.993527 | 0.0         |      | 0.993527 | False       |
| interrog_liwc22                                                                                                                                             | 15.0  | 23.0     | 520.0  | omnibus    | 0.0162    | 0.991929 | 0.0         |      | 0.993527 | False       |

Post-hoc Pairwise Mann-Whitney U (only BH-surviving omnibus features)  
effect\_size = rank-biserial correlation; yellow = significant

| feature             | n_pre | n_during | n_post | comparison     | statistic | p_raw    | effect_size | note | p_bh     | significant |
|---------------------|-------|----------|--------|----------------|-----------|----------|-------------|------|----------|-------------|
| absolutist_liwc2015 |       | 23.0     | 520.0  | during_vs_post | 5948.5    | 0.966418 | 0.0053      |      | 0.987427 | False       |
| absolutist_liwc2015 | 15.0  | 23.0     |        | pre_vs_during  | 84.0      | 0.008588 | 0.513       |      | 0.047486 | True        |
| absolutist_liwc2015 | 15.0  |          | 520.0  | pre_vs_post    | 1862.0    | 0.000557 | 0.5226      |      | 0.016734 | True        |
| filler_liwc2015     |       | 23.0     | 520.0  | during_vs_post | 4590.5    | 0.059245 | 0.2324      |      | 0.192034 | False       |
| filler_liwc2015     | 15.0  | 23.0     |        | pre_vs_during  | 126.0     | 0.16952  | 0.2696      |      | 0.328197 | False       |
| filler_liwc2015     | 15.0  |          | 520.0  | pre_vs_post    | 2249.0    | 0.00517  | 0.4233      |      | 0.030373 | True        |
| filler_liwc22       |       | 23.0     | 520.0  | during_vs_post | 4438.0    | 0.036306 | 0.2579      |      | 0.16537  | False       |
| filler_liwc22       | 15.0  | 23.0     |        | pre_vs_during  | 126.0     | 0.169497 | 0.2696      |      | 0.328197 | False       |
| filler_liwc22       | 15.0  |          | 520.0  | pre_vs_post    | 2247.0    | 0.005116 | 0.4238      |      | 0.030373 | True        |
| informal_liwc2015   |       | 23.0     | 520.0  | during_vs_post | 3816.0    | 0.003301 | 0.3619      |      | 0.02387  | True        |
| informal_liwc2015   | 15.0  | 23.0     |        | pre_vs_during  | 119.0     | 0.113468 | 0.3101      |      | 0.248047 | False       |
| informal_liwc2015   | 15.0  |          | 520.0  | pre_vs_post    | 1909.0    | 0.000745 | 0.5105      |      | 0.016734 | True        |
| informal_liwc22     |       | 23.0     | 520.0  | during_vs_post | 3678.0    | 0.001774 | 0.3849      |      | 0.020863 | True        |
| informal_liwc22     | 15.0  | 23.0     |        | pre_vs_during  | 127.0     | 0.178986 | 0.2638      |      | 0.336494 | False       |
| informal_liwc22     | 15.0  |          | 520.0  | pre_vs_post    | 1938.0    | 0.00089  | 0.5031      |      | 0.016734 | True        |
| netspeak_liwc2015   |       | 23.0     | 520.0  | during_vs_post | 3751.0    | 0.002474 | 0.3727      |      | 0.020863 | True        |
| netspeak_liwc2015   | 15.0  | 23.0     |        | pre_vs_during  | 182.5     | 0.776515 | -0.058      |      | 0.923954 | False       |
| netspeak_liwc2015   | 15.0  |          | 520.0  | pre_vs_post    | 2693.5    | 0.041031 | 0.3094      |      | 0.16537  | False       |
| netspeak_liwc22     |       | 23.0     | 520.0  | during_vs_post | 3767.5    | 0.002663 | 0.37        |      | 0.020863 | True        |
| netspeak_liwc22     | 15.0  | 23.0     |        | pre_vs_during  | 183.5     | 0.753715 | -0.0638     |      | 0.908324 | False       |
| netspeak_liwc22     | 15.0  |          | 520.0  | pre_vs_post    | 2724.0    | 0.046421 | 0.3015      |      | 0.16537  | False       |

# Per-User Rates by Period — Significant Features (BH-corrected)



## **Section 4 — Sensitivity: Single-Period Users Only**

Same Design, Restricted to Users Appearing in Exactly One Period  
Removes the between-group dependence of multi-period users; compare significance patterns with Section 3

| feature               | n_pre | n_during | n_post | comparison | statistic | p_raw | effect_size | note                        | p_bh | significant |
|-----------------------|-------|----------|--------|------------|-----------|-------|-------------|-----------------------------|------|-------------|
| adj_liwc2015          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| adverb_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| affect_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| anger_liwc2015        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| anx_liwc2015          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| article_liwc2015      | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| assent_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| auxverb_liwc2015      | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| compare_liwc2015      | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| conj_liwc2015         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| filler_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| focusfuture_liwc2015  | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| focuspast_liwc2015    | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| focuspresent_liwc2015 | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| function_liwc2015     | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| i_liwc2015            | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| informal_liwc2015     | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| interrog_liwc2015     | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| ipron_liwc2015        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| negate_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| negemo_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| netspeak_liwc2015     | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| nonflu_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| number_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| posemo_liwc2015       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| ppron_liwc2015        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| prep_liwc2015         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| pronoun_liwc2015      | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| quant_liwc2015        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| sad_liwc2015          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| shehe_liwc2015        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| swear_liwc2015        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| they_liwc2015         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| verb_liwc2015         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| we_liwc2015           | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| you_liwc2015          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| absolutist_liwc2015   | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| function_liwc22       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| pronoun_liwc22        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| ppron_liwc22          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| i_liwc22              | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| we_liwc22             | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| you_liwc22            | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| shehe_liwc22          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| they_liwc22           | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| ipron_liwc22          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| article_liwc22        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| prep_liwc22           | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| auxverb_liwc22        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| adverb_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| conj_liwc22           | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| negate_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| verb_liwc22           | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| adj_liwc22            | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| compare_liwc22        | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| interrog_liwc22       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| number_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| quant_liwc22          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| affect_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| posemo_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| negemo_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| anx_liwc22            | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| anger_liwc22          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| sad_liwc22            | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| focuspast_liwc22      | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| focuspresent_liwc22   | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| focusfuture_liwc22    | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| informal_liwc22       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| swear_liwc22          | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| netspeak_liwc22       | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| assent_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| nonflu_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |
| filler_liwc22         | 3     | 2        | 503    | omnibus    |           |       |             | skipped: <2 periods with ≥5 |      | False       |